

Instrumentation, Message Center and Warning Chimes



Special Tool(s)

 ST3093-A	Fluke 77-1V Digital Multimeter FLU77-4 or equivalent
 ST1473-A	Instrument Gauge System Tester 014-R1063 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

The Instrument Panel Cluster (IPC) utilizes a microprocessor to control the gauge, indicator, message center and warning chime functions. Data is sent to the IPC over the High Speed Controller Area Network (HS-CAN) and through hardwired circuitry from individual components. The IPC uses each input to output an action to the gauges, indicators, message center or chime arbitrator.

When diagnosing an IPC concern, it is very important to understand:

- where the input originates.
- all of the information necessary for a feature to operate.
- which module(s) receive(s) the input or command message.
- which module controls the output of the feature.
- whether the module that receives the input controls the output of the feature, or whether it outputs a message over the communication network to another module.

Instrument Panel Cluster (IPC)

IPC Network Messages

The IPC uses input messages from other modules to control the gauges, informational indicators, warning indicators and message center indicators over the HS-CAN. Messages that originate within or are sent to other modules on the Medium

Speed Controller Area Network (MS-CAN) are sent to the Body Control Module (BCM) and converted to an HS-CAN message. For additional information on the communication networks or the BCM gateway function, refer to Section 418-00. If a required message is missing or invalid for less than 5 seconds, the gauge or indicator that requires the message remains at the last commanded state based upon the last message received. For example, if the brake status message is missing for less than 5 seconds and the brake warning indicator was illuminated, the indicator remains in the ON state until the next message is received. If the message remains missing or invalid for greater than 5 seconds, the IPC sets a network U-code DTC and the IPC output becomes a default action for the indicator or gauge. Each indicator or gauge utilizes a different default strategy depending on the nature of the indication. Refer to the normal operation descriptions located in the individual pinpoint test for further description of the default action specific to each indicator or gauge. If the missing messaged input to the cluster returns at any time, the normal function of the gauge or indicator resumes.

NOTE: *Whenever a message is suspected as missing and confirmed by a missing message DTC, it is important to look for other symptoms that may also be present in the IPC and throughout the vehicle. Once a DTC sets in the IPC, it may be helpful to review the complete message list available in Section 418-00 to see what other modules also rely on the same message and run the self-test for those modules. If the message is missing from other modules, the same DTC may also be set in those modules. Confirmation of missing messages common to multiple modules may indicate that the originating module is the source of the concern or the communication network may be experiencing some problems.*

IPC Configuration

The IPC contains items that are configurable. Configurable items include customer preference items, which can also be set with a scan tool. The remaining configurable items can only be set by the Programmable Module Installation (PMI) procedure or by using As-Built data. Refer to Section 418-01.

The configurable IPC items are as follows:

IPC Configuration Parameters

As-Built Parameter	State Description
ABS	ABS
	No ABS
Traction control/interactive vehicle dynamics/Roll Stability Control (RSC®)	No traction control/interactive vehicle dynamics/RSC®
	Traction control
	Interactive vehicle dynamics RSC®
Tire Pressure Monitoring System (TPMS)	Off
	On
Oil pressure input	Hardwired High Speed Controller Area Network (HS-CAN)
Belt-Minder®	Off
	On
Oil life	Off
	On
RSC® warning chime	Enabled
	Disabled

As-Built Parameter	State Description
Compass	Enabled
	Disabled
Rear park aid	Rear park aid
	No rear park aid
Front park aid	Front park aid
	No front park aid
MyKey®	Enabled
	Disabled
Autolamp	Enabled
	Disabled
Auto lock	Enabled
	Disabled
Auto unlock	Enabled
	Disabled
Easy entry/easy exit	Enabled
	Disabled
Rain sensing wipers	Yes
	No
Trailer Brake Control (TBC) gain	TBC and gain warnings
	No TBC and gain warnings
Trailer sway	Enabled
	Disabled
Default language	English
	French
	Spanish
Speedometer bias	European and gulf coast countries
	North American
Number of fuel senders	One sender
	Two sender
Number of fuel tanks	One fuel tank
	Two fuel tanks
Overspeed warning chime	Gulf coast countries
	Non-gulf coast countries
Reverse warning chime	Japan

As-Built Parameter	State Description
	Non-Japan
Key-in-ignition chime	Enabled
	Disabled
Electronic Power Assist Steering (EPAS)	Enabled
	Disabled
Two-wheel drive/Four-Wheel Drive (4WD) /All-Wheel Drive (AWD)	Two-wheel drive 4WD AWD
	Undefined
Power Running Board (PRB)	PRB
	No PRB
Electronic Locking Differential (ELD)	ELD
	No ELD
Welcome	Yes
	No
Tank configuration	Single tank
	Dual tanks
	Saddle tank
Tank selection	Tank 1 (26 gallon)
	Tank 2 (36 gallon)
Fuel capacity	Fuel tank capacity in tenths of a liter (0-409.5)
Flex-fuel	Flex fuel vehicle
	Not a flex-fuel vehicle
Fuel economy metric display	L/100km
	km/L
Fuel economy gallon display	US gallon
	UK gallon
Welcome start	Blank in start
	Not blank in start
Rear camera feature	Yes
	No
Rear camera delay	Yes
	No
Rear camera visual park aid and auto-zoom	Yes
	No
Rear camera active guidelines	Yes

As-Built Parameter	State Description
	No
Courtesy wipe after wash	Yes
	No
One/two stage unlocking	Yes
	No
Remote start feature	Yes
	No
Remote start — climate settings	Yes
	No
Remote start — driver seat temperature control	Yes
	No
Remote start — passenger seat temperature control	Yes
	No
Remote start — rear defrost	Yes
	No
Splash screen	Built Ford Concrete
	Lincoln
	Built Ford Desert
Background	No background
	F150
	Lincoln
Mexico	Mexico a IPC
	Not a Mexico IPC
Rear camera auto zoom	Yes
	No
Front camera	Yes
	No
Neutral tow	Yes
	No
Steering gear ratio	Low 17 to 1
	High 20 to 1
ControlTrac 4WD display	Yes
	No

IPC Prove-Out

The IPC and other vehicle modules carry out a display prove-out to verify all module controlled warning/indicator lamps and monitored systems are functioning correctly within the IPC . When the ignition is cycled to ON with the engine off, the indicators illuminate to prove-out according to the following table:

Indicator	Indicator Type	Prove-Out Duration
ABS	Warning	3 seconds
Air bag	Warning	6 seconds
Brake	Warning	3 seconds
Cruise control	Informational	No prove-out
High beam	Informational	No prove-out
Malfunction Indicator Lamp (MIL)	Warning	Engine startup
RH/LH turn signals	Informational	No prove-out
Safety belt	Warning	65 seconds if the safety belt is unbuckled, turns off when the safety belt is buckled
Tire Pressure Monitoring System (TPMS)	Warning	3 seconds
Stability/traction control	Informational	3 seconds
Stability/traction control disabled	Informational	3 seconds

Information And Message Center

The message center is an integral part of the IPC that receives and acts upon much of the same information that is input and used to operate the IPC gauges, indicators, warning indicators and message center indicators.

Whenever conditions are present that require a warning message, the message center replaces the last selected display with the new warning display. Once the message is reset or cleared, the message center returns to the last selected display. If multiple warnings are present, the message center displays each warning for approximately 4 seconds. Warning messages are also generally associated with other observable IPC indications. For example, when the LH front door is opened, the message center displays the message DRIVER DOOR AJAR along with the door ajar message center warning indicator. This allows the message center to be a more informative supplement to the IPC gauges and indicators.

Another function of the message center is to display message center warning indicators. The message center often displays a warning message and the message center warning indicator at the same time. For example, if the PCM detects low engine oil pressure, the PCM sends a request to the IPC to illuminate the low oil pressure message center indicator and to display LOW OIL PRESSURE in the message center. If multiple warnings are present requiring multiple message center warning indicator displays, the message center cycles or rotates through the displays in the same manner the message center cycles the text messages.

Warning Chimes

Each warning chime has unique characteristics. The warning chimes use volume, chime frequency, length of time the chime sounds and the number of chime tones to identify which chime is sounding. The IPC prioritizes the chimes according to a preset hierarchy programmed into the IPC software. When more than one chime request is received by the IPC , the highest priority chime sounds. If a lower priority chime is currently sounding, the higher priority request takes over and replaces the lower priority chime.

There are 3 different chimes as listed below:

- Repetitive
- Single tone
- Tick-tock

The following table provides a summary of the chimes:

IPC Chime Characteristics

Chime Name	Chime Type	Condition Description
Air bag	Repetitive	Five sequences of 1 second on and 1 second off (10 seconds total)
Belt-Minder® A and B	Repetitive	The chime sounds 6 chime tones/safety belt warning indicator flash sequences for as long as the IPC receives the Belt-Minder® chime request.
Door ajar	Single tone	The chime sounds a single 1.6 second chime tone when any door changes from the closed to the open state.
Headlamps on	Repetitive	On for as long as the headlamps are on and the IPC receives the chime on command. Shuts off when battery saver has expired (approximately 30 minutes).
Key-in-ignition	Repetitive	On for as long as the key is in the ignition with the driver door ajar and the IPC receives the chime on command. Shuts off when battery saver has expired (approximately 30 minutes).
Memory feedback	Single tone	The chime sounds for 0.1 second when the IPC receives a memory state change message.
Message center	Single tone	The chime is a single 0.1 second tone chime that sounds once per ignition cycle whenever the IPC receives a new message commanding on one of the message center warnings.
Parking brake on	Repetitive	On for as long as the parking brake is applied and the IPC receives the chime on command.
Perimeter alarm	Repetitive	On for 12 seconds when the driver door is opened using a key and the alarm is armed. The chime sounds 2 chime tones for the first 6 seconds followed by 3 chime tones for the next 4 seconds and finally 4 chime tones for the last 3 seconds.
Safety belt	Repetitive	Six cycles of 0.5 second on/0.5 second off for as long as the safety belt is unbuckled and the IPC receives the chime on command.
Service AdvanceTrac®	Repetitive	The chime sounds three 1.0 second chime tones for as long as the IPC receives the chime request on command.
Turn signal/hazard on	Tick-tock	On for as long as the RH or LH turn signal or hazards are on and the IPC receives the chime on command.
Turn signal left on	Repetitive	On for as long as the RH or LH turn signal is on and the IPC receives the chime on command (approximately 3.2 km [2 miles]).

Air Bag Secondary Warning Chime

The air bag secondary warning chime warns that the air bag warning indicator light does not work correctly. The IPC monitors the air bag warning indicator status internally and provides an air bag indication status report to the RCM . When a fault is present in the Supplemental Restraint System (SRS) with the RCM sending the IPC an air bag warning indicator on request and a fault is detected in the air bag warning indicator, the RCM sends the IPC a secondary request to sound the air bag warning chime.

The air bag secondary warning chime inputs are:

- Ignition state
- Air bag warning indicator request communicated by the RCM to the IPC
- Air bag warning indicator chime request communicated by the RCM to the IPC

Message Center Warning Chime

The message center warning chime feature draws the driver's attention to the message center display to view each new warning message. The IPC provides a single one second tone whenever a new warning message is displayed in the message center. If multiple warning messages are present, the IPC sounds a chime for each of the warning messages that are present.

Perimeter Alarm Warning Chime

The perimeter alarm warning chime alerts the driver that the perimeter alarm is armed when the driver door is unlocked with a key. The warning chime is only functional when a key is used to unlock the driver door. If the key fob or keypad (if equipped) is used to unlock the door, the perimeter alarm is disarmed and the warning chime does not sound. The warning chime sounds for 12 seconds when the driver door is opened and turns off when the perimeter alarm is disarmed (either by using the key fob, keypad or turning the key on). After the 12 second warning chime duration, the warning chime stops sounding and the perimeter alarm activates, sounding the horn and flashing the turn signal lamps.

Turn/Hazard On Chime

The turn/hazard on chime provides an audible sound to indicate the RH/LH turn or hazard lamps are on. When the LH turn lamps, RH turn lamps or hazard lamps are on, the IPC provides a tick-tock tone along with the visual turn signal indicators.

The turn/hazard on tone inputs are:

- Left and right turn indicator signal from the BCM to the IPC through the HS-CAN
- Hazard signal from the BCM to the IPC through the HS-CAN

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Accessory drive belt • Brake fluid level • Correct tire size • Courtesy lamps • Engine coolant level • Engine oil level • Fuel tank • Fuel evaporative system • Tire pressure	• Body Control Module (BCM) fuse(s): ▪ 11 (10A) ▪ 32 (15A) • Wiring, terminals or connectors • Auto-dimming interior mirror • Message center switch (part of the LH steering wheel switch) • Instrument Panel Cluster (IPC)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** Make sure to use the latest scan tool software release.

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:

- verify the ignition key is in the ON position.
- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
- If the network test passes, retrieve and record the continuous memory DTCs.

8. Clear the continuous DTCs and carry out the self-test diagnostics for the IPC , PCM, Restraints Control Module (RCM) , ABS module, Trailer Brake Control (TBC) module, Parking Aid Module (PAM) and the BCM .

9. If the DTCs retrieved are related to the concern, go to DTC Charts in this section.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart — Instrument Panel Cluster \(IPC\)](#), GO to [Symptom Chart — Information And Message Center](#) or GO to [Symptom Chart — Warning Chimes](#).

Instrument Panel Cluster (IPC) Dealer Test Mode — Base IPC

To enter the IPC dealer test mode, press and hold the message center RESET button. Place the ignition in RUN and continue to hold the RESET button until the display indicates ENGINEERING TEST MODE, usually within 3 to 5 seconds. Press the RESET button once to advance through each stage of the self-test.

To exit the IPC dealer test mode, press and hold the RESET button for more than 3 seconds or simply turn the ignition off.

NOTE: The displays listed in the table below use xxx's to represent a numeric or alpha-numeric value or a text display. The value may display the same amount of characters represented by the xxx's or there may be more/less depending on the type of display. For example, VEHICLE SPEED xxx.x mph may display VEHICLE SPEED 45 mph.

First Line Display	Second Line Display	Description
ENGINEERING	TEST MODE	Initial entry display into dealer test mode.
GAUGE	TEST	Carries out a gauge sweep of all the gauges, then displays the present gauge values.
All segments illuminated	—	Illuminates all odometer segments.
TELLTALE	TEST	Illuminates all the micro-controlled lamps and LEDs.
PART NUMBER	xxxx-xxx	Displays the alpha-numeric part number prefix and suffix.
ROM LEVEL	\$xxxx	Returns all <u>IPC</u> controlled lamps/LEDs to the normal operation and displays the Read-Only Memory (ROM) level stored in hexadecimal format.

First Line Display	Second Line Display	Description
EEPROM LEVEL	\$xx	Displays the EE level in hexadecimal format.
MANUFACTURE	DATE \$xxxxx	Displays final manufacturing test date in hexadecimal format.
AUTOCONFIG	BYTE 1 \$xx	Displays all bytes of the IPC configuration settings in hexadecimal format.
B&A CONFIG	BYTE 1-2 \$xx	Displays all bytes of the IPC configuration settings in hexadecimal format.
MANU CONFIG	BYTE 1 \$xx	Displays the Non-Volatile Memory (NVM) As-Built configuration in hexadecimal format.
CONTINUOUS	DTC \$xxxx	Displays a 16-bit DTC in hexadecimal format. DTCs displayed are those detected in continuous operation not during self-test.
VEHICLE SPEED	xxx.x mph	Displays the messaged English speed value received. The speedometer indicates the present speed. If the vehicle speed message is missing, MISSING DATA is displayed.
VEHICLE SPEED	xxx.x km/h	Displays the messaged metric speed value received. The speedometer indicates the present speed.
TACHOMETER	xxxx rpm	Displays the messaged engine speed signal value received. The tachometer indicates present rpm.
FUEL LEVEL	xxx INSTANT	Displays the present instantaneous unfiltered fuel level. The fuel gauge indicates the filtered level.
FUEL LEVEL%	STATUS xxx	Displays the present fuel level percent status in decimal input from the secondary fuel level sensor in decimal. The fuel gauge indicates the present level.
INSTANT DTE	xxx MILES	Provides the current Distance To Empty (DTE) .
RAFE DTE	xx.x MPG	Displays the running average fuel economy in decimal.
FUEL ON/OFF/SZ	xxx xxx xxx	Displays the key on initialization (left xxx), key off reference (middle xxx), and zero speed reference (right xxx) fuel level values in fuel level percent status.
F#S ON/OFF/SZ	xxx xxx xxx	Displays the number of valid fuel samples taken at key on initialization (left xxx), key off convergence (middle xxx), and zero speed convergence (right xxx) in decimal counts, no leading zeros.
COOLANT TEMP	xxx DEG C	Displays the last messaged engine coolant temperature value message received. The temperature gauge indicates the present temperature.
ODOMETER	xxx COUNTS	Displays the messaged odometer rolling count input received in decimal.
TRIP ODOMETER	xxx.x mi	Displays the trip odometer value stored in Random Access Memory (RAM) in miles.
BATTERY	xx.x VOLTS	Displays the present battery voltage in 10th of volts.
DIMMING CMD	xxx	Displays the messaged dimming value input from the Body Control Module (BCM) .
DIMMING STEP	xx	Displays the messaged dimming level step (0 thru 21) received in decimal from the BCM .
MC SWITCH	xxx A/D COUNTS	Displays the present message center switch input analog/digital counts in decimal.
LWF OPS OAT	x x xxx	Displays the low washer fluid switch input (left x), oil pressure switch input (middle x) and the outside air temperature sensor input (right xxx).
COMPASS	xxx	Displays the present compass input in decimal.
LAST CHIME	SOUNDED xx	Displays the last chime that sounded.
GAUGE	TEST	Repeats the sweep of all gauges and then displays present gauge values.

Instrument Panel Cluster (IPC) Dealer Test Mode — Optional IPC

To enter the IPC dealer test mode, press and hold the message center OK button. Place the ignition in RUN and continue to hold the OK button until the display indicates ENGINEERING TEST MODE, usually within 3 to 5 seconds. Press the OK button once to advance through each stage of the self-test.

To exit the IPC dealer test mode, press and hold the OK button for more than 3 seconds or simply turn the ignition off.

NOTE: The displays listed in the table below use xxx's to represent a numeric or alpha-numeric value or a text display. The value may display the same amount of characters represented by the xxx's or there may be more/less depending on the type of display. For example, VEHICLE SPEED xxx.x mph may display VEHICLE SPEED 45 mph.

First Line Display	Second Line Display	Description
ENGINEERING	TEST MODE	Initial entry display into dealer test mode.
GAUGE	TEST	Carries out a gauge sweep of all the gauges, then displays the present gauge values.
All segments illuminated	—	Illuminates all message center LEDs in the different colors.
ALL MICRO CONT	BULBS ON	Illuminates all the micro-controlled lamps and LEDs.
PART NUMBER	xxxx-xxx	Displays the alpha-numeric part number prefix and suffix.
ROM LEVEL	\$xxxx	Returns all IPC controlled lamps/LEDs to normal operation and displays the Read-Only Memory (ROM) level in hexadecimal format.
NVM ROM LEVEL	\$xxxx	Displays the Non-Volatile Memory (NVM) level in hexadecimal format.
EEPROM LEVEL	\$xx	Displays the EE level in hexadecimal format.
MANUFACTURE	DATE \$xxxx	Displays final manufacturing test date in hexadecimal format.
MANU CONFIG	BYTE 1-2 \$xx	Displays the Non-Volatile Memory (NVM) As-Built configuration in hexadecimal format.
VOPS CONFIG	BYTE 1-5 \$xx	Displays all bytes of the IPC configuration settings in hexadecimal format.
VOPS CONFIG	TANK CAP xx.x	Displays all bytes of the IPC configuration settings in hexadecimal format.
VOPS CONFIG	BYTE 7-8 LN \$xx	Displays all bytes of the IPC configuration settings in hexadecimal format.
CONTINUOUS	DTC \$xxxx	Displays a 16-bit DTC in hexadecimal format. DTCs displayed are those detected in continuous operation not during self-test.
VEHICLE SPEED	xxx.x mph	Displays the messaged English speed value received. The speedometer indicates the present speed. If the vehicle speed message is missing, MISSING DATA is displayed.
VEHICLE SPEED	xxx.x km/h	Displays the messaged metric speed value received. The speedometer indicates the present speed.
SPEEDOMETER	DC xxxx	Displays corresponding gauge driver counts output for present vehicle speed value.
TACHOMETER	xxxx rpm	Displays the messaged engine speed signal value received. The tachometer indicates present rpm.

First Line Display	Second Line Display	Description
TACHOMETER	DC xxxx	Displays corresponding gauge driver counts output for present rpm value.
FUEL LEVEL	A/D INPUT xxxx	Displays the present ratio'd analog/digital fuel level input in decimal.
FUEL LEVEL	FLPS IN xxxx	Displays the present input for fuel level percent status in decimal.
FUEL LVL FILT	FLPS IN xxxx	Displays the present filtered fuel level percent status in decimal (used by gauge).
FUEL FILT	% STATUS xxxx	Displays the present fuel filter percent status in decimal.
FUEL FILT HYST	% STATUS xxxx	Displays the present fuel filter delay percent status in decimal.
FUEL GAUGE	DC xxxx	Displays the corresponding fuel gauge driver counts output for filtered fuel level percent status value.
FUEL LEVEL	INIT REF \$xxxx	Displays the fuel initial reference type used at key on initiation in hexadecimal followed by the fuel level percent status value of the initial reference in hexadecimal.
FUEL RAFE	xx.x MPG	Displays the running average fuel economy in decimal in miles.
FUEL RAFE	xx.x L/100km	Displays the running average fuel economy in decimal in kilometers.
INSTANT DTE	xxx MILES	Displays the instantaneous Distance To Empty (DTE) in miles.
INSTANT DTE	xxx km	Displays the instantaneous Distance To Empty (DTE) in kilometers.
FUEL ON/OFF/SZ	xxx xxx xxx	Displays the key on initialization (left xxx), key off reference (middle xxx), and zero speed reference (right xxx) fuel level values in fuel level percent status.
F#S ON/OFF/SZ	xxx xxx xxx	Displays the number of valid fuel samples taken at key on initialization (left xxx), key off convergence (middle xxx), and zero speed convergence (right xxx) in decimal counts, no leading zeros.
COOLANT TEMP	xxx C	Displays the last messaged engine coolant temperature value message received. The temperature gauge indicates the present temperature.
TEMP GAUGE	DC xxxx	Displays corresponding gauge driver counts output for present engine temperature value.
FAIL SAFE	COOL MODE xx	Displays the fail safe mode in hexadecimal.
OIL PRESSURE	LAMP xxx	Displays the low oil pressure indicator status.
OIL PRS GAUGE	DC xxxx	Displays corresponding gauge driver counts output for present oil pressure gauge value.
TRANS TEMP	xxx C	Displays the transmission temperature input in degrees Celsius.
TRANS TEMP	DC xxxx	Displays corresponding gauge driver counts output for present transmission temperature gauge value.
BATTERY	xx.x VOLTS	Displays the present battery voltage in 10th of volts.
BATTERY A/D	COUNTS xxxx	Displays the present battery analog/digital counts in decimal.
ODOMETER	xxx COUNTS	Displays the messaged odometer rolling count input received in decimal.
TRIP ODOMETER	A xxx.x mi	Displays the trip odometer value stored in Random Access Memory (RAM) in miles.
TRIP ODOMETER	B xxx.x mi	Displays the trip odometer value stored in Random Access Memory (RAM) in miles.
PDS IN	SENSE x	Displays the current state of the park detect switch input.

First Line Display	Second Line Display	Description
MC SW IN	SENSE x	Displays the current state of the message center switch input.
HDC SW IN	SENSE x	Displays the current state of the hill descent switch input.
HDC LED OUT	CMD xx	Displays the current state of the hill descent indicator.
DIMMING CMD	xxxx	Displays the messaged dimming value input from the Body Control Module (BCM) .
DIMMING STEP	D xx	Displays the messaged dimming level step (0 thru 21) received in decimal from the BCM .
MCSW IN	xxxx A/D COUNTS	Displays the present message center switch analog/digital counts in decimal.
MC SWITCH	xxx A/D COUNTS	Displays the present message center switch input analog/digital counts in decimal.
LAST CHIME	SOUNDED xx	Displays the last chime that sounded.
A/D PORT 2-9	\$ xxxx	Displays 10 bit hexadecimal value of the analog port readings.
DIGITAL INPUTS	PORT A, C, A/D \$xx	Displays 8 bit hexadecimal value of the digital port readings.
DIGITAL INPUTS	PORT A/D \$xxxx	Displays 8 bit hexadecimal value of the digital port readings.
DIGITAL INPUTS	HET1 \$xxxxxxxx	Displays 8 bit hexadecimal value of the digital port readings.
GAUGE	TEST	Repeats the sweep of all gauges and then displays present gauge values.

DTC Charts

Instrument Panel Cluster (IPC) DTC Chart

DTC	Description	Action
B1A82:11	External Trip Switch: Circuit Short To Ground	GO to Pinpoint Test AF.
B1A82:13	External Trip Switch: Circuit Open	GO to Pinpoint Test AF.
B1A82:23	External Trip Switch: Signal Stuck Low	INSTALL a new LH steering wheel switch. REFER to Section 211-05. CLEAR the DTCs. REPEAT the self-test.
P0460:11	Fuel Level Sensor A Circuit: Circuit Short to Ground	GO to Pinpoint Test B.
P0460:13	Fuel Level Sensor A Circuit: Circuit Open	GO to Pinpoint Test B.
P062F:46	Internal Control Module EEPROM Error: Calibration/Parameter Memory Failure	CLEAR the DTCs. REPEAT the self-test. If DTC P062F:46 returns, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
P1534:01	Restraint Deployment Indicator Circuit: General Electrical Failure	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
P1706:00	High Vehicle Speed Observed in Park: No Sub Type Information	GO to Pinpoint Test X.

DTC	Description	Action
U0100:00	Lost Communication With ECM /PCM "A": No Sub-Type Information	GO to Pinpoint Test AT.
U0102:00	Lost Communication With Transfer Case Control Module: No Sub-Type Information	GO to Pinpoint Test AU.
U0121:00	Lost Communication With Anti-Lock Brake System (ABS) Control Module: No Sub-Type Information	GO to Pinpoint Test AV.
U0131:00	Lost Communication With Power Steering Control Module: No Sub-Type Information	GO to Pinpoint Test AW.
U0137:00	Lost Communication With Trailer Brake Control Module: No Sub-Type Information	GO to Pinpoint Test AX.
U0140:00	Lost Communication With Body Control Module: No Sub-Type Information	GO to Pinpoint Test AY.
U0151:00	Lost Communication With Restraints Control Module: No Sub-Type Information	GO to Pinpoint Test AZ.
U0154:00	Lost Communication With Restraints Occupant Classification System Module: No Sub-Type Information	GO to Pinpoint Test BA.
U0159:00	Lost Communication With Parking Assist Control Module "A": No Sub-Type Information	GO to Pinpoint Test BB.
U0212:00	Lost Communication With Steering Column Control Module: No Sub-Type Information	GO to Pinpoint Test BC.
U0401:00	Invalid Data Received from ECM /PCM A: No Sub-Type Information	This DTC sets when the IPC receives invalid network data from the PCM for the vehicle speed input. RETRIEVE and REPAIR all non-network DTCs in the PCM and other modules on the network. REFER to Section 419-10 for the correct navigation of all DTCs. TEST the system for normal operation.
U0401:55	Invalid Data Received from ECM /PCM A: Not Configured	This DTC sets when the IPC receives invalid network data from the PCM for Trailer Brake Control (TBC) when the IPC is not configured for TBC . CARRY OUT the Programmable Module Installation (PMI) (when the original module is not available) for the IPC . REFER to Section 418-01. TEST the system for normal operation.
U0401:81	Invalid Data Received from ECM /PCM A: Invalid Serial Data Received	This DTC sets when the IPC receives invalid network data from the PCM for the engine temperature data. RETRIEVE and REPAIR all non-network DTCs in the PCM and other modules on the network. REFER to Section 419-10 for the correct navigation of all DTCs. TEST the system for normal operation.
U0401:86	Invalid Data Received from ECM /PCM A: Signal Invalid	This DTC sets when the IPC receives invalid network data from the PCM for the odometer data. RETRIEVE and REPAIR all non-network DTCs in the PCM and other modules on the network. REFER to Section 419-10 for the correct navigation of all DTCs. TEST the system for normal operation.

DTC	Description	Action
U0415:56	Invalid Data Received from Anti-Lock Brake System (ABS) Control Module: Invalid/Incompatible Configuration	This DTC sets when the IPC receives invalid data from the ABS module. DETERMINE which module (IPC or ABS module) has been replaced. CARRY OUT the Programmable Module Installation (PMI) (when the original module is not available) for the module in question. REFER to Section 418-01. TEST the system for normal operation.
U0422:00	Invalid Data Received from Body Control Module: No Sub-Type Information	This DTC sets when the IPC receives invalid Body Control Module (BCM) data for the Tire Pressure Monitoring System (TPMS) function. RETRIEVE and REPAIR all non-network DTCs in the BCM . REFER to Section 419-10 for the correct navigation of all DTCs. TEST the system for normal operation.
U0422:56	Invalid Data Received from Body Control Module: Invalid/Incompatible Configuration	This DTC sets when the IPC receives BCM data for the autolamp function and the IPC is not configured for autolamps. DETERMINE which module (IPC or BCM) has been replaced. CARRY OUT the Programmable Module Installation (PMI) (when the original module is not available) for the module in question. REFER to Section 418-01. TEST the system for normal operation.
U045A:55	Invalid Data Received From Parking Assist Control Module "A": Not Configured	This DTC sets when the IPC receives rear parking aid data when the IPC is not configured for parking aid. CARRY OUT the PMI (when the original module is not available) for the IPC . REFER to Section 418-01. TEST the system for normal operation.
U045A:68	Invalid Data Received From Parking Assist Control Module "A": Event Information	This DTC sets when the IPC receives invalid rear parking aid data. RETRIEVE and REPAIR all non-network DTCs in the Parking Aid Module (PAM) . REFER to Section 413-13 for a list of all DTCs. If there are no DTCs present, INSTALL a new PAM . REFER to Section 413-13. TEST the system for normal operation.
U0462:82	Invalid Data Received from Compass Module: Alive/Sequence Counter Incorrect/Not Updated	If equipped with a 4.2" message center and a 4.2" FCDIM , GO to Pinpoint Test AG. For all other configurations, Refer to the appropriate section in Group 415 for the procedure.
U0545:56	Invalid Data Received From Running Board Control Module: Invalid/Incompatible Configuration	This DTC sets when the IPC receives Power Running Board (PRB) data when the IPC is not configured for PRB s. CARRY OUT the PMI (when the original module is not available) for the IPC . REFER to Section 418-01. TEST the system for normal operation.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	CARRY OUT the PMI (when the original module is not available) for the IPC . REFER to Section 418-01. TEST the system for normal operation. After the PMI , CLEAR the DTC. REPEAT the self-test.
U3000:04	Control Module: System Internal Failure	CLEAR the DTCs. REPEAT the self-test. If the DTC returns, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the

DTC	Description	Action
		system for normal operation.
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test BD.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test BE.
U300A:64	Ignition Switch: Signal Plausibility Failure	This DTC sets on-demand and in continuous memory when the IPC detects a messaged START input for 15 seconds or longer. Once the fault condition is detected, the IPC operates with limited functionality for 15 seconds then returns to full functionality. REFER to Section 211-05 to diagnose the ignition start circuit to the BCM .

Body Control Module (BCM) DTC Chart

DTC	Description	Action
B1048:01	Brake Fluid Level Switch: General Electrical Failure	GO to Pinpoint Test N.
B1048:7B	Brake Fluid Level Switch: Low Fluid Level	GO to Pinpoint Test N.
B10F1:14	Key In Switch: Circuit Short To Ground or Open	If the chime sounds with the door ajar and no key in the ignition, GO to Pinpoint Test AM. If the key-in-ignition chime is inoperative, GO to Pinpoint Test AO.
C1D00:11	Park Brake Apply Switch: Circuit Short to Ground	GO to Pinpoint Test N.
All other DTCs	—	REFER to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

Powertrain Control Module DTC Chart

DTC	Description	Action
P0457	Evaporative Emission System Leak Detected (Fuel Cap Loose/Off)	REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual.
P0460	Fuel Level Sensor A Circuit	REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual first. If sent here from the PC/ED manual, GO to Pinpoint Test B.
P0461	Fuel Level Sensor A Circuit Range/Performance	REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual first.

DTC	Description	Action
		If sent here from the PC/ED manual, GO to Pinpoint Test B .
P0462	Fuel Level Sensor A Circuit Low	REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual first. If sent here from the PC/ED manual, GO to Pinpoint Test B .
P0463	Fuel Level Sensor A Circuit High	REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual first. If sent here from the PC/ED manual, GO to Pinpoint Test B .
All other DTCs	—	REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual.

Symptom Charts

Symptom Chart — Instrument Panel Cluster (IPC)

Condition	Possible Causes	Action
No communication with the Instrument Panel Cluster (IPC)	- Fuse - Wiring, terminals or connectors - IPC	REFER to Section 418-00.
The Instrument Panel Cluster (IPC) is inoperative	- Communication concern - IPC	GO to Pinpoint Test A.
Incorrect fuel gauge indication	- Wiring, terminals or connectors - Fuel tank - Fuel level sender (float and resistance card) - Fuel pump module - IPC	GO to Pinpoint Test B.
Incorrect oil pressure gauge indication	- Communication network concern - Wiring, terminals or connectors - Engine oil pressure switch	GO to Pinpoint Test C.

	• Base engine oil pressure concern • PCM • IPC	
• The temperature gauge is inoperative	• PCM • IPC	• GO to Pinpoint Test D.
• Incorrect temperature gauge indication	• PCM concern • IPC	• GO to Pinpoint Test E.
• The speedometer is inoperative	• PCM • IPC	• GO to Pinpoint Test F.
• Incorrect speedometer indication	• Tire size configuration • Axle ratio configuration • PCM concern • IPC	• GO to Pinpoint Test G.
• The odometer is inoperative	• PCM • IPC	• GO to Pinpoint Test H.
• The tachometer is inoperative	• PCM • IPC	• GO to Pinpoint Test I.
• Incorrect tachometer indication	• PCM concern • IPC	• GO to Pinpoint Test J.
• The transmission temperature gauge is inoperative	• PCM • IPC	• GO to Pinpoint Test K.
• Incorrect transmission temperature gauge indication	• Transmission concern • IPC	• GO to Pinpoint Test L.
• The ControlTrac 4WD gauge is inoperative	• 4WD concern • IPC	• GO to Pinpoint Test BG.
• The brake warning indicator is never on	• Wiring, terminals or connectors	• GO to Pinpoint Test M.

	• Brake fluid level switch • Parking brake warning indicator switch • Body Control Module (BCM) • IPC	
• The brake warning indicator is always on	• Wiring, terminals or connectors • Parking brake warning indicator switch • Brake fluid level switch • ABS concern • BCM • IPC	• GO to Pinpoint Test N.
• The ABS warning indicator is never/always on	• ABS module • IPC	• GO to Pinpoint Test O.
• The stability/traction control indicator is never/always on	• Stability/traction control concern • ABS module • IPC	• GO to Pinpoint Test P.
• The stability/traction control disabled indicator is never/always on	• A MyKey® programmed key is in use • Stability/traction control cannot be disabled • ABS module • IPC	• GO to Pinpoint Test Q.
• The LH/RH turn or high beam indicator is never/always on	• Turn signal concern • High beam concern • BCM • IPC	• GO to Pinpoint Test R.

• The Tire Pressure Monitoring System (TPMS) warning indicator is never/always on	• Tire pressure • TPMS concern • BCM • IPC	• GO to Pinpoint Test S.
• The air bag warning indicator is never/always on	• SRS concern • Restraints Control Module (RCM) • IPC	• GO to Pinpoint Test T.
• The safety belt warning indicator is never/always on	• Safety belt buckle concern • RCM • IPC	• GO to Pinpoint Test U.
• The Malfunction Indicator Lamp (MIL) is never/always on	• PCM • IPC	• GO to Pinpoint Test V.
• The cruise control indicator is never/always on	• PCM • IPC	• GO to Pinpoint Test W.
• The PRNDL indicator is never on/always displays P	• Wiring, terminals or connectors • Brake shift interlock actuator • PCM concern • IPC	• GO to Pinpoint Test X.
• The powertrain malfunction (wrench) message center indicator is always on	• Message center concern • Electronic Throttle Control (ETC) concern • 4WD concern (if equipped) • Transmission concern	• GO to Pinpoint Test Y.
• The engine over-temperature message center warning indicator is never/always on	• PCM concern • IPC	• OPEN then CLOSE the driver door and VERIFY the door ajar message center warning indicator functions correctly.

		• If the door ajar message center warning indicator does not function correctly, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation. • If the door ajar message center warning indicator functions correctly, GO to Pinpoint Test D to diagnose the temperature gauge.
• The charging system message center warning indicator is never/always on	• Message center concern • Charging system concern • PCM	• GO to Pinpoint Test Z.
• The low oil pressure message center warning indicator is never/always on	• Wiring, terminals or connectors • Message center concern • Communication concern • Tachometer concern • Engine oil pressure switch • Base engine oil pressure concern • PCM	• GO to Pinpoint Test AA.
• The 4X4 auto, 4X4 low, 4X4 high or 4X2 message center indicator is never/always on (if equipped)	• Message center concern • Four-Wheel Drive (4WD) concern • Transfer Case Control Module (TCCM)	• GO to Pinpoint Test AB.
• The tow haul message center indicator is never/always on	• Message center concern • PCM	• GO to Pinpoint Test AC.
• The Electronic Locking Differential (ELD) message	• Message center concern	• GO to Pinpoint Test AD.

center indicator is never/always on	• ELD concern • TCCM	
• The door ajar message center warning indicator is never/always on	• Message center concern • Interior lighting concern • BCM	• GO to Pinpoint Test AE.
• The low fuel message center indicator is never/always on	• Fuel gauge indication concern • IPC	• GO to Pinpoint Test BF.

Symptom Chart — Information And Message Center

Condition	Possible Causes	Action
• The message center display is blank	• Message center set to blank display mode • IPC	• On the base IPC , PRESS the INFO button to make sure the message center is not set to a blank display mode. On the optional IPC , PRESS the message center button left several times to navigate from the blank message center display. • If the message center display turns on, the system is operating normally at this time. The message center display was set to a blank display mode. • If the message center display is still blank, GO to Pinpoint Test A.
• Not all segments of the display illuminate — base IPC only	• IPC	• INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
• Not all display pixels illuminate — optional IPC only	• IPC	• NOTE: <i>In order to condemn the IPC , the display failure must meet all of the following criteria:</i> • <i>There must be 3 or more pixels inoperative.</i> • <i>The inoperative pixels must be within 5 mm (0.196 in) or less from each other.</i> • <i>The inoperative pixels must be inside of the area that is 19 mm (0.748 in) from each outside border.</i> • VERIFY the number of pixels are inoperative and the location of the

		inoperative pixels in the display screen. • If the does not meet the above criteria, the display is operating within design specifications. • If display meets the above criteria, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
• The message center is not operating correctly	• Wiring, terminals or connectors • Message center switch (part of the LH steering wheel switch) • Clockspring • IPC	• GO to Pinpoint Test AF.
• The compass is inoperative	• Communication concern • GPSM	• GO to Pinpoint Test AG.
• The compass is inoperative (without 4.2" message center and 4.2" FCDIM)	• Fuse • Wiring, terminals or connectors • Communication concern • GPSM	• Refer to the appropriate section in Group 415 for the procedure.
• The compass is inaccurate	• Zone setting • Calibration • Compass module • Auto-dimming interior mirror	• Refer to the appropriate section in Group 415 for the procedure.
• The factory mode message is always on	• Vehicle still in factory mode	• NOTE: <i>This vehicle automatically defaults to normal operation mode after 60 ignition cycles.</i> • CARRY OUT the procedure to place the vehicle in normal operation mode. REFER to Disable Transport or Factory Mode in Section 419-10.

The transport mode message is always on	Vehicle still in transport mode	NOTE: <i>The vehicle automatically defaults to normal operation mode when the mileage reaches 125 miles (200 km).</i> - CARRY OUT the procedure to place the vehicle in normal operation mode. REFER to Disable Transport or Factory Mode in Section 419-10.
The ENGINE OIL CHANGE SOON, OIL CHANGE REQUIRED message is inoperative	- Odometer - IPC	- Using the system check in the message center, VERIFY the oil life is between 1% and 5% for the ENGINE OIL CHANGE SOON message or at 0% for the OIL CHANGE REQUIRED message. - If the oil life is between 1% and 5%, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation. - If the oil life is above 5%, the message center is operating correctly. INFORM the customer how the message center functions and how the oil life messages are set.
The Distance To Empty (DTE) /average fuel economy display is inoperative	IPC	- CHECK the operation of the fuel gauge, the speedometer and the odometer. - If the speedometer, the fuel gauge or the odometer do not operate correctly, GO to Pinpoint Test B (fuel gauge), GO to Pinpoint Test F (speedometer) or GO to Pinpoint Test H (odometer). - If the speedometer, the fuel gauge and the odometer operate correctly, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
Incorrect or fluctuating Distance To Empty (DTE)	- Fuel gauge concern - Change in driving conditions - High idle times - Parking on any grades - Refueling patterns	GO to Pinpoint Test BH.

	• IPC	
• The outside air temperature display is inoperative/incorrect	• Communication concern • PCM concern	• GO to Pinpoint Test AI.
• A hill descent message is inoperative/always on	• Message center concern • ABS module	• GO to Pinpoint Test AJ.
• The CHECK FUEL FILL INLET message is inoperative/always on	• Evaporative fuel system concern • PCM • IPC	• GO to Pinpoint Test AK.
• The CHECK BRAKE SYSTEM, PARK BRAKE ENGAGED or BRAKE FLUID LEVEL LOW warning is inoperative/always on	• Brake warning indication concern • IPC	• OPEN then CLOSE the driver door and VERIFY the DRIVER DOOR AJAR warning functions correctly. • If the DRIVER DOOR AJAR warning does not function correctly, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation. • If the DRIVER DOOR AJAR warning functions correctly, GO to Pinpoint Test M (never on) or GO to Pinpoint Test N (always on) to diagnose the brake warning indicator.
• The DRIVER, PASSENGER, REAR LEFT or REAR RIGHT DOOR AJAR warning is inoperative/always on	• Interior lighting concern • IPC	• DISCONNECT the brake fluid level switch and VERIFY the message center displays CHECK BRAKE SYSTEM. • If the CHECK BRAKE SYSTEM warning turns on, GO to Pinpoint Test AE to diagnose the door ajar message center warning indicator. • If the CHECK BRAKE SYSTEM warning does not turn on, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
• The TIRE PRESSURE SENSOR FAULT, LOW TIRE PRESSURE or TIRE PRESSURE MONITOR	• BCM • IPC	• OPEN then CLOSE the driver door and VERIFY the DRIVER DOOR AJAR warning functions correctly.

FAULT warning is inoperative/always on		- If the DRIVER DOOR AJAR warning does not function correctly, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation. - If the DRIVER DOOR AJAR warning functions correctly, GO to Pinpoint Test S to diagnose the TPMS warning indicator.
The TRAIN LEFT FRONT, RIGHT FRONT, RIGHT REAR or LEFT REAR message is inoperative/always on	- TPMS warning indicator concern - TPMS sensor training	NOTE: <i>These messages only appear during the TPMS sensor training procedure. REFER to Section 204-04.</i> - VERIFY the TPMS warning indicator functions. - If the TPMS warning indicator does not operate, GO to Pinpoint Test S. - If the TPMS warning indicator operates, CARRY OUT the TPMS sensor training. REFER to Section 204-04.
The TIRES NOT TRAINED-REPEAT message is always on	TPMS sensor training	CARRY OUT the TPMS sensor training. REFER to Section 204-04.
The TRAINING COMPLETE message is inoperative	TPMS sensor training	CARRY OUT the TPMS sensor training. REFER to Section 204-04.
A MyKey® menu is not available or incorrect	- MyKey® programmed key in use - Number of keys set to unlimited mode - Non-Ford approved remote start system installed	REFER to Section 419-01B to diagnose the MyKey® system.
The WIRING FAULT ON TRAILER, TRAILER BRAKE MODULE FAULT or TRAILER DISCONNECTED warning is inoperative/always on	- IPC configuration - Trailer brake control system	- VERIFY the vehicle is equipped with the trailer brake control system. - If the trailer brake warning is inoperative, VERIFY the IPC is correctly configured for trailer brake using the PMI (when the original module is not available). REFER to Section 418-01. - If the trailer brake warning is always on, REFER to Section 206-10.

• A 4X4 system message is inoperative/always on	• Message center concern • 4WD system concern • TCCM	• OPEN then CLOSE the driver door and VERIFY the DRIVER DOOR AJAR warning functions correctly. • If the DRIVER DOOR AJAR warning does not function correctly, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation. • If the DRIVER DOOR AJAR warning functions correctly, GO to Pinpoint Test AB to diagnose the 4X4 auto, 4X4 low, 4X4 high or 4X2 message center warning indicator.
• An off road mode message is inoperative/always on (Raptor only)	• Message center concern • 4WD concern • TCCM	• GO to Pinpoint Test AL.
• The shift to park message is never or always displayed	• PRNDL display concern • Communication concern • IPC	• GO to Pinpoint Test BI.

Symptom Chart — Warning Chimes

Condition	Possible Causes	Action
• All the chimes are inoperative	• Instrument Panel Cluster (IPC)	• VERIFY the IPC gauges and indicators are functioning correctly. • If the IPC gauges and indicators do not function correctly, GO to Pinpoint Test A. • If the IPC gauges and indicators function correctly, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
• The air bag secondary warning chime is inoperative	• IPC	• RETRIEVE the recorded IPC DTCs from the self-test. • If the IPC DTC P1534:01 is not present, the system is operating correctly. The air bag secondary warning chime sounds only when a fault in the air bag warning indicator is present and DTC P1534:01 is recorded. • If the IPC DTC P1534:01 is present, INSTALL a new IPC. Refer to the appropriate Removal and Installation

		procedure in Section 413-01. TEST the system for normal operation.
• The air bag secondary warning chime has activated	• IPC	• INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
• The Belt-Minder® feature does not operate correctly	• Safety belt warning indication concern • Belt-Minder® deactivated • Speedometer concern • RCM • IPC	• GO to Pinpoint Test AR.
• The chime sounds when the driver door is ajar (no key in ignition and headlamps off)	• Exterior lighting concern • Interior lighting concern • Wiring, terminals or connectors • Key-in-ignition warning switch (part of the ignition switch) • Body Control Module (BCM)	• GO to Pinpoint Test AM.
• The door ajar warning chime is inoperative	• Interior lighting concern • BCM • IPC	• VERIFY the key-in-ignition warning chime functions correctly. • If the key-in-ignition warning chime does not function correctly, GO to Pinpoint Test AO. • If the key-in-ignition warning chimes functions correctly, GO to Pinpoint Test AE to diagnose the door ajar message center warning indicator.
• The headlamps on warning chime is inoperative	• Interior lighting concern • BCM • IPC	• GO to Pinpoint Test AN.
• The key-in-ignition warning chime is	• Interior lighting concern	• GO to Pinpoint Test AO.

inoperative	• Wiring, terminals or connectors • Key-in-ignition warning switch (part of the ignition switch) • BCM • IPC	
• The message center warning chime is inoperative	• IPC	• DISCONNECT the brake fluid level switch and VERIFY the message center warning chime sounds when the message center displays CHECK BRAKE SYSTEM. • If the message center warning chime sounds when the CHECK BRAKE SYSTEM warning appears, the system is operating correctly at this time. The message center warning chime only sounds when the message center displays a new warning message. • If the message center warning chime does not sound when the CHECK BRAKE SYSTEM warning appears, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
• The memory feedback warning chime is inoperative	• Driver Seat Module (DSM) • IPC	• GO to Pinpoint Test AP.
• The parking brake warning chime is inoperative	• Brake warning indication concern • BCM • IPC	• GO to Pinpoint Test AQ.
• The perimeter alarm warning chime is inoperative	• Perimeter alarm concern • IPC	• VERIFY the key-in-ignition warning chime functions correctly. • If the key-in-ignition warning chime does not function correctly, GO to Pinpoint Test AO. • If the key-in-ignition warning chimes functions correctly, REFER to Section 419-01A to diagnose the perimeter alarm.

• The safety belt warning chime is inoperative	• Safety belt warning indication concern • IPC	• GO to Pinpoint Test AR.
• The service AdvanceTrac® warning chime is inoperative	• Message center traction control warning concern • IPC	• VERIFY the SERVICE ADVANCETRAC message is displayed in the message center and the stability/traction control indicator operates correctly. • If the SERVICE ADVANCETRAC message is not displayed in the message center and the stability/traction control indicator is not illuminated, the system is operating correctly. • If the SERVICE ADVANCETRAC message is displayed and the stability/traction control indicator is illuminated, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
• The turn/hazard on chime (tick-tock) is inoperative	• RH/LH turn indicator • IPC	• VERIFY the RH/LH turn indicators operate correctly. • If the RH/LH turn indicators do not operate correctly, GO to Pinpoint Test R. • If the RH/LH turn indicators operate correctly, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
• The turn signal on warning chime is inoperative	• Turn signal indication concern • Odometer operation • BCM • IPC	• GO to Pinpoint Test AS.

Pinpoint Tests

Pinpoint Test A: The Instrument Panel Cluster (IPC) Is Inoperative

Normal Operation

The Instrument Panel Cluster (IPC) receives a hot at all times voltage from the Body Control Module (BCM) fuse 11 (10A). The ~~IPC~~ uses a run/start input messaged from the ~~BCM~~ over the High Speed Controller Area Network (HS-CAN) .

This pinpoint test is intended to diagnose the following:

- Communication concern
- ~~IPC~~

PINPOINT TEST A : THE IPC IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

A1 VERIFY THE IPC PASSES THE NETWORK TEST

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- In the LH pane of theIntegrated Diagnostic System (IDS) network test display screen, verify whether the **IPC** passed the network test.

Did the **IPC pass the network test?**

Yes	GO to A2 .
No	REFER to Section 418-00 to diagnose the IPC . does not respond to the scan tool.

A2 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded **BCM** DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to Section 419-10.
No	GO to A3 .

A3 CHECK FOR CORRECT IPC OPERATION

- Disconnect the **IPC** connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the **IPC** connector and make sure it seats correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test B: Incorrect Fuel Gauge Indication

Refer to Wiring Diagrams Cell [60](#) , Instrument Cluster for schematic and connector information.

Normal Operation

The fuel level sender (part of the fuel pump module) is hardwired to the Instrument Panel Cluster (IPC) between the signal and return circuits. The fuel level sender is a variable resistor that ranges from 10 ohms $\pm$ 2 ohms at full (F) and 180 ohms $\pm$ 4 ohms at empty (E). The IPC provides a reference voltage to the fuel pump module. As the fuel level changes, a float actuates a variable resistor on the fuel pump sender and raises or lowers the fuel level signal voltage depending on the resistance of the fuel level sender. The IPC monitors the changes in voltage and commands the fuel gauge with a corresponding movement of the pointer.

The IPC uses 4 different operating modes to calculate the fuel level:

- Anti-slosh (default mode)
- Key off fueling
- Key on fueling
- Recovery

After a fuel fill up, the time for the fuel gauge to move from empty (E) to full (F) ranges from 2 seconds to 55 minutes depending on which operating mode the fuel gauge is in.

Anti-Slosh Mode

The default fuel gauge mode is called the anti-slosh mode. To prevent fuel gauge changes from fuel slosh (gauge instability due to changes in fuel sender readings caused by fuel moving around in the tank), the fuel gauge takes approximately 55 minutes to go from empty (E) to full (F).

Key Off Fueling Mode

The key off fueling mode (2 seconds to read empty [E] to full [F]) requires the following conditions be met:

- The ignition must be off when refueling the vehicle.
- At least 15% of the vehicle's fuel capacity must be added to the fuel tank.
- The IPC must receive a valid key on fuel sender reading within 1 second of the ignition being put into run. The key on sample readings are considered valid if the fuel sender reading is between 10 ohms $\pm$ 2 ohms and 180 ohms $\pm$ 4 ohms.

If these conditions are not met, the fuel gauge stays in the default anti-slosh mode, which results in a slow to read full (F) event.

Key On Fueling Mode

The key on fueling mode (approximately 60 seconds to read empty [E] to full [F]) requires the following conditions be met:

- Engine speed input greater than 0 rpm.
- The transmission is in park.
- The ignition is in run.
- At least 15% of the vehicle's fuel capacity must be added to the fuel tank.

In key on fueling mode, a 30 second timer activates after the transmission is put into the PARK position. When the 30 second time has elapsed and at least 15% of the vehicle's fuel capacity has been added, the fuel gauge response time is 60 seconds to read from empty (E) to full (F). When the transmission is shifted out of park, the fuel gauge strategy reverts to the anti-slosh mode. The key on fueling mode prevents slow to read full events from happening if the customer refuels the vehicle with the ignition in RUN.

Recovery Mode

Recovery mode is incorporated into the IPC strategy to recover from a missing fuel level input after a refueling event. Missing fuel level inputs result from intermittent opens in the fuel sender or its circuits. Recovery mode (empty [E] to full [F] approximately 20 minutes) is initiated when the following 2 conditions are met:

- The **IPC** is in the anti-slosh (default) mode.
- The actual fuel level in the tank is greater than what is being displayed by the fuel gauge.

Instrument Panel Cluster (IPC) DTCs

DTC Description	Fault Trigger Conditions
• P0460:11 — Fuel Level Sensor A Circuit: Circuit Short to Ground	A continuous and on-demand DTC that sets in the IPC if the IPC detects the fuel sender is out of range on the fuel level input circuit with a short to ground for more than 33 seconds. The IPC defaults the fuel gauge to empty (E), once the IPC detects a fault and sets DTC P0460:11.
• P0460:13 — Fuel Level Sensor A Circuit: Circuit Open	A continuous and on-demand DTC that sets in the IPC if the IPC detects the fuel sender is out of range on the fuel level input circuit with an open or short to voltage for more than 33 seconds. The IPC defaults the fuel gauge to empty (E), once the IPC detects a fault and sets DTC P0460:13.

PCM DTCs

DTC Description	Fault Trigger Condition
• P0460 — Fuel Level Sensor A Circuit	This DTC sets when the PCM determines the value of the fuel level input signal is stuck, the fuel level input signal does not change or does not correspond with the calculated fuel usage.
• P0461 — Fuel Level Sensor A Circuit Range/Performance	This DTC sets when the PCM determines the fuel level input signal repeatedly moves in and out of range, exceeding the minimum or maximum allowable calibrated parameters for a specified fuel fill percentage in the fuel tank.
• P0462 — Fuel Level Sensor A Circuit Low	This DTC sets in the PCM when the PCM detects a short to ground on the fuel pump module signal circuit based on the messaged input received from the IPC .
• P0463 — Fuel Level Sensor A Circuit High	This DTC sets in the PCM when the PCM detects an open or a short to voltage on the fuel pump module signal circuit based on the messaged input received from the IPC .

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Fuel tank
- Fuel level sender (float and resistance card)
- Fuel pump module
- **IPC**

PINPOINT TEST B : INCORRECT FUEL GAUGE INDICATION

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
B1 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Ignition OFF.
- Check for recorded **IPC** DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC P0460:11, GO to B2 . For DTC P0460:13, GO to B5 . For all other DTCs, REFER to DTC Charts in this section.
No	GO to B9 .

B2 CHECK THE FUEL PUMP MODULE FOR A SHORT TO GROUND

- Disconnect: Fuel Pump Module [C4330](#).
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC** Self-Test .
- Wait 1 minute.
- **NOTE:** *DTC P0460:11 may also be present when carrying out this test step and should be ignored.*

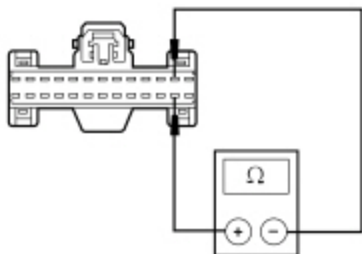
Clear the **IPC** DTCs. Repeat the **IPC** self-test.

Is DTC P0460:13 retrieved?

Yes	GO to B13 .
No	GO to B3 .

B3 CHECK THE FUEL PUMP MODULE SIGNAL AND RETURN CIRCUITS FOR A SHORT TOGETHER

- Ignition OFF.
- Disconnect: **IPC** [C220](#) .
- Measure the resistance between the **IPC** [C220](#) Pin 15, circuit VMC11 (YE/VT), harness side and the **IPC** [C220](#) Pin 2, circuit RMC32 (GN/BU), harness side.



• N0094472

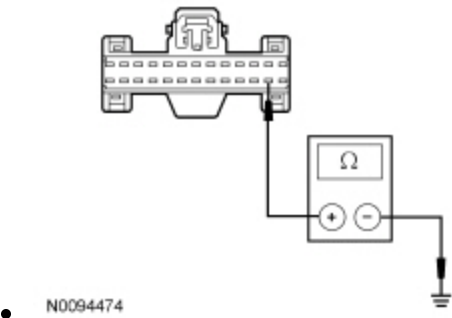
Is the resistance greater than 10,000 ohms?

Yes	GO to B4 .
------------	----------------------------

No	REPAIR the circuits for a short together. CLEAR the DTCs. REPEAT the self-test.
----	---

B4 CHECK THE FUEL LEVEL SIGNAL CIRCUIT FOR A SHORT TO GROUND

- Measure the resistance between the IPC [C220](#) Pin 15, circuit VMC11 (YE/VT), harness side and ground.

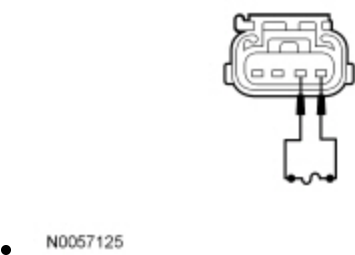


Is the resistance greater than 10,000 ohms?

Yes	GO to B15 .
No	REPAIR circuit VMC11 (YE/VT) for a short to ground. CLEAR the DTCs. REPEAT the self-test.

B5 CHECK THE FUEL PUMP MODULE FOR AN OPEN

- Disconnect: Fuel Pump Module [C4330](#).
- Connect a fused jumper wire between the fuel pump module [C4330](#) Pin 2, circuit VMC11 (YE/VT), harness side and the fuel pump module [C4330](#) Pin 1, circuit RMC32 (GN/BU), harness side.



- Ignition ON.
- Wait 1 minute.
- Enter the following diagnostic mode on the scan tool: IPC Self-Test .
- NOTE:** *DTC P0460:13 may also be present when carrying out this test step and should be ignored.*

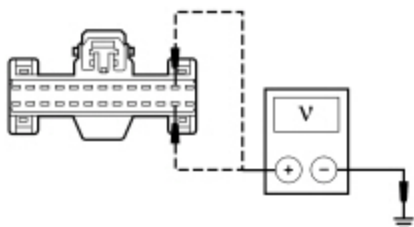
Clear the IPC DTCs. Repeat the IPC self-test.

Is DTC P0460:11 retrieved?

Yes	REMOVE the jumper wire. GO to B13 .
No	REMOVE the jumper wire. GO to B6 .

B6 CHECK THE FUEL LEVEL SIGNAL AND RETURN CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: IPC [C220](#) .
- Ignition ON.
- Measure the voltage between the IPC [C220](#) Pin 15, circuit VMC11 (YE/VT), harness side and ground; and between the IPC [C220](#) Pin 2, circuit RMC32 (GN/BU), harness side and ground.



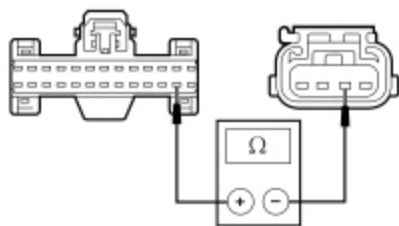
• N0094467

Is any voltage present?

Yes	REPAIR the circuit in question for a short to voltage. CLEAR the DTCs. REPEAT the self-test.
No	GO to B7 .

B7 CHECK THE FUEL LEVEL SIGNAL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between the IPC [C220](#) Pin 15, circuit VMC11 (YE/VT), harness side and the fuel pump module [C4330](#) Pin 2, circuit VMC11 (YE/VT), harness side:



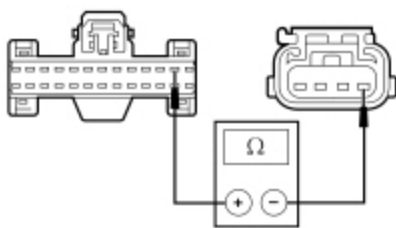
• N0120192

Is the resistance less than 5 ohms?

Yes	GO to B8 .
No	REPAIR circuit VMC11 (YE/VT) for an open. CLEAR the DTCs. REPEAT the self-test.

B8 CHECK THE FUEL LEVEL RETURN CIRCUIT FOR AN OPEN

- Measure the resistance between the IPC [C220](#) Pin 2, circuit RMC32 (GN/BU), harness side and the fuel pump module [C4330](#) Pin 1, circuit RMC32 (GN/BU), harness side.



- N0120193

Is the resistance less than 5 ohms?

Yes	GO to B15 .
No	REPAIR circuit RMC32 (GN/BU) for an open. CLEAR the DTCs. REPEAT the self-test.

B9 CARRY OUT THE IPC FUEL GAUGE ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: IPC DataLogger .
- Select the IPC fuel gauge (FUELLEVEL) active command. Monitor the fuel gauge while scrolling the fuel level at 0%, 50%, and 100%.

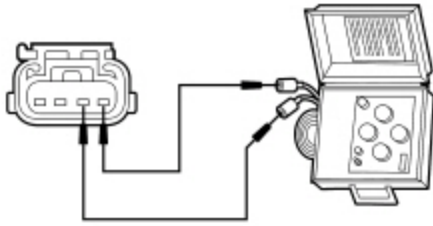
Does the fuel gauge display below E with 0%, half with 50%, and full stop with 100%?

Yes	GO to B10 .
No	GO to B15 .

B10 CHECK THE FUEL LEVEL INPUT TO THE IPC

NOTE: Since the IPC may be in anti-slosh mode, the PID values may not match the fuel gauge readings. Disregard the actual gauge indication during this test step.

- Ignition OFF.
- Disconnect: Fuel Pump Module [C4330](#).
- Connect one lead of the instrument gauge system tester to the fuel pump module [C4330](#) Pin 2, circuit VMC11 (YE/VT), harness side and the other lead to the fuel pump module assembly [C4330](#) Pin 1, circuit RMC32 (GN/BU), harness side.



- N0057130
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC** DataLogger .
- **NOTE:** It is important to confirm the gauge tester settings with an ohmmeter to make sure the gauge tester is in the correct position. Failure to follow this check may result in inaccurate test results.

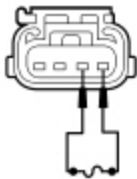
Monitor the **IPC** fuel level PID (FUELLEVEL) with the instrument gauge system tester set at 180 ohms, 90 ohms, 55 ohms, 30 ohms, and 10 ohms.

Does the PID begin at approximately 0%, move to 25%, 50%, 75% then 100%?

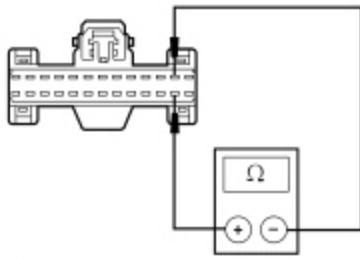
Yes	DISCONNECT the instrument gauge system tester. GO to B12 .
No	DISCONNECT the instrument gauge system tester. GO to B11 .

B11 CHECK THE FUEL PUMP MODULE SIGNAL AND RETURN CIRCUITS FOR A HIGH RESISTANCE

- Ignition OFF.
- Disconnect: **IPC** [C220](#) .
- Connect a fused jumper wire between the fuel pump module [C4330](#) Pin 2, circuit VMC11 (YE/VT), harness side and the fuel pump module [C4330](#) Pin 1, circuit RMC32 (GN/BU), harness side.



- N0057125
- Measure the resistance between the **IPC** [C220](#) Pin 15, circuit VMC11 (YE/VT), harness side and the **IPC** [C220](#) Pin 2, circuit RMC32 (GN/BU), harness side.



• N0094472

Is the resistance less than 5 ohms?

Yes	GO to B15 .
No	REPAIR the circuits for high resistance. TEST the system for normal operation.

B12 CHECK THE FUEL TANK

- Check the fuel tank for any damage or deformation.

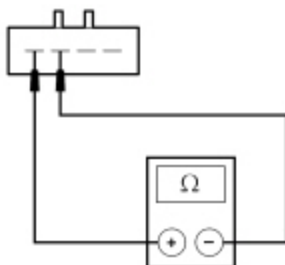
Is the fuel tank OK?

Yes	GO to B13 .
No	VERIFY the fuel level sensor and fuel pump module are OK. INSTALL a new fuel tank. REFER to Section 310-01. CLEAR the DTCs. REPEAT the self-test.

B13 CHECK THE FUEL PUMP MODULE

NOTE: The fuel pump module resistance measures between 10 ohms $\pm$ 2 ohms at the upper stop position and 180 ohms $\pm$ 4 ohms at the lower stop position.

- Remove the fuel pump module. Refer to Section 310-01.
- Measure the resistance between the fuel pump module pin 1, component side, and the fuel pump module pin 2, component side, while moving the float arm from the lower stop position to the upper stop position.



• N0027636

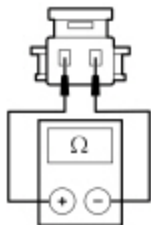
Does the resistance slowly decrease from approximately 180 ohms to 10 ohms?

Yes	CHECK the fuel pump module connector for corrosion, damaged or pushed out pins and REPAIR as necessary.
No	GO to B14 .

B14 CHECK THE FUEL LEVEL SENDER

NOTE: The fuel level sender resistance measures between 10 ohms $\pm$ 2 ohms at the upper stop position and 180 ohms $\pm$ 4 ohms at the lower stop position.

- Disconnect the fuel level sender input wire from the fuel pump module.
- Measure the resistance between the fuel level sender connector pin 1, harness side and the fuel level sender connector pin 2, harness side while slowly moving the float arm between the lower and upper stop position.



• N0112563

Does the resistance slowly decrease from approximately 180 ohms to 10 ohms?

Yes	INSTALL a new fuel pump module. REFER to Section 310-01. CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new fuel level sender (float and card). CLEAR the DTCs. REPEAT the self-test.

B15 CHECK FOR CORRECT IPC OPERATION

- Disconnect the IPC connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the IPC connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test C: Incorrect Oil Pressure Gauge Indication

Normal Operation

The engine oil pressure switch is hardwired to the PCM. The Instrument Panel Cluster (IPC) receives the engine oil pressure status from the PCM over the High Speed Controller Area Network (HS-CAN) .

The PCM provides a reference voltage to the engine oil pressure switch when the ignition key is in RUN. When there is low or no oil pressure, the engine oil pressure switch remains open. The PCM detects no change in the reference voltage and provides engine oil pressure status to **IPC** to keep the oil pressure gauge in the low oil pressure area of the gauge. With the engine running and sufficient oil pressure, the engine oil pressure switch closes, pulling the reference voltage low. The PCM detects the low reference voltage and provides engine oil pressure status to the **IPC** to command the oil pressure gauge to the normal oil pressure range.

If the **IPC** fails to receive the oil pressure status from the PCM for 5 seconds or less, the **IPC** defaults the oil pressure gauge to the last setting, based upon the last known good oil pressure status message. If the **IPC** fails to receive the oil pressure status from the PCM for more than 5 seconds, the **IPC** sets DTC U0100:00 in continuous memory and defaults the oil pressure gauge to low (L).

NOTE: *If DTC U0100:00 sets in the **IPC** , other observable symptoms may be an inoperative speedometer, tachometer, temperature gauge or odometer.*

This pinpoint test is intended to diagnose the following:

- Communication network concern
- Wiring, terminals or connectors
- Engine oil pressure switch
- Base engine oil pressure concern
- PCM
- **IPC**

PINPOINT TEST C : INCORRECT OIL PRESSURE GAUGE INDICATION

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

C1 CARRY OUT THE IPC OIL PRESSURE GAUGE ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC** DataLogger .
- Select the **IPC** oil pressure gauge active command (OIL_PRESS) on and off to sweep the gauge.

Does the engine oil pressure gauge sweep to the normal range when commanded on and return to low (L) when commanded off?

Yes	If the oil pressure gauge always reads in the normal range, GO to C2 . If the oil pressure gauge always reads low (L) (does not register any oil pressure), GO to C4 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

C2 CHECK FOR A STUCK CLOSED ENGINE OIL PRESSURE SWITCH

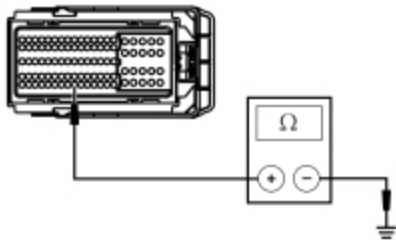
- Ignition OFF.
- Disconnect: Engine Oil Pressure Switch [C103](#).
- Ignition ON.
- Observe the engine oil pressure gauge.

Does the engine oil pressure gauge indicate low (L) oil pressure?

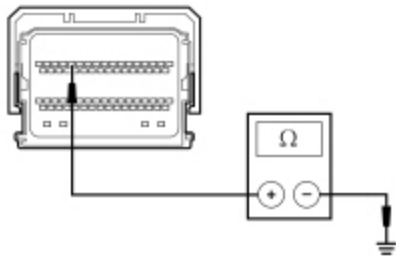
Yes	INSTALL a new engine oil pressure switch. Refer to the appropriate section in Group 303 for the procedure.. TEST the system for normal operation.
No	GO to C3 .

C3 CHECK THE OIL PRESSURE INPUT CIRCUIT FOR A SHORT TO GROUND

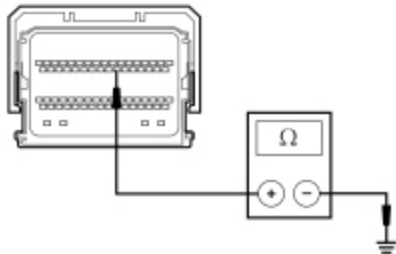
- Ignition OFF.
- Disconnect: PCM [C175E](#) (3.7L), [C1381E](#) (5.0L and 6.2L), or [C1551E](#) (3.5L) .
- For 3.5L, measure the resistance between the PCM [C1551E](#) Pin 9, circuit CMC24 (GY), harness side and ground.



- For 3.7L, measure the resistance between the PCM [C175E](#) Pin 13, circuit CMC24 (GY), harness side and ground.



- For 5.0L and 6.2L, measure the resistance between the PCM [C1381E](#) Pin 24, circuit CMC24 (GY), harness side and ground.



Is the resistance greater than 10,000 ohms?

Yes	GO to C8 .
-----	----------------------------

No	REPAIR circuit CMC24 (GY) for a short to ground. TEST the system for normal operation.
----	--

C4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

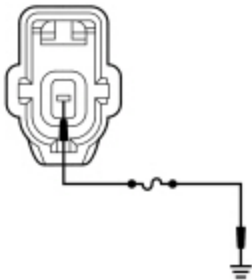
- Check for recorded IPC DTCs from the self-test.

Is DTC U0100:00 recorded?

Yes	GO to Pinpoint Test AT.
No	GO to C5 .

C5 CHECK FOR A STUCK OPEN ENGINE OIL PRESSURE SWITCH

- Ignition OFF.
- Disconnect: Engine Oil Pressure Switch [C103](#).
- Connect a fused jumper wire between the engine oil pressure switch [C103](#) Pin 1, circuit CMC24 (GY), harness side and ground.



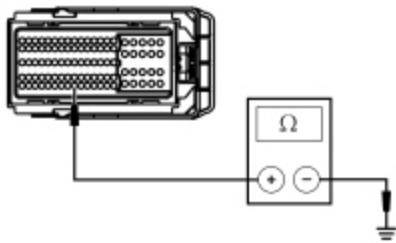
- N0002760
- Ignition ON.
- Observe the engine oil pressure gauge.

Does the engine oil pressure gauge indicate normal oil pressure?

Yes	REMOVE the jumper wire. GO to C7 .
No	LEAVE the jumper wire connected. GO to C6 .

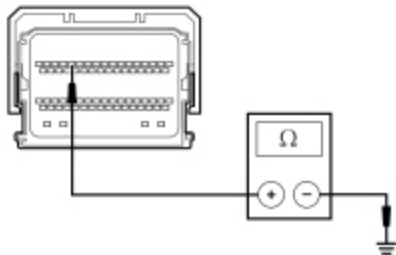
C6 CHECK THE ENGINE OIL PRESSURE INPUT CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: PCM [C175E](#) (3.7L), [C1381E](#) (5.0L and 6.2L), or [C1551E](#) (3.5L) .
- For 3.5L, measure the resistance between the PCM [C1551E](#) Pin 9, circuit CMC24 (GY), harness side and ground.



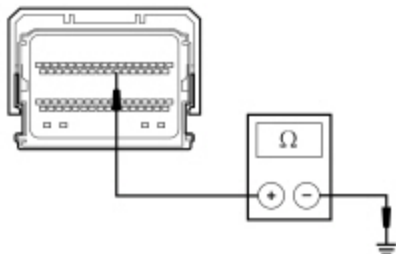
• N0122305

- For 3.7L, measure the resistance between the PCM [C175E](#) Pin 13, circuit CMC24 (GY), harness side and ground.



• N0120194

- For 5.0L and 6.2L, measure the resistance between the PCM [C1381E](#) Pin 24, circuit CMC24 (GY), harness side and ground.



• N0112570

Is the resistance less than 5 ohms?

Yes	REMOVE the jumper wire. GO to C8 .
No	REMOVE the jumper wire. REPAIR circuit CMC24 (GY) for an open. TEST the system for normal operation.

C7 CHECK THE ENGINE OIL PRESSURE

- Carry out the engine oil pressure test. Refer to Section 303-00.

Is the engine oil pressure within specification?

Yes	INSTALL a new engine oil pressure switch. Refer to the appropriate section in Group 303 for the procedure. TEST the system for normal operation.
No	REFER to Section 303-00 to continue diagnosis of the engine oil pressure.

C8 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Connect: Engine Oil Pressure Switch [C103](#).
- Connect: PCM [C175E](#) (3.7L), [C1381E](#) (5.0L and 6.2L) or [C1551E](#) (3.5L) .
- Check the recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Are any DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to C9 .

C9 CHECK FOR CORRECT PCM OPERATION

- Disconnect the PCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the PCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test D: The Temperature Gauge Is Inoperative

Normal Operation

The PCM uses the Cylinder Head Temperature (CHT) sensor to measure the engine temperature. The Instrument Panel Cluster (IPC) receives the engine temperature data from the PCM over the High Speed Controller Area Network (HS-CAN) . The IPC monitors the engine temperature data received from the PCM and commands the temperature gauge indication with a corresponding movement of the pointer.

If the IPC receives invalid engine temperature data from the PCM for 5 seconds or less, the IPC defaults the temperature gauge to the last setting, based upon the last known good engine temperature status message. If the IPC does not receive the engine temperature status message from the PCM or if the data received is deemed invalid for more than 5 seconds, the IPC sets DTC U0100:00 in continuous memory and defaults the temperature gauge to cold (C). If the engine temperature data is deemed invalid by the IPC , the IPC sets DTC U0401:00 and defaults the temperature gauge to cold (C).

NOTE: *If DTC U0100:00 sets in the IPC , other observable symptoms may be an inoperative speedometer, tachometer or odometer.*

This pinpoint test is intended to diagnose the following:

- PCM
- IPC

PINPOINT TEST D : THE TEMPERATURE GAUGE IS INOPERATIVE

D1 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check the recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Are any DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to D2 .

D2 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCs, REFER to DTC Charts in this section.
No	GO to D3 .

D3 CARRY OUT THE TEMPERATURE GAUGE ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: IPC DataLogger .
- **NOTE:** *Make sure the temperature gauge is commanded to full cold before beginning this step.*

Select the IPC temperature gauge (ENGCOOLNT) active command. Command the temperature gauge on and off while monitoring the gauge.

Does the temperature gauge start at cold (C), move to full hot (H) and return to cold (C)?

Yes	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test E: Incorrect Temperature Gauge Indication

Normal Operation

The PCM uses the Cylinder Head Temperature (CHT) sensor to measure the engine temperature. The Instrument Panel Cluster (IPC) receives the engine temperature data from the PCM over the High Speed Controller Area Network (HS-CAN) . The IPC monitors the engine temperature data received from the PCM and commands the temperature gauge indication with a corresponding movement of the pointer.

If the **IPC** receives invalid engine temperature data from the PCM for 5 seconds or less, the **IPC** defaults the temperature gauge to the last setting, based upon the last known good engine temperature status message. If the **IPC** does not receive the engine temperature status message from the PCM or if the data received is deemed invalid for more than 5 seconds, the **IPC** sets DTC U0100:00 in continuous memory and defaults the temperature gauge to cold (C). If the engine temperature data is deemed invalid by the **IPC** , the **IPC** sets DTC U0401:00 and defaults the temperature gauge to cold (C).

NOTE: *If DTC U0100:00 sets in the **IPC** , other observable symptoms may be an inoperative speedometer, tachometer or odometer.*

This pinpoint test is intended to diagnose the following:

- PCM concern
- **IPC**

PINPOINT TEST E : INCORRECT TEMPERATURE GAUGE INDICATION

E1 CHECK FOR CORRECT OPERATION OF THE COOLING SYSTEM

- Verify the engine cooling system and thermostat are functioning correctly.

Do the engine cooling system and thermostat operate correctly?

Yes	GO to E2 .
No	REFER to Section 303-03.

E2 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check the recorded PCM DTCs from the **KOEO** self-test.

Are any DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to E3 .

E3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded **IPC** DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCs, REFER to DTC Charts in this section.
No	GO to E4 .

E4 CARRY OUT THE TEMPERATURE GAUGE ACTIVE COMMAND USING THE SCAN TOOL

NOTE: *The coolant temperature should range between 0-59°C (32-138°F) with the gauge in the cold (C) range, between 60-120°C (140-248°F) with the gauge in the normal or mid-range and above 121°C (250°F) with the gauge in the hot (H) range.*

- Ignition ON.

- Enter the following diagnostic mode on the scan tool: **IPC** DataLogger .
- **NOTE:** *Make sure the temperature gauge is commanded to full cold before beginning this step.*

Select the **IPC** temperature gauge (ENGCOOLNT) active command. Command the temperature gauge according to the following table:

Temperature Command	Gauge Position
20%	Beginning of the normal band
50%	Center of the gauge (mid-range)
90%	End of the normal band

Does the temperature gauge operate according to the above specifications?

Yes	GO to E5 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

E5 CHECK THE PCM TEMPERATURE PID

NOTE: *The coolant temperature should range between 0-59°C (32-138°F) with the gauge in the cold (C) range, between 60-120°C (140-248°F) with the gauge in the normal or mid-range and above 121°C (250°F) with the gauge in the hot (H) range.*

- Enter the following diagnostic mode on the scan tool: **PCM** DataLogger .
- Ignition ON.
- Select the PCM temperature PID (CHT). With the engine running, monitor the PCM temperature PID.

Does the PCM PID agree with the position of the temperature gauge indication?

Yes	The temperature gauge is operating correctly at this time.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test F: The Speedometer Is Inoperative

Normal Operation

The Instrument Panel Cluster (IPC) receives the vehicle speed signal from the PCM over the High Speed Controller Area Network (HS-CAN) . The PCM receives the vehicle speed signal input from the transmission Output Shaft Speed (OSS) sensor. The **IPC** monitors the vehicle speed signal input from the PCM and commands the speedometer with a corresponding movement of the pointer.

If the **IPC** receives invalid vehicle speed data from the PCM for 5 seconds or less, the **IPC** defaults the speedometer to the last setting, based upon the last known good vehicle speed message. If the **IPC** does not receive the vehicle speed data from the PCM for more than 5 seconds, the **IPC** sets DTC U0100:00 in continuous memory and defaults the speedometer to 0 km/h (0 mph). If the vehicle speed data received is deemed invalid by the **IPC** , the **IPC** sets DTC U0401:00 and defaults the speedometer to 0 km/h (0 mph).

NOTE: *If DTC U0100:00 sets in the **IPC** , other observable symptoms may be an inoperative tachometer, temperature gauge or odometer.*

This pinpoint test is intended to diagnose the following:

- PCM
- **IPC**

PINPOINT TEST F : THE SPEEDOMETER IS INOPERATIVE

F1 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST	
• Check the recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.	
Are any DTCs recorded?	
Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to F2 .
F2 CARRY OUT THE SPEEDOMETER ACTIVE COMMAND USING THE SCAN TOOL	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: IPC DataLogger . • Select the IPC speedometer (SPDOMETER) active command. Command the speedometer on and off while monitoring the speedometer.	
Does the speed begin at 0 km/h (0 mph) move to 193 km/h (120 mph) and return to 0 km/h (0 mph)?	
Yes	GO to F3 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
F3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST	
• Check for recorded IPC DTCs from the self-test.	
Are any DTCs recorded?	
Yes	For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCs, REFER to DTC Charts in this section.
No	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.

Pinpoint Test G: Incorrect Speedometer Indication

Normal Operation

The Instrument Panel Cluster (IPC) receives the vehicle speed signal from the PCM over the High Speed Controller Area Network (HS-CAN) . The PCM receives the vehicle speed signal input from the transmission Output Shaft Speed (OSS) sensor. The IPC monitors the vehicle speed signal input from the PCM and commands the speedometer with a corresponding movement of the pointer.

The IPC provides a tolerance which allows the gauge to display between 3% lower and 7% higher than the actual vehicle speed. This means that with an actual vehicle speed of 96.6 km/h (60 mph), the speedometer may indicate between 93.7-103.3 km/h (58.2-64.1 mph). Factors that could potentially affect the speedometer accuracy are incorrect tire size configuration and axle ratio configuration.

If the IPC receives invalid vehicle speed data from the PCM for 5 seconds or less, the IPC defaults the speedometer to the last setting, based upon the last known good vehicle speed message. If the IPC does not receive the vehicle speed data from the PCM for more than 5 seconds, the IPC sets DTC U0100:00 in continuous memory and defaults the speedometer to 0 km/h (0 mph). If the vehicle speed data received is deemed invalid by the IPC , the IPC sets DTC U0401:00 and defaults the speedometer to 0 km/h (0 mph).

NOTE: If DTC U0100:00 sets in the IPC , other observable symptoms may be an inoperative tachometer, temperature gauge or odometer.

This pinpoint test is intended to diagnose the following:

- Tire size configuration
- Axle ratio configuration
- PCM concern
- IPC

PINPOINT TEST G : INCORRECT SPEEDOMETER INDICATION

G1 CHECK THE TIRE SIZE CONFIGURATION

- Enter the following diagnostic mode on the scan tool: Module Programming .
- **NOTE:** The correct tire size can be found on the Vehicle Certification (VC) label on the LH B-pillar. Refer to Section 100-01.

Verify the size of the tires installed on the vehicle matches the VC label.

- Verify the tire size configuration matches the VC label.

Is the tire size correct and is tire size correctly configured?

Yes	GO to G2 .
No	INSTALL the correct size tires or CONFIGURE the correct tire size as required. TEST the system for normal operation.

G2 CHECK THE AXLE RATIO CONFIGURATION

- Enter the following diagnostic mode on the scan tool: Module Programming .
- **NOTE:** The correct axle ratio can be found on the Vehicle Certification (VC) label on the LH B-pillar. Refer to Section 100-01.

Verify the axle ratio configuration matches the VC label.

Is the axle ratio correctly configured?

Yes	GO to G3 .
No	CONFIGURE the correct axle ratio. TEST the system for normal operation.

G3 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check the recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Are any DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to G4 .

G4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCs, REFER to DTC Charts in this section.
No	GO to G5 .

G5 CARRY OUT THE SPEEDOMETER ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: ~~IPC~~ DataLogger .
- Select the ~~IPC~~ speedometer (SPDOMETER) active command. Command the speedometer in 10% increments while monitoring the speedometer. The speed indicated should increase in increments of approximately 19 km/h (12 mph) for each 10% change.

Command	Indication Range
10% (19 km/h [12 mph])	19-21 km/h (12-13 mph)
20% 39 km/h [24 mph])	37-42 km/h (23-26 mph)
30% (58 km/h [36 mph])	56-63 km/h (35-39 mph)
40% (77 km/h [48 mph])	76-82 km/h (47-51 mph)
50% (97 km/h [60 mph])	93-103 km/h (58-64 mph)
60% (116 km/h [72 mph])	113-124 km/h (70-77 mph)
70% (135 km/h [84 mph])	130-145 km/h (81-90 mph)
80% (154 km/h [96 mph])	150-166 km/h (93-103 mph)
90% (174 km/h [108 mph])	169-187 km/h (105-116 mph)

Command	Indication Range
100% (193 km/h [120 mph])	116-206 km/h (116-128 mph)

Does the speed indicated increase according to the above specifications?

Yes	GO to G6 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

G6 CHECK THE PCM VSS PIDS

- Enter the following diagnostic mode on the scan tool: PCM DataLogger .
- Select the PCM vehicle speed (VSS) PID and monitor the PID while driving the vehicle at 32 km/h (20 mph), 64 km/h (40 mph) and 97 km/h (60 mph).

Does the speedometer indicate between 31-34 km/h (19-21 mph), 63-69 km/h (39-43 mph) and 93-103 km/h (58-64 mph) at the 3 PCM PID values?

Yes	GO to G7 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

G7 OBSERVE THE SPEEDOMETER OPERATION

- Ignition ON.
- Observe the speedometer while driving the vehicle at various speeds and stopping frequently.

Does the speedometer begin at 0 km/h (0 mph) and fully return to the 0 km/h (0 mph) position when the vehicle is stopped?

Yes	The speedometer is operating correctly at this time.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test H: The Odometer Is Inoperative

Normal Operation

The Instrument Panel Cluster (IPC) receives the odometer rolling count data from the PCM over the High Speed Controller Area Network (HS-CAN) . The IPC monitors the odometer rolling count input from the PCM and commands the odometer with a display in the message center display.

If the **IPC** does not receive the odometer rolling count data from the PCM for more than 5 seconds, the **IPC** sets DTC U0100:00 in continuous memory and defaults the odometer display to all dashes (-----). If the odometer rolling count data is deemed invalid by the **IPC**, the **IPC** sets DTC U0401:86 and defaults the odometer display to all dashes (-----).

If the Non-Volatile Memory (NVM) becomes corrupted or if any **NVM** failure occurs, the **IPC** sets DTC U3000:04 and displays ERROR in the message center display.

NOTE: If DTC U0100:00 sets in the **IPC**, other observable symptoms may be an inoperative speedometer, tachometer, temperature gauge.

This pinpoint test is intended to diagnose the following:

- PCM
- **IPC**

PINPOINT TEST H : THE ODOMETER IS INOPERATIVE

H1 CHECK FOR CORRUPTED NVM	
• Ignition ON. • Observe the message center display area.	
Does the odometer display ERROR?	
Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
No	GO to H2 .
H2 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST	
• Check the recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.	
Are any DTCs recorded?	
Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to H3 .
H3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST	
• Check for recorded IPC DTCs from the self-test.	
Are any DTCs recorded?	
Yes	For DTC U0100:00, GO to Pinpoint Test AT . For DTC U3000:04, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation. For all other DTCs, REFER to DTC Charts in this section.
No	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.

Pinpoint Test I: The Tachometer Is Inoperative

Normal Operation

The PCM uses the Crankshaft Position (CKP) sensor to measure the engine rpm. The PCM sends the rpm data to the Instrument Panel Cluster (IPC) over the High Speed Controller Area Network (HS-CAN) to command the tachometer according to the data.

If the IPC does not receive the engine rpm data from the PCM for 5 seconds or less, the IPC defaults the tachometer to the last setting, based upon the last known good rpm message. If the IPC does not receive the engine rpm data from the PCM for more than 5 seconds, the IPC sets DTC U0100:00 in continuous memory and defaults the tachometer to 0 rpm.

NOTE: If DTC U0100:00 sets in the IPC, other observable symptoms may be an inoperative speedometer, temperature gauge or odometer.

This pinpoint test is intended to diagnose the following:

- PCM
- IPC

PINPOINT TEST I : THE TACHOMETER IS INOPERATIVE

I1 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check the recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Are any DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to I2.

I2 CARRY OUT THE TACHOMETER ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: IPC DataLogger .
- Select the IPC tachometer (TACH_IND) active command. Command the tachometer on and off while monitoring the tachometer.

Does the tachometer begin at 0 rpm, increase to 7,000 rpm then return to 0 rpm?

Yes	GO to I3.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

I3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT. For all other DTCs, REFER to DTC Charts in this section.
-----	---

No	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.
-----------	---

Pinpoint Test J: Incorrect Tachometer Indication

Normal Operation

The PCM uses the Crankshaft Position (CKP) sensor to measure the engine rpm. The PCM sends the rpm data to the Instrument Panel Cluster (IPC) over the High Speed Controller Area Network (HS-CAN) to command the tachometer gauge according to the data.

If the IPC does not receive the engine rpm data from the PCM for 5 seconds or less, the IPC defaults the tachometer to the last setting, based upon the last known good rpm message. If the IPC does not receive the engine rpm data from the PCM for more than 5 seconds, the IPC sets DTC U0100:00 in continuous memory and defaults the tachometer to 0 rpm.

NOTE: If DTC U0100:00 sets in the IPC, other observable symptoms may be an inoperative speedometer, temperature gauge or odometer.

This pinpoint test is intended to diagnose the following:

- PCM concern
- IPC

PINPOINT TEST J : INCORRECT TACHOMETER INDICATION

J1 RETRIEVE THE RECORDED DTCS FROM PCM KOEO SELF-TEST

- Check the recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Are any DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to J2 .

J2 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCs, REFER to DTC Charts in this section.
No	GO to J3 .

J3 CARRY OUT THE TACHOMETER ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: IPC DataLogger .
- Select the IPC tachometer (TACH_IND) active command. Command the tachometer in 10% increments while monitoring the tachometer.

Does the tachometer indication increase by 700 rpm at each 10% increment?

Yes	GO to J4 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

J4 CHECK THE PCM RPM PIDS

- Enter the following diagnostic mode on the scan tool: PCM DataLogger .
- Select the PCM engine rpm (RPM) PID and monitor the tachometer while operating the engine at 1,000, 1,500, 2,000 and 2,500 rpm using the PCM PID to set the vehicle speed.

Command	Indication Range
1,000 rpm	922-1,078 rpm
1,500 rpm	1,422-1,578 rpm
2,000 rpm	1,922-2,078 rpm
2,500 rpm	2,422-2,578 rpm

Does the tachometer indication increase according to the above specifications?

Yes	The tachometer is operating correctly at this time.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test K: The Transmission Temperature Gauge Is Inoperative

Normal Operation

The Instrument Panel Cluster (IPC) receives the transmission temperature data from the PCM over the High Speed Controller Area Network (HS-CAN) . The PCM filters the transmission temperature input and sends the filtered data to the IPC . The IPC then further dampens the temperature data to eliminate minor fluctuations in the gauge indication.

If the transmission temperature data is missing or invalid from the PCM for 5 seconds or less, the IPC defaults the transmission temperature gauge to the last setting, based upon the last known good message received. If the IPC does not receive the transmission temperature data from the PCM for more than 5 seconds, the IPC sets DTC U0100:00 in continuous memory and defaults the transmission temperature gauge to the cold (C) stop position. If the transmission temperature data is deemed invalid by the IPC for more than 5 seconds, the IPC sets DTC U0401:00 and defaults the transmission temperature gauge to the cold (C) stop position.

NOTE: If DTC U0100:00 sets in the IPC , other observable symptoms may be the powertrain malfunction (wrench) warning indicator always on.

This pinpoint test is intended to diagnose the following:

- PCM

- IPC

PINPOINT TEST K : THE TRANSMISSION TEMPERATURE GAUGE IS INOPERATIVE

K1 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check the recorded PCM DTCs from the self-test.

Are any transmission temperature DTCs recorded?

Yes	REFER to Section 307-01. For all other DTCs, REFER to Section 303-14. TEST the system for normal operation.
No	GO to K2 .

K2 CARRY OUT THE TRANSMISSION TEMPERATURE GAUGE ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: IPC DataLogger .
- Select the IPC transmission temperature gauge (TRANS_TEMP) active command. Command the transmission temperature gauge on and off while monitoring the transmission temperature gauge.

Does the transmission temperature gauge move to hot (H) and return to cold (C)?

Yes	GO to K3 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

K3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCs, REFER to DTC Charts in this section.
No	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.

Pinpoint Test L: Incorrect Transmission Temperature Gauge Indication

Normal Operation

The Instrument Panel Cluster (IPC) receives the transmission temperature data from the PCM over the High Speed Controller Area Network (HS-CAN) . The PCM filters the transmission temperature input and sends the filtered data to the IPC . The IPC then further dampens the temperature data to eliminate minor fluctuations in the gauge indication.

If the transmission temperature data is missing or invalid from the PCM for 5 seconds or less, the IPC defaults the transmission temperature gauge to the last setting, based upon the last known good message received. If the IPC does not

receive the transmission temperature data from the PCM for more than 5 seconds, the **IPC** sets DTC U0100:00 in continuous memory and defaults the transmission temperature gauge to the cold (C) stop position. If the transmission temperature data is deemed invalid by the **IPC** for more than 5 seconds, the **IPC** sets DTC U0401:00 and defaults the transmission temperature gauge to the cold (C) stop position.

This pinpoint test is intended to diagnose the following:

- Transmission concern
- **IPC**

PINPOINT TEST L : INCORRECT TRANSMISSION TEMPERATURE GAUGE INDICATION

L1 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check the recorded PCM DTCs from the self-test.

Are any transmission temperature DTCs recorded?

Yes	REFER to Section 307-01. For all other DTCs, REFER to Section 303-14.
No	GO to L2 .

L2 CARRY OUT THE TRANSMISSION TEMPERATURE GAUGE ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC** DataLogger .
- **NOTE:** *Make sure the gauge begins at full cold (C). If necessary, adjust the scan tool to command the gauge to full cold (C) before beginning this step.*

Select the **IPC** transmission temperature gauge (TRANS_TEMP) active command. Command the transmission temperature gauge as follows:

Scan Tool Setting	Gauge Indication	Temperature
18%	Beginning of normal range	27°C (81°F)
45 and 54%	Center of normal range (just below or above center respectively)	28-106°C (82-223°F)
81%	End of normal range	107°C (225°F)

Does the transmission temperature gauge indication match the temperature specification?

Yes	GO to L3 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

L3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded **IPC** DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCs, REFER to DTC Charts in this section.
No	GO to L4 .

L4 CHECK THE PCM TEMPERATURE PID

NOTE: This step may require the use of an assistant to monitor the transmission temperature PID and the transmission temperature gauge indication.

- Enter the following diagnostic mode on the scan tool: PCM DataLogger .
 - Ignition ON.
 - **NOTE:** The transmission temperature should range below 10°C (50°F) with the gauge in the cold (C) range, between 27 to 107°C (81 to 225°F) with the gauge in the normal or mid-range and above 135°C (275°F) with the gauge in the hot (H) range.
- Select the PCM transmission temperature gauge PID (TFT).
- Beginning with a cold transmission, drive the vehicle while monitoring the PCM transmission temperature PID.

Does the PID agree with the position of the transmission temperature gauge indication at different temperature ranges?

Yes	The transmission temperature gauge is operating correctly at this time.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation. TEST the system for normal operation.

Pinpoint Test M: The Brake Warning Indicator is Never On

Refer to Wiring Diagrams Cell [60](#) , Instrument Cluster for schematic and connector information.

Normal Operation

The Instrument Panel Cluster (IPC) uses 3 basic messaged inputs to control the brake warning indicator. The first two are the parking brake warning indicator switch and the brake fluid level switch. The third is for the Electronic Brake Distribution (EBD) function controlled by the ABS module. The parking brake warning indicator switch and the brake fluid level switch status are sent from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The EBD message is sent from the ABS module over the HS-CAN .

The parking brake warning indicator switch is hardwired to the BCM through a single signal circuit while using a separate ground to control the input. The brake fluid level switch is hardwired to the BCM through separate signal and return circuits while using a separate ground to control the input.

The BCM provides a reference voltage to both the parking brake warning indicator switch and the brake fluid level switch. When the parking brake is applied, the parking brake warning indicator switch closes to ground, pulling the parking brake signal circuit low. When a low brake fluid level condition exists, the brake fluid level switch closes to ground, pulling the signal and return circuits low. If the brake fluid level switch is disconnected, the BCM reference voltage sent to the brake

fluid level switch is sent high and the BCM sets DTC B1048:7B. When the BCM detects either the brake fluid level signal or the parking brake signal circuits low, or the brake fluid level signal and return circuit high, the BCM sends the IPC a brake warning indicator on request.

When the ABS module detects a base brake system concern or other ABS related concerns that affect the EBD function, the ABS module sends a brake warning indicator on request to the IPC to illuminate the brake warning indicator and the ABS warning indicator.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Brake fluid level switch
- Parking brake warning indicator switch
- BCM
- IPC

PINPOINT TEST M : THE BRAKE WARNING INDICATOR IS NEVER ON

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Failure to disconnect the battery when instructed, results in false resistance readings. Refer to Section 414-01.

M1 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCS from the self-test.

Are any DTCS recorded?

Yes	REFER to DTC Charts in this section.
No	GO to M2 .

M2 DETERMINE IF THE BRAKE WARNING INDICATOR OPERATES WITH THE PARKING BRAKE

- Ignition ON.
- Apply the parking brake while monitoring the brake warning indicator.

Does the brake warning indicator illuminate with the parking brake applied?

Yes	RELEASE the parking brake. GO to M8 .
No	RELEASE the parking brake. GO to M3 .

M3 CARRY OUT THE IPC INDICATOR LAMP CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: IPC DataLogger .
- Select the IPC brake warning indicator (PBA_LMP) active command. Command the brake warning indicator on and observe the brake warning indicator.

Does the brake warning indicator illuminate when commanded on?

Yes	GO to M4 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

M4 CHECK THE PARKING BRAKE WARNING INDICATOR SWITCH PID

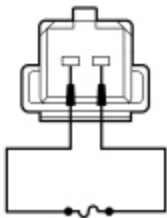
- Enter the following diagnostic mode on the scan tool: BCM DataLogger .
- Select the BCM parking brake (PARK_BRK_SW) PID. Monitor the parking brake PID while applying the parking brake.

Does the PID agree with the parking brake position?

Yes	GO to M12 .
No	GO to M5 .

M5 CHECK THE PARKING BRAKE WARNING INDICATOR SWITCH

- Disconnect: Parking Brake Warning Indicator Switch [C2015](#).
- Connect a fused jumper wire between the parking brake warning indicator switch [C2015](#) Pin 2, circuit GD120 (BK/GN), harness side and the parking brake warning indicator switch [C2015](#) Pin 1, circuit CMC25 (WH/VT), harness side.



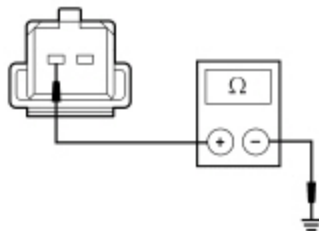
• N0002774

Does the brake warning indicator illuminate?

Yes	REMOVE the jumper wire. INSTALL a new parking brake warning indicator switch. TEST the system for normal operation.
No	REMOVE the jumper wire. GO to M6 .

M6 CHECK THE PARKING BRAKE WARNING INDICATOR SWITCH GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the parking brake warning indicator switch [C2015](#) Pin 2, circuit GD120 (BK/GN), harness side and ground.



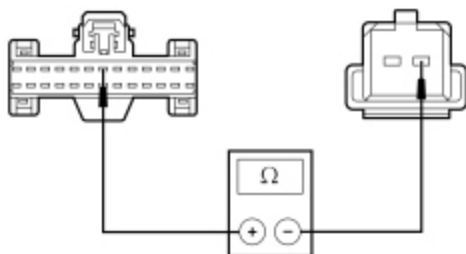
• N0057133

Is the resistance less than 5 ohms?

Yes	GO to M7 .
No	REPAIR circuit GD120 (BK/GN) for an open. TEST the system for normal operation.

M7 CHECK THE PARKING BRAKE WARNING INDICATOR SIGNAL CIRCUIT FOR AN OPEN

- Disconnect: BCM [C2280C](#) .
- Measure the resistance between the BCM [C2280C](#) Pin 7, circuit CMC25 (WH/VT), harness side and the parking brake warning indicator switch [C2015](#) Pin 1, circuit CMC25 (WH/VT), harness side and ground.



• N0112572

Is the resistance less than 5 ohms?

Yes	GO to M12 .
No	REPAIR circuit CMC25 (WH/VT) for an open. TEST the system for normal operation.

M8 CHECK THE BRAKE FLUID LEVEL SWITCH OPERATION

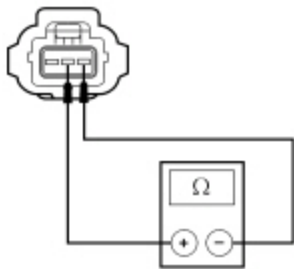
- Ignition OFF.
- Disconnect: Brake Fluid Level Switch [C124](#).
- Ignition ON.

Does the brake warning indicator illuminate?

Yes	GO to M10 .
No	GO to M9 .

M9 CHECK THE BRAKE FLUID LEVEL SIGNAL AND RETURN CIRCUITS FOR A SHORT TOGETHER

- Ignition OFF.
- Disconnect: BCM [C2280F](#) .
- Measure the resistance between the brake fluid level switch [C124](#) Pin 2, circuit CMC19 (GY/VT), harness side and the brake fluid level switch [C124](#) Pin 3, circuit RMC19 (YE/GY), harness side.



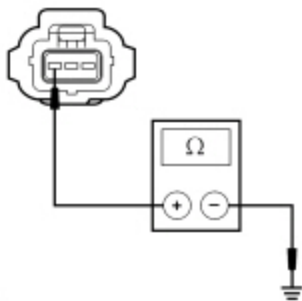
• N0066349

Is the resistance greater than 10,000 ohms?

Yes	GO to M12 .
No	REPAIR the circuits. TEST the system for normal operation.

M10 CHECK THE BRAKE FLUID LEVEL SWITCH GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the brake fluid level switch [C124](#) Pin 1, circuit GD120 (BK/GN), harness side and ground.



• N0027596

Is the resistance less than 5 ohms?

Yes	GO to M11 .
No	REPAIR circuit GD120 (BK/GN) for an open. TEST the system for normal operation.

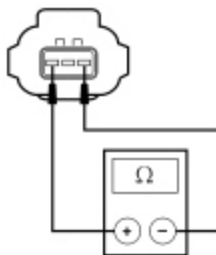
M11 CHECK THE BRAKE FLUID LEVEL SWITCH

- Ignition ON.

- **NOTICE: Do not spill brake fluid onto painted or plastic surfaces. If spilled, wipe up immediately before damage to the painted or plastic surface occurs.**

Remove the brake fluid from the reservoir to allow the brake fluid level switch to fall to the bottom of the reservoir.

- Measure the resistance between the brake fluid level switch pin 3, component side and the brake fluid level switch pin 1, component side.



- N0057135

Is the resistance less than 5 ohms?

Yes	GO to M12 .
No	INSTALL a new brake fluid level switch. TEST the system for normal operation.

M12 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test N: The Brake Warning Indicator is Always On

Normal Operation

The Instrument Panel Cluster (IPC) uses 3 basic messaged inputs to control the brake warning indicator. The first two are the parking brake warning indicator switch and the brake fluid level switch. The third is for the Electronic Brake Distribution (EBD) function controlled by the ABS module. The parking brake warning indicator switch and the brake fluid

level switch status are sent from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The EBD message is sent from the ABS module over the HS-CAN .

The parking brake warning indicator switch is hardwired to the BCM through a single signal circuit while using a separate ground to control the input. The brake fluid level switch is hardwired to the BCM through separate signal and return circuits while using a separate ground to control the input.

The BCM provides a reference voltage to both the parking brake warning indicator switch and the brake fluid level switch. When the parking brake is applied, the parking brake warning indicator switch closes to ground, pulling the parking brake signal circuit low. When a low brake fluid level condition exists, the brake fluid level switch closes to ground, pulling the signal and return circuits low. If the brake fluid level switch is disconnected, the BCM reference voltage sent to the brake fluid level switch is sent high and the BCM sets DTC B1048:7B. When the BCM detects either the brake fluid level signal or the parking brake signal circuits low, or the brake fluid level signal and return circuit high, the BCM sends the IPC a brake warning indicator on request.

When the ABS module detects a base brake system concern or other ABS related concerns that affect the EBD function, the ABS module sends a brake warning indicator on request to the IPC to illuminate the brake warning indicator and the ABS warning indicator.

If the brake fluid level or parking brake status messages are missing for less than 5 seconds, the IPC defaults the brake warning indicator to its last state (on or off) based upon the last known good message received. If the brake fluid level or parking brake status messages are missing for more than 5 seconds, the IPC sets DTC U0140:00 and defaults the brake warning indicator on. If the IPC does not receive the ABS status message or the EBD message from the ABS module for more than 5 seconds, the IPC sets DTC U0121:00 and defaults the ABS warning indicator on.

DTC Description	Fault Trigger Conditions
DTC B1048:01 — Brake Fluid Level Switch: General Electrical Failure	A continuous DTC that sets when the BCM detects an open or short to ground on the brake fluid level switch input.
DTC B1048:7B — Brake Fluid Level Switch: Low Fluid Level	A continuous DTC that sets when the BCM detects an open or short to ground on the brake fluid level switch input.
DTC C1D00:11 — Park Brake Apply Switch: Circuit Short to Ground	A continuous DTC that sets in the BCM if the BCM detects a short to ground on the parking brake input circuit.

NOTE: If DTC U0140:00 sets in the IPC , other symptoms may also be present such as Tire Pressure Monitoring System (TPMS) , instrument panel backlighting, and door ajar indication. If DTC U0121:00 is set, other symptoms may also be present such as ABS warning indicator and stability/traction control or stability/traction control off indicators always on.

NOTE: Both the ABS warning indicator and the brake warning indicator illuminate if the input to illuminate the brake warning indicator is received from the ABS module.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Parking brake warning indicator switch
- Brake fluid level switch
- ABS concern
- BCM
- IPC

PINPOINT TEST N : THE BRAKE WARNING INDICATOR IS ALWAYS ON

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

N1 CARRY OUT THE IPC INDICATOR LAMP CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC** DataLogger .
- Select the **IPC** brake warning indicator (PBA_LMP) active command. Command the brake warning indicator on and off. Observe the brake warning indicator.

Does the brake warning indicator turn off when commanded off?

Yes	GO to N2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

N2 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded **IPC** DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0121:00, GO to Pinpoint Test AV . For DTC U0140:00, GO to Pinpoint Test AY . For all other DTCs, REFER to DTC Charts in this section.
No	GO to N3 .

N3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded **BCM** DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC B1048:7B, VERIFY the brake fluid level is low. If the brake fluid level is low, CORRECT the cause of the low brake fluid condition. If the brake fluid level is not low, INSTALL a new brake fluid level switch. CLEAR the DTCs. REPEAT the self-test. For DTC B1048:01, GO to N8 . For DTC C1D00:11, GO to N5 . For all other DTCs, Section 419-10.
No	GO to N4 .

N4 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE SELF-TEST

- Check for recorded ABS module DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to Section 206-09.
No	GO to N5 .

N5 CHECK THE PARKING BRAKE WARNING INDICATOR SWITCH PID

- Enter the following diagnostic mode on the scan tool: **BCM** DataLogger .
- Select the **BCM** parking brake (PARK_BRK_SW) PID.
- Monitor the parking brake PID while applying and releasing the parking brake.

Does the PID agree with the parking brake position?

Yes	GO to N12 .
No	GO to N6 .

N6 CHECK THE PARKING BRAKE WARNING INDICATOR SWITCH

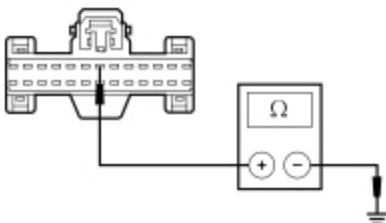
- Disconnect: Parking Brake Warning Indicator Switch [C2015](#).
- Enter the following diagnostic mode on the scan tool: **BCM** DataLogger .
- Select the **BCM** parking brake (PARK_BRK_SW) PID.
- Monitor the parking brake PID.

Does the PID indicate the parking brake is released or off?

Yes	INSTALL a new parking brake warning indicator switch. CLEAR the DTCs. REPEAT the self-test.
No	GO to N7 .

N7 CHECK THE PARKING BRAKE WARNING INDICATOR SWITCH INPUT CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: **BCM** [C2280C](#) .
- Measure the resistance between the **BCM** [C2280C](#) Pin 7, circuit CMC25 (WH/VT), harness side and ground.

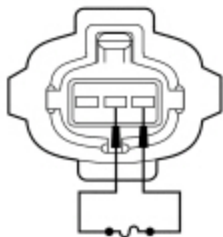


Is the resistance greater than 10,000 ohms?

Yes	GO to N12 .
No	REPAIR circuit CMC25 (WH/VT) for a short to ground. CLEAR the DTCs. REPEAT the self-test.

N8 CHECK THE BRAKE WARNING INDICATOR USING THE FLUID LEVEL SWITCH

- Ignition OFF.
- Disconnect: Brake Fluid Level Switch [C124](#).
- Connect a fused jumper wire between the brake fluid level switch [C124](#) Pin 2, circuit CMC19 (GY/VT), harness side and the brake fluid level switch [C124](#) Pin 3, circuit RMC19 (YE/GY), harness side.



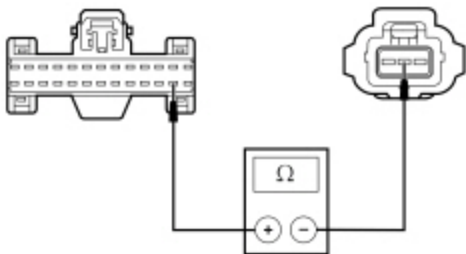
- N0002771
- Ignition ON.

Does the brake warning indicator turn off?

Yes	REMOVE the jumper wire. INSTALL a new brake fluid level switch. CLEAR the DTCs. REPEAT the self-test.
No	REMOVE the jumper wire. GO to N9 .

N9 CHECK THE BRAKE FLUID LEVEL SWITCH INPUT CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: BCM [C2280F](#) .
- Measure the resistance between the BCM [C2280F](#) Pin 15, circuit CMC19 (GY/VT), harness side and the brake fluid level switch [C124](#) Pin 2, circuit CMC19 (GY/VT), harness side.



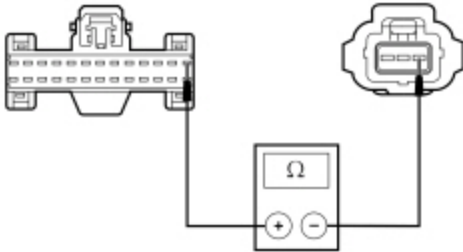
- N0112574

Is the resistance less than 5 ohms?

Yes	GO to N10 .
No	REPAIR circuit CMC19 (GY/VT) for an open. TEST the system for normal operation.

N10 CHECK THE BRAKE FLUID LEVEL SWITCH RETURN CIRCUIT FOR AN OPEN

- Measure the resistance between the BCM [C2280F](#) Pin 1, circuit RMC19 (YE/GY), harness side and the brake fluid level switch [C124](#) Pin 3, circuit RMC19 (YE/GY), harness side.



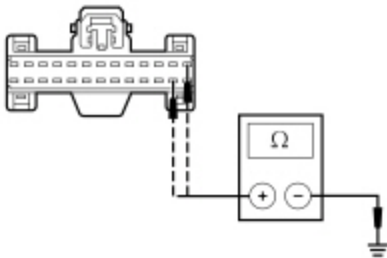
• N0112575

Is the resistance less than 5 ohms?

Yes	GO to N11 .
No	REPAIR circuit RMC19 (YE/GY) for an open. TEST the system for normal operation.

N11 CHECK THE BRAKE FLUID LEVEL SWITCH INPUT AND RETURN CIRCUITS FOR A SHORT TO GROUND

- Measure the resistance between the BCM [C2280F](#) Pin 15, circuit CMC19 (GY/VT), harness side and ground; and between the BCM [C2280F](#) Pin 1, circuit RMC19 (YE/GY), harness side and ground.



• N0112576

Are the resistances greater than 10,000 ohms?

Yes	GO to N12 .
No	REPAIR the circuit in question. TEST the system for normal operation.

N12 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion

- damaged pins
- pushed-out pins

- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test O: The ABS Warning Indicator Is Never/Always On

Normal Operation

The ABS module provides the ABS warning indicator request to the Instrument Panel Cluster (IPC) over the High Speed Controller Area Network (HS-CAN) . If a fault condition exists in the ABS, the ABS module sends the **IPC** a request to illuminate the ABS warning indicator.

If the ABS warning indicator status message is missing for less than 5 seconds, the **IPC** defaults the ABS warning indicator to its last state (on or off) based upon the last known good message received. If the ABS warning indicator request message is missing for more than 5 seconds, the **IPC** sets DTC U0121:00 and defaults the ABS warning indicator on.

This pinpoint test is intended to diagnose the following:

- ABS module
- **IPC**

PINPOINT TEST O : THE ABS WARNING INDICATOR IS NEVER/ALWAYS ON

O1 CARRY OUT THE IPC INDICATOR LAMP CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC** DataLogger .
- Select the **IPC** ABS warning indicator (ABS_LAMP) active command. Command the ABS warning indicator on then off. Observe the ABS warning indicator.

Does the ABS warning indicator turn on when commanded on and turn off when commanded off?

Yes	GO to O2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

O2 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded **IPC** DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0121:00, GO to Pinpoint Test AV . For all other DTCs, REFER to DTC Charts in this section.
No	GO to O3 .

O3 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE SELF-TEST

- Check for recorded ABS module DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to Section 206-09.
No	If the ABS warning indicator is always on, INSTALL a new ABS module. REFER to Section 206-09. TEST the system for normal operation. If the ABS warning indicator is never on after the bulb prove-out, the system is operating correctly at this time.

Pinpoint Test P: The Stability/Traction Control Indicator Is Never/Always On

Normal Operation

The Instrument Panel Cluster (IPC) receives the stability/traction control indicator request from the ABS module over the High Speed Controller Area Network (HS-CAN) . The stability/traction control indicator operates according to the following chart:

Indication Mode	Condition
Flashes rapidly	The vehicle stability/traction control is in active mode or is being controlled by the ABS module
Flashes slowly	A configuration conflict or concern is present in the ABS module
On steady	A fault condition exists in the stability/traction control system

The IPC monitors the stability/traction control message from the ABS module and either flashes the stability/traction control indicator or turns it on steady depending on the condition. The stability/traction control indicator is also used to indicate a trailer sway condition (if equipped).

If the stability/traction control indicator status message is missing for less than 5 seconds, the IPC defaults the stability/traction control indicator to its last state (on or off) based upon the last known good message received. If the stability/traction control indicator status message is missing for more than 5 seconds, the IPC sets DTC U0121:00 in continuous memory and defaults the stability/traction control indicator on.

This pinpoint test is intended to diagnose the following:

- Stability/traction control concern
- ABS module

- IPC

PINPOINT TEST P : THE STABILITY/TRACTION CONTROL INDICATOR IS NEVER/ALWAYS ON

P1 CARRY OUT THE IPC INDICATOR LAMP CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: IPC DataLogger .
- Select the IPC stability/traction control indicator (TC/IVD/RSCLMP) active command. Command the stability/traction control active indicator on and off while observing the stability/traction control indicator.

Does the stability/traction control indicator illuminate when commanded on and turn off when commanded off?

Yes	GO to P2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

P2 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the IPC self-test.

Are any DTCS recorded?

Yes	For DTC U0121:00, GO to Pinpoint Test AV . For all other DTCS, REFER to DTC Charts in this section.
No	GO to P3 .

P3 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE SELF-TEST

- Check for recorded DTCS from the ABS module self-test.

Are any DTCS recorded?

Yes	REFER to Section 206-09.
No	If the stability/traction control indicator is never on after the bulb prove-out, the system is operating correctly at this time. If the stability/traction control indicator is always on, INSTALL a new ABS module. REFER to Section 206-09. TEST the system for normal operation.

Pinpoint Test Q: The Stability/Traction Control Disabled Indicator Is Never/Always On

Normal Operation

The traction control switch is hardwired to the Instrument Panel Cluster (IPC) . When the traction control switch is pressed, the IPC sends a message to the ABS module over the High Speed Controller Area Network (HS-CAN) indicating the

traction control switch has been pressed. The ABS module either disables or enables the stability/traction control system and sends a stability/traction control disabled indicator request back to the **IPC** to illuminate or turn off the stability/traction control disabled indicator based upon the system state.

When a MyKey® programmed key is in use and the AdvanceTrac® on feature is configured always on, the stability/traction control system cannot be disabled and the stability/traction control indicator does not illuminate when the traction control disable button is pressed. The stability/traction control indicator still functions normally to indicate a stability/traction control system fault and a stability/traction control active event.

If the stability/traction control disabled indicator request message is missing for less than 5 seconds, the stability/traction control disabled indicator remains in the last state (on or off) based upon the last known good message received. If the stability/traction control disabled indicator request message is missing for more than 5 seconds, the **IPC** sets DTC U0121:00 in continuous memory and defaults the stability/traction control disabled indicator off.

This pinpoint test is intended to diagnose the following:

- A MyKey® programmed key is in use
- Stability/traction control cannot be disabled
- ABS module
- **IPC**

PINPOINT TEST Q : THE STABILITY/TRACTION CONTROL DISABLED INDICATOR IS NEVER/ALWAYS ON

Q1 CARRY OUT THE IPC INDICATOR LAMP CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC** DataLogger .
- Select the **IPC** stability/traction control disabled indicator (STAB_TC_LMP) active command. Command the stability/traction control disabled indicator on and off while observing the stability/traction control disabled indicator.

Does the stability/traction control disabled indicator illuminate when commanded on and turn off when commanded off?

Yes	GO to Q2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Q2 CHECK FOR THE KEY TYPE

- Check if the key is a MyKey® programmed key or an administrator key. Refer to Section 419-01B.

Is the key an administrator key?

Yes	GO to Q3 .
No	If the MyKey® AdvanceTrac® on feature is configured always on, the system is normal. The stability/traction control system cannot be disabled when the AdvanceTrac® on feature is configured always on. If the MyKey® AdvanceTrac® on feature is not configured always on, GO to Q3 .

Q3 CHECK THE TRACTION CONTROL INPUT TO THE IPC

- Enter the following diagnostic mode on the scan tool: **IPC** DataLogger .
- Select the **IPC** traction control switch PID (STAB_CTRL_SW).
- Monitor the traction control PID while pressing and releasing the traction control disable switch.

Does the traction control PID read RELEASED when the button is released and DEPRESSED when the button is pressed?

Yes	GO to Q4 .
No	REFER to Section 206-09 to diagnose the stability/traction control system cannot be disabled.

Q4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the **IPC** self-test.

Are any DTCS recorded?

Yes	For DTC U0121:00, GO to Pinpoint Test AV . For all other DTCS, REFER to DTC Charts in this section.
No	GO to Q5 .

Q5 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE SELF-TEST

- Check for recorded DTCS from the ABS module self-test.

Are any DTCS recorded?

Yes	REFER to Section 206-09.
No	INSTALL a new ABS module. REFER to Section 206-09. TEST the system for normal operation.

Pinpoint Test R: The LH/RH Turn Or High Beam Indicator Is Never/Always On

Normal Operation

When the multifunction switch is in the LH or RH TURN position, a turn signal on/off request is sent to the Instrument Panel Cluster (IPC) from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . Upon receipt of the applicable turn signal on/off request, the **IPC** flashes the turn signal indicator on and off.

When the high beams are turned on, the **BCM** sends a high beam indicator on request to the **IPC** through the **HS-CAN** to illuminate the high beam indicator.

If the high beam indicator request status message is missing for less than 5 seconds, the **IPC** defaults the high beam indicator to its last state (on or off) based upon the last known good message received. If the high beam indicator request status message is missing for more than 5 seconds, the **IPC** sets DTC U0140:00 in continuous memory and defaults the high beam indicator off.

This pinpoint test is intended to diagnose the following:

- Turn signal concern
- High beam concern
- BCM
- IPC

PINPOINT TEST R : THE LH/RH TURN OR HIGH BEAM INDICATOR IS NEVER/ALWAYS ON

R1 DETERMINE THE FAULT CONDITION

- Turn the high beams on then off again.
- Ignition ON.
- Observe the high beam indicator.

Does the high beam indicator illuminate when the high beams are on and turn off when the high beams are off?

Yes	GO to R4 .
No	LEAVE the headlamp switch in the HIGH BEAM position. GO to R2 .

R2 CHECK THE HIGH BEAM HEADLAMPS OPERATION

- Observe the high beam headlamps.

Do the high beam headlamps operate correctly?

Yes	GO to R3 .
No	REFER to Section 417-01.

R3 CARRY OUT THE IPC INDICATOR LAMP CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: IPC DataLogger .
- Select the IPC high beam indicator (HIGH_BEAM).
- Command the high beam indicator on and off. Observe the high beam indicator.

Does the high beam indicator illuminate when commanded on and turn off when commanded off?

Yes	GO to R6 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

R4 CHECK THE LH OR RH TURN SIGNAL LAMPS OPERATION

- Ignition ON.

- Operate the LH or RH turn signal. Observe the exterior LH or RH turn signal lamps.

Do the exterior turn signal lamps operate correctly?

Yes	GO to R5 .
No	REFER to Section 417-01.

R5 CARRY OUT THE IPC INDICATOR LAMP CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Enter the following diagnostic mode on the scan tool: **IPC DataLogger** .
- Select the **IPC** RH or LH turn signal indicator (RH_TURN_L) or (LH_TURN_L) active command. Command the RH or LH turn signal indicator on and off. Observe the LH or RH turn signal indicator.

Does the LH or RH turn signal indicator illuminate when commanded on and turn off when commanded off?

Yes	GO to R6 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

R6 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded **IPC** DTCs from the self-test.

Is DTC U0140:00 recorded?

Yes	GO to Pinpoint Test AY .
No	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

Pinpoint Test S: The Tire Pressure Monitoring System (TPMS) Warning Indicator Is Never/Always On

NOTE: *The Tire Pressure Monitor (TPM) is also known as the Radio Transmitter Module (RTM).*

Normal Operation

The Instrument Panel Cluster (IPC) receives the Tire Pressure Monitoring System (TPMS) warning indicator request from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . If the **BCM** determines the tire pressure has exceeded the low tire pressure limits, a request is sent to the **IPC** to illuminate the **TPMS** warning indicator. If a **TPMS** fault condition exists, the **BCM** sends a request to the **IPC** to flash the **TPMS** warning indicator.

If the **TPMS** status message is missing for more than 5 seconds, the **IPC** sets DTC U0140:00 and flashes the **TPMS** indicator for approximately 75 seconds then turns the **TPMS** indicator on.

NOTE: *If DTC U0140:00 sets in the **BCM** , other symptoms may also be present such as high beam indicator, RH/LH turn indicators, instrument panel backlighting, and door ajar indication inoperative.*

This pinpoint test is intended to diagnose the following:

- Tire pressure
- TPMS concern
- BCM
- IPC

PINPOINT TEST S : THE TPMS WARNING INDICATOR IS NEVER/ALWAYS ON

S1 CHECK THE TIRE PRESSURE

- Verify the tire pressure in all tires meets the recommended tire pressures on the Vehicle Certification (VC) label. Refer to Section 100-01.

Do all the tires meet the recommended tire pressures?

Yes	GO to S2 .
No	CORRECT the tire pressures. TEST the system for normal operation.

S2 RETRIEVE THE RECORDED DTCS FROM IPC SELF-TEST

- Check for recorded ~~IPC~~ DTC from the self-test.

Are any DTCs recorded?

Yes	For DTC U0140:00, GO to Pinpoint Test AY . For all other DTCs, REFER to DTC Charts in this section.
No	GO to S3 .

S3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded ~~BCM~~ DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to the Body Control Module (BCM) DTC Chart in Section 419-10.
No	GO to S4 .

S4 RETRIEVE THE RECORDED DTCS FROM THE TPM MODULE SELF-TEST

NOTE: *The Tire Pressure Monitor (TPM) is also known as the Radio Transmitter Module (RTM).*

- Check for recorded Tire Pressure Monitor (TPM) module DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to Section 204-04.
No	GO to S5 .

S5 CARRY OUT THE IPC INDICATOR LAMP CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC DataLogger** .
- Select the **IPC TPMS** warning indicator (TPMWARN_LMP) active command. Command the **TPMS** warning indicator on and off. Observe the **TPMS** warning indicator.

Does the TPMS warning indicator illuminate when commanded on and turn off when commanded off?

Yes	If the TPMS warning indicator is never on after the bulb prove-out, the system is operating correctly at this time. If the TPMS warning indicator is always on, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test T: The Air Bag Warning Indicator Is Never/Always On

Normal Operation

The Instrument Panel Cluster (IPC) receives the Supplemental Restraint System (SRS) warning indicator request from the Restraints Control Module (RCM) over the High Speed Controller Area Network (HS-CAN) . If an **SRS** concern is detected, the **RCM** sets a DTC and sends a request to the **IPC** to illuminate the air bag warning indicator.

If the **SRS** status message is missing for less than 5 seconds, the **IPC** defaults the air bag warning indicator to its last state (on or off) based upon the last known good message received. If the **SRS** status message is missing or is deemed invalid for more than 5 seconds, the **IPC** sets DTC U0151:00 in continuous memory and defaults the air bag warning indicator on.

This pinpoint test is intended to diagnose the following:

- **SRS** concern
- **RCM**
- **IPC**

PINPOINT TEST T : THE AIR BAG WARNING INDICATOR IS NEVER/ALWAYS ON

T1 CARRY OUT THE IPC RESTRAINTS OUTPUT CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC DataLogger** .
- Select the **IPC** air bag warning indicator (AIRBAG_LMP) active command. Command the air bag warning indicator on and off. Observe the air bag warning indicator.

Does the air bag warning indicator illuminate when commanded on and turn off when commanded off?

Yes	GO to T2 .
------------	----------------------------

No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
-----------	---

T2 RETRIEVE THE RECORDED DTCS FROM THE RCM SELF-TEST

- Check for recorded **RCM** DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to Section 501-20B.
No	GO to T3 .

T3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded **IPC** DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0151:00, GO to Pinpoint Test AZ . For all other DTCs, REFER to DTC Charts in this section.
No	If the air bag warning indicator is never on after the bulb prove-out, the system is operating correctly at this time. If the air bag warning indicator is always on, INSTALL a new RCM . REFER to Section 501-20B. TEST the system for normal operation.

Pinpoint Test U: The Safety Belt Warning Indicator Is Never/Always On

Normal Operation

The Restraints Control Module (RCM) monitors the safety belt position through the safety belt buckle switch. The **RCM** provides the safety belt buckle status to the Instrument Panel Cluster (IPC) over the High Speed Controller Area Network (HS-CAN) .

If the safety belt buckle status message is missing or invalid for less than 5 seconds, the **IPC** defaults the safety belt warning indicator to its last state (on or off) based upon the last known good message received. If the **IPC** does not receive the safety belt buckle switch status message from the **RCM** or the message is deemed invalid by the **IPC** for greater than 5 seconds, the **IPC** sets DTC U0151:00 and defaults the safety belt warning indicator off.

This pinpoint test is intended to diagnose the following:

- Safety belt buckle concern
- RCM**
- IPC**

PINPOINT TEST U : THE SAFETY BELT WARNING INDICATOR IS NEVER/ALWAYS ON

U1 RETRIEVE THE RECORDED RCM DTCS FROM THE SELF-TEST

- Check for recorded **RCM** DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to Section 501-20B.
No	GO to U2 .

U2 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded ~~IPC~~ DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0151:00, GO to Pinpoint Test AZ . For all other DTCs, REFER to DTC Charts in this section.
No	GO to U3 .

U3 CARRY OUT THE IPC INDICATOR LAMP CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: ~~IPC~~ DataLogger .
- Select the ~~IPC~~ all warning indicators (SBLT_LAMP) active command. Command all the warning indicators on and off. Observe the safety belt warning indicator.

Does the safety belt warning indicator illuminate when commanded on and turn off when commanded off?

Yes	If the safety belt warning indicator is never on after the bulb prove-out, the system is operating correctly at this time. If the safety belt warning indicator is always on, INSTALL a new RCM . REFER to Section 501-20B. TEST the system for normal operation.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test V: The Malfunction Indicator Lamp (MIL) Is Never/Always On

Normal Operation

The Malfunction Indicator Lamp (MIL) is controlled by the Instrument Panel Cluster (IPC) using data sent from the PCM over the High Speed Controller Area Network (HS-CAN) .

If the ~~MIL~~ status message is missing or invalid for less than 5 seconds, the ~~IPC~~ defaults the ~~MIL~~ to its last state (on or off) based upon the last known good message received. If the ~~MIL~~ status message is missing for more than 5 seconds, the ~~IPC~~ sets DTC U0100:00 in continuous memory and defaults the ~~MIL~~ on.

NOTE: *If DTC U0100:00 sets in the ~~IPC~~ , other observable symptoms may be an inoperative speedometer, tachometer or odometer.*

This pinpoint test is intended to diagnose the following:

- PCM
- IPC

PINPOINT TEST V : THE MIL IS NEVER/ALWAYS ON

V1 CARRY OUT THE IPC MIL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC** DataLogger .
- Select the **IPC MIL** active command (MIL). Command the **MIL** on and off. Observe the **MIL** .

Does the **MIL illuminate when selected on and turn off when selected off?**

Yes	GO to V2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

V2 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check the recorded PCM DTCs from the **KOEO** self-test.

Are any PCM DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to V3 .

V3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded **IPC** DTCs from the self-test.

Is DTC U0100:00 recorded?

Yes	GO to Pinpoint Test AT .
No	If the MIL is never on with the KOEO , the system is operating correctly at this time. If the MIL is always on, INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.

Pinpoint Test W: The Cruise Control Indicator Is Never/Always On

Normal Operation

The Instrument Panel Cluster (IPC) receives the cruise control indicator on request from the PCM over the High Speed Controller Area Network (HS-CAN) . When the cruise control switch is placed in the ON position, the PCM sends the **IPC** a standby mode request and the **IPC** illuminates the cruise control indicator to amber. When the cruise control is engaged, the PCM sends a cruise control on request message to the **IPC** and the **IPC** changes the cruise control indicator illumination from amber to green.

If the cruise control message is missing or if the data is deemed invalid by the **IPC** for more than 5 seconds, the **IPC** sets DTC U0100:00 in continuous memory and defaults the cruise control off.

NOTE: *If DTC U0100:00 sets in the **IPC** , other observable symptoms may be an inoperative speedometer, tachometer or odometer.*

This pinpoint test is intended to diagnose the following:

- PCM
- **IPC**

PINPOINT TEST W : THE CRUISE CONTROL INDICATOR IS NEVER/ALWAYS ON

W1 CHECK THE CRUISE CONTROL OPERATION

- Operate the cruise control.

Does the cruise control operate correctly?

Yes	GO to W2 .
No	REFER to Section 419-03.

W2 CARRY OUT THE IPC INDICATOR LAMP CONTROL ACTIVE COMMAND USING THE SCAN TOOL

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC** DataLogger .
- Select the **IPC** cruise control indicator (CRUISE) active command. Command the cruise control indicator on and off. Observe the cruise control indicator.

Does the cruise control indicator illuminate when commanded on and turn off when commanded off?

Yes	GO to W3 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

W3 RETRIEVE AND RECORD THE DTCS FROM THE PCM KOEO SELF-TEST

- Check for recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Are any PCM DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to W4 .

W4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded **IPC** DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCs, REFER to DTC Charts in this section.
No	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.

Pinpoint Test X: The PRNDL Indicator is Never On/Always Displays P

Normal Operation

The Instrument Panel Cluster (IPC) receives the transmission range status and the select shift status from the PCM over the High Speed Controller Area Network (HS-CAN) . The IPC also uses a park detect switch (part of the brake shift interlock actuator) to signal the IPC the shift lever is fully seated in the PARK (P) position. The IPC compares the park detect switch input with the transmission range status sent from the PCM and looks for vehicle speed above 16 km/h (10 mph). The IPC provides a reference voltage to the brake shift interlock. When the selector lever is in PARK (P), the brake shift indicator routes the reference voltage to ground pulling the circuit low to the IPC . When the selector lever is moved out of PARK (P), the brake shift interlock opens the ground, sending the reference voltage high.

When in select shift mode, the select shift indication is illuminated along with the PRNDM21 display. For example, when M is selected and select shift is selected, the PRNM21 illuminates the M and the select shift display illuminates 1-6 (base IPC) or M1-M6 (optional IPC) simultaneously.

If the transmission range information is missing from the PCM, the IPC sets DTC U0100:00.

NOTE: If DTC U0100:00 sets in the IPC , other observable symptoms may be an inoperative speedometer, tachometer, temperature gauge or odometer. If DTC U0101:00 sets in the IPC , other observable symptoms may be the powertrain malfunction (wrench) warning indicator always on.

- DTC P1706:00 (High Vehicle Speed Observed In Park: No Sub Type Information) — a continuous memory DTC that sets if the IPC receives a short to ground on the park detect switch input circuit, pulling the circuit low with a Vehicle Speed Sensor (VSS) input over 16 km/h (10 mph) for greater than 5 seconds (indicating the vehicle is in park and in motion).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Brake shift interlock actuator
- PCM concern
- IPC

PINPOINT TEST X : THE PRNDL INDICATOR IS NEVER ON/ALWAYS DISPLAYS P

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

X1 CHECK THE PRNDL DISPLAY PARK OPERATION

- Ignition ON.
- Shift the selector lever through the different gear positions while observing the PRNDL display.
- Place the selector lever in the PARK (P) position.
- Ignition OFF.
- Observe the PRNDL display immediately after turning the key off.

Does the PRNDL display the different ranges according to the selector lever position then display P for approximately 5 seconds when the key is turned off?

Yes	The system is functioning correctly at this time.
No	GO to X2 .

X2 RETRIEVE AND RECORD THE IPC DTCS FROM THE SELF-TEST

- Check for recorded IPC DTCS from the self-test.

Are any DTCS recorded?

Yes	For DTC P1706:00, GO to X3 . For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCS, REFER to DTC Charts in this section.
No	GO to X4 .

X3 CONFIRM DTC P1706:00 IS STILL PRESENT

- CLEAR the DTCS. DRIVE the vehicle above 16 km/h (10 mph) for more than 5 seconds. REPEAT the self-test.

Is DTC P1706:00 present?

Yes	GO to X5 .
No	The PRNDL is operating correctly at this time. The fault may be caused by an intermittent concern with the park detect input to the IPC .

X4 RETRIEVE AND RECORD THE DTCS FROM THE PCM KOEO SELF-TEST

- Check for recorded PCM DTCS from the Key ON Engine OFF (KOEO) self-test.

Are any PCM DTCS recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

X5 CHECK THE BRAKE SHIFT INTERLOCK OPERATION

- Apply the park brake and shift the transmission into NEUTRAL (N).
- Ignition OFF.
- Disconnect: Brake Shift Interlock [C2008](#) (Column Shift) .
- Disconnect: Brake Shift Interlock [C3245](#) (Floor Shift) .

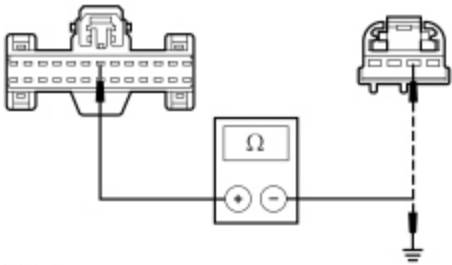
- Clear the DTCs.
- Disengage the park brake and drive the vehicle above 16 km/h (10 mph) for more than 5 seconds.
- Retrieve the **IPC** continuous memory DTCs.

Is DTC P1706:00 still present?

Yes	GO to X6 .
No	INSTALL a new brake shift interlock actuator. REFER to Section 307-05. TEST the system for normal operation.

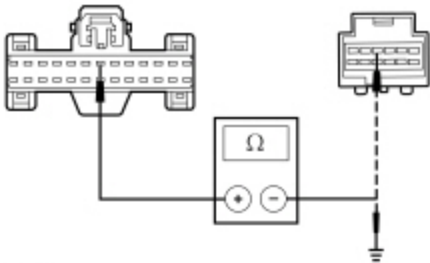
X6 CHECK THE PARK DETECT INPUT CIRCUIT FOR AN OPEN OR SHORT TO GROUND

- Ignition OFF.
- Disconnect: **IPC** [C220](#) .
- For column shift, measure the resistance between the **IPC** [C220](#) Pin 7, circuit CET52 (GN), harness side and the brake shift interlock [C2008](#) Pin 2, circuit CET52 (GN), harness side; and between the **IPC** [C220](#) Pin 7, circuit CET52 (GN), harness side and ground.



N0120195

- For floor shift, measure the resistance between the **IPC** [C220](#) Pin 7, circuit CET52 (GN), harness side and the selector lever [C3245](#) Pin 3, circuit CET52 (GN), harness side; and between the **IPC** [C220](#) Pin 7, circuit CET52 (GN), harness side and ground.



N0121734

Is the resistance less than 5 ohms between the **IPC and the brake shift interlock or selector lever, and greater than 10,000 ohms between the **IPC** and ground?**

Yes	GO to X7 .
No	REPAIR circuit CET52 (GN) for an open or short to ground. REPEAT the self-test. CLEAR the DTCs.

X7 CHECK FOR CORRECT IPC OPERATION

- Disconnect the **IPC** connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the **IPC** connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test Y: The Powertrain Malfunction (Wrench) Message Center Indicator Is Always On

Normal Operation

The Instrument Panel Cluster (IPC) receives requests from the PCM and Transfer Case Control Module (TCCM) to control the powertrain malfunction (wrench) message center indicator. The PCM provides an indicator request for Electronic Throttle Control (ETC) faults and transmission faults over the High Speed Controller Area Network (HS-CAN) . The TCCM provides an indicator request for Four-Wheel Drive (4WD) faults over the **HS-CAN** .

If the indicator request message for **ETC** or transmission faults is missing or if the data is deemed invalid for less than 5 seconds, the powertrain malfunction (wrench) message center indicator remains in the last state (on or off) based upon the last known good message received. If the indicator request message for **ETC** or transmission faults is missing or if the data is deemed invalid by the **IPC** for greater than 5 seconds, the **IPC** sets DTC U0100:00 and defaults the powertrain malfunction (wrench) message center warning indicator active (on).

If the indicator request message for the **4WD** faults is missing or deemed invalid for less than 5 seconds, the powertrain malfunction (wrench) message center indicator remains in the last state (on or off) based upon the last known good message received. If the indicator request message is missing or if the data is deemed invalid by the **IPC** for greater than 5 seconds, the **IPC** sets DTC U0102:00 and defaults the powertrain malfunction (wrench) message center warning indicator active (on).

NOTE: *If DTC U0100:00 sets in the **IPC** , other symptoms may also be present such as an inoperative speedometer, tachometer or temperature gauge. If DTC U0102:00 sets in the **IPC** , other symptoms may also be present such as an inoperative 4X4 high or 4X4 low indicator when equipped with Electronic Shift-On-The-Fly (ESOF) .*

This pinpoint test is intended to diagnose the following:

- Message center concern
- **ETC** concern
- **4WD** concern (if equipped)
- Transmission concern

PINPOINT TEST Y : THE POWERTRAIN MALFUNCTION (WRENCH) MESSAGE CENTER INDICATOR IS ALWAYS ON

Y1 VERIFY THE OPERATION OF THE DOOR AJAR MESSAGE CENTER WARNING INDICATOR

- Close all doors then open the LH front door.
 - Ignition ON.
 - **NOTE:** *If multiple warnings are present, the message center cycles through each of the warning indicators. Allow the displays to cycle through each display while determining whether the door ajar message center warning indicator is active.*
- Observe the message center.

Is the door ajar message center warning indicator active?

Yes	GO to Y2 .
No	GO to Pinpoint Test AF .

Y2 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded **IPC** DTCs from the self-test.

Is DTC U0100:00 or DTC U0102:00 recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT . For DTC U0102:00, GO to Pinpoint Test AU .
No	GO to Y3 .

Y3 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check for recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Are any DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	REFER to Section 308-07A to diagnose TCCM DTCs.

Pinpoint Test Z: The Charging System Message Center Warning Indicator Is Never/Always On

Normal Operation

The charging system indicator is controlled by the Instrument Panel Cluster (IPC) based upon data received from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The **BCM** receives the charging system status from the PCM over the **HS-CAN** . When a fault is detected in the charging system, the **BCM** sends the **IPC** a request to illuminate the charging system warning indicator.

If the charging system warning indicator request message is missing for less than 5 seconds, the charging system indicator remains in the last state (on or off). If the charging system message is missing for more than 5 seconds, the **IPC** sets DTC U0140:00 in continuous memory and defaults the charging system indicator on.

NOTE: If DTC U0140:00 sets in the IPC, other observable symptoms may be an inoperative key-in-ignition chime, door ajar message center indicator or RH/LH turn indicators.

This pinpoint test is intended to diagnose the following:

- Message center concern
- Charging system concern
- PCM

PINPOINT TEST Z : THE CHARGING SYSTEM MESSAGE CENTER WARNING INDICATOR IS NEVER/ALWAYS ON

Z1 VERIFY THE OPERATION OF THE DOOR AJAR MESSAGE CENTER WARNING INDICATOR

- Close all doors then open the LH front door.
 - Ignition ON.
 - **NOTE:** If multiple warnings are present, the message center cycles through each of the warning indicators. Allow the displays to cycle through each display while determining whether the door ajar message center warning indicator is active.
- Observe the message center.

Is the door ajar message center warning indicator active?

Yes	GO to Z2 .
No	GO to Pinpoint Test AF .

Z2 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check the recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Are any charging system DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to Z3 .

Z3 CHECK THE CHARGING SYSTEM OPERATION

- Check the charging system operation. Refer to Section 414-00.

Does the charging system operate correctly?

Yes	GO to Z4 .
No	REFER to Section 414-00.

Z4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Is DTC U0140:00 recorded?

Yes	GO to Pinpoint Test AY.
No	GO to Z5 .

Z5 MONITOR THE PCM CHARGING SYSTEM INDICATOR LAMP INPUT

- Using a diagnostic scan tool, monitor the PCM charging system fault indicator lamp PID (GENFIL) with the KOEO and with the KOER .

Does the PCM PID indicate ON with the engine not running and OFF with the engine running?

Yes	GO to Z6 .
No	If the charging system warning indicator is never on with the KOER , the system is operating correctly at this time. If the charging system warning indicator is always on, DIAGNOSE all BCM Diagnostic Trouble Codes (DTCs). Section 419-10.

Z6 CHECK FOR CORRECT PCM OPERATION

- Disconnect the PCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the PCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PCM . REFER to Section 303-14. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test AA: The Low Oil Pressure Message Center Warning Indicator Is Never/Always On

Refer to Wiring Diagrams Cell [60](#) , Instrument Cluster for schematic and connector information.

Normal Operation

The engine oil pressure switch is hardwired to the PCM. The Instrument Panel Cluster (IPC) receives the engine oil pressure status from the PCM over the High Speed Controller Area Network (HS-CAN) . The IPC uses both the messaged oil pressure status and engine rpm to control the low oil pressure message center warning indicator. The PCM provides a reference voltage to the engine oil pressure switch when the ignition key is in RUN. With the engine running and low or no

oil pressure, the engine oil pressure switch remains open. The PCM detects no change in the reference voltage and provides the IPC a request to illuminate the low oil pressure warning indicator. With the engine running and sufficient oil pressure, the engine oil pressure switch closes, pulling the reference voltage low. The PCM detects the low reference voltage and provides the IPC a request to turn off the low oil pressure warning indicator.

If the IPC fails to receive the oil pressure status from the PCM for 5 seconds or less, the IPC defaults the low oil pressure warning indicator to the last state (on or off), based upon the last known good oil pressure status message. If the IPC fails to receive the oil pressure status from the PCM for more than 5 seconds, the IPC sets DTC U0100:00 in continuous memory and defaults the low oil pressure warning indicator active (on).

NOTE: *If DTC U0100:00 sets in the IPC, other observable symptoms may be an inoperative speedometer, tachometer, temperature gauge or odometer.*

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Message center concern
- Communication concern
- Tachometer concern
- Engine oil pressure switch
- Base engine oil pressure concern
- PCM

PINPOINT TEST AA : THE LOW OIL PRESSURE MESSAGE CENTER WARNING INDICATOR IS NEVER/ALWAYS ON

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

AA1 VERIFY THE OPERATION OF THE TACHOMETER

- Ignition ON.
- With the engine running, verify the tachometer is operating correctly.

Does the tachometer operate correctly?

Yes	GO to AA2 .
No	For an inoperative tachometer, GO to Pinpoint Test I . For an inaccurate tachometer, GO to Pinpoint Test J .

AA2 VERIFY THE OPERATION OF THE DOOR AJAR MESSAGE CENTER WARNING INDICATOR

- Close all doors then open the LH front door.
 - **NOTE:** *If multiple warnings are present, the message center cycles through each of the warnings. Allow the displays to cycle through each display while determining whether the door ajar message center warning indicator is active.*
- Observe the message center.

Is the door ajar message center warning indicator active?

Yes	If the low oil pressure message center warning indicator is never on, GO to AA3 .
-----	---

If the low oil pressure message center warning indicator is always on, GO to [AA5](#).

No [GO to Pinpoint Test AF](#).

AA3 CHECK FOR A STUCK CLOSED ENGINE OIL PRESSURE SWITCH

- Ignition OFF.
- Disconnect: Engine Oil Pressure Switch [C103](#).
- Start the engine.
- Observe the low oil pressure warning indicator.

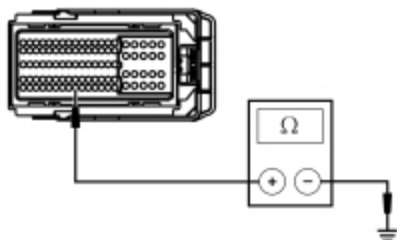
Does the low oil pressure warning indicator illuminate?

Yes INSTALL a new engine oil pressure switch. Refer to the appropriate section in Group 303 for the procedure. TEST the system for normal operation.

No GO to [AA4](#).

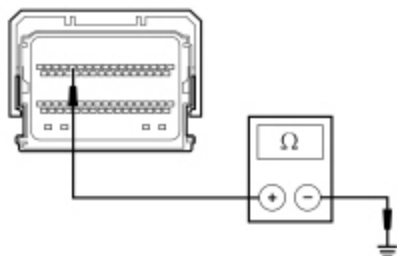
AA4 CHECK THE OIL PRESSURE INPUT CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: PCM [C175E](#) (3.7L), [C1381E](#) (5.0L and 6.2L), or [C1551E](#) (3.5L) .
- For 3.5L, measure the resistance between the PCM [C1551E](#) Pin 9, circuit CMC24 (GY), harness side and ground.



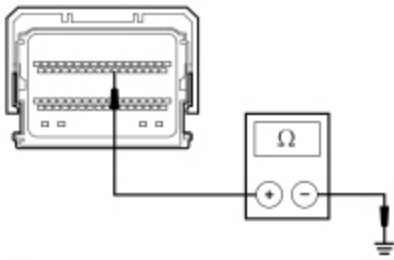
N0122305

- For 3.7L, measure the resistance between the PCM [C175E](#) Pin 13, circuit CMC24 (GY), harness side and ground.



N0120194

- For 5.0L and 6.2L, measure the resistance between the PCM [C1381E](#) Pin 24, circuit CMC24 (GY), harness side and ground.



• N0112570

Is the resistance greater than 10,000 ohms?

Yes	GO to AA9 .
No	REPAIR circuit CMC24 (GY) for a short to ground. TEST the system for normal operation.

AA5 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

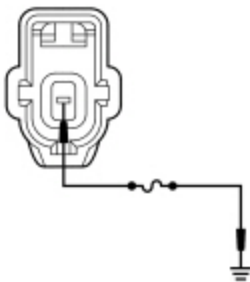
- Check for recorded ~~IPC~~ DTCs from the self-test.

Is DTC U0100:00 recorded?

Yes	GO to Pinpoint Test AT .
No	GO to AA6 .

AA6 CHECK FOR A STUCK OPEN ENGINE OIL PRESSURE SWITCH

- Ignition OFF.
- Disconnect: Engine Oil Pressure Switch [C103](#).
- Connect a fused jumper wire between the engine oil pressure switch [C103](#) Pin 1, circuit CMC24 (GY), harness side and ground.



• N0002760

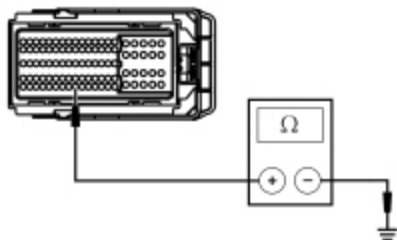
- Start the engine.
- Observe the low oil pressure warning indicator.

Does the low oil pressure indicator turn off?

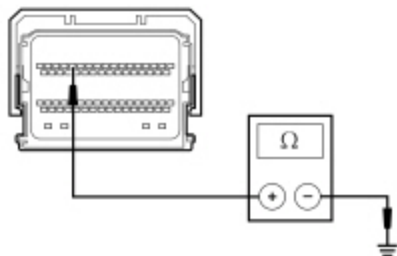
Yes	REMOVE the jumper wire. GO to AA8 .
No	LEAVE the jumper wire connected. GO to AA7 .

AA7 CHECK THE ENGINE OIL PRESSURE INPUT CIRCUIT FOR AN OPEN

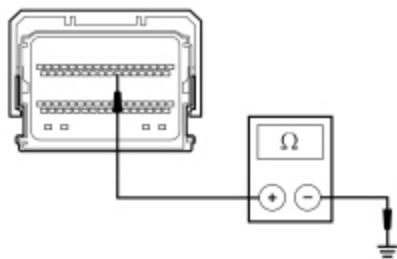
- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Disconnect: PCM [C175E](#) (3.7L), [C1381E](#) (5.0L and 6.2L), or [C1551E](#) (3.5L) .
- For 3.5L, measure the resistance between the PCM [C1551E](#) Pin 9, circuit CMC24 (GY), harness side and ground.



- N0122305
- For 3.7L, measure the resistance between the PCM [C175E](#) Pin 13, circuit CMC24 (GY), harness side and ground.



- N0120194
- For 5.0L and 6.2L, measure the resistance between the PCM [C1381E](#) Pin 24, circuit CMC24 (GY), harness side and ground.



- N0112570

Is the resistance less than 5 ohms?

Yes	REMOVE the jumper wire. GO to AA9 .
No	REMOVE the jumper wire. REPAIR circuit CMC24 (GY) for an open. TEST the system for normal operation.

AA8 CHECK THE OIL PRESSURE

- Carry out the engine oil pressure test. Refer to Section 303-00.

Is the oil pressure within specification?

Yes	INSTALL a new engine oil pressure switch. Refer to the appropriate section in Group 303 for the procedure. TEST the system for normal operation.
No	REFER to Section 303-00 to continue diagnosis of the engine oil pressure.

AA9 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Connect: Engine Oil Pressure Switch [C103](#).
- Connect: PCM [C175E](#) (3.7L), [C1381E](#) (5.0L and 6.2L) or [C1551E](#) (3.5L) .
- Check the recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Are any DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to AA10 .

AA10 CHECK FOR CORRECT PCM OPERATION

- Disconnect the PCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the PCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test AB: The 4X4 Auto, 4X4 Low, 4X4 High Or 4X2 Message Center Indicator Is Never/Always On (If Equipped)

Refer to Wiring Diagrams Cell [60](#) , Instrument Cluster for schematic and connector information.

Normal Operation

The Instrument Panel Cluster (IPC) receives the Four-Wheel Drive (4WD) status from the Transfer Case Control Module (TCCM) over the High Speed Controller Area Network (HS-CAN) . If the vehicle is shifted into 4X4 auto, 4WD low or high, the TCCM sends the IPC a message to illuminate the 4X4 auto, 4X4 low or 4X4 high message center indicator. If the vehicle is shifted out of either 4X4 auto, 4X4 low or 4X4 high into 4X2 mode, the TCCM sends the IPC a message to illuminate the 4X2 message center indicator, informing the driver the vehicle has shifted into 4X2 mode.

If the 4X4 status message is missing for less than 5 seconds, the IPC defaults the applicable 4X4 message center indicator to the last on/off state, based upon the last known good ABS status message received.

If the 4X4 status message is missing for greater than 5 seconds, the IPC sets DTC U0102:00 and defaults the applicable 4X4 message center indicator off.

This pinpoint test is intended to diagnose the following:

- Message center concern
- 4WD concern
- TCCM

PINPOINT TEST AB : THE 4X4 AUTO, 4X4 LOW, 4X4 HIGH OR 4X2 MESSAGE CENTER INDICATOR IS NEVER/ALWAYS ON (IF EQUIPPED)

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

AB1 VERIFY THE OPERATION OF THE DOOR AJAR MESSAGE CENTER WARNING INDICATOR

- Close all doors then open the LH front door.
 - Ignition ON.
 - **NOTE:** *If multiple warnings are present, the message center cycles through each of the warning indicators. Allow the displays to cycle through each display while determining whether the door ajar message center warning indicator is active.*
- Observe the message center.

Is the door ajar message center warning indicator active?

Yes	GO to AB2 .
No	GO to Pinpoint Test AF .

AB2 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Is DTC U0102:00 recorded?

Yes	GO to Pinpoint Test AU .
No	GO to AB3 .

AB3 RETRIEVE THE RECORDED DTCS FROM THE TCCM SELF-TEST

- Check for recorded DTCs from the TCCM self-test.

Are any DTCs recorded?

Yes	REFER to Section 308-07A.
No	GO to AB4 .

AB4 CHECK THE 4WD SYSTEM OPERATION

- Test drive the vehicle and operate the 4WD system.

Does the 4WD system operate correctly?

Yes	INSTALL a new TCCM . REFER to Section 308-07A. TEST the system for normal operation.
No	REFER to Section 308-07A.

Pinpoint Test AC: The Tow Haul Message Center Indicator Is Never/Always On

Normal Operation

The tow haul indicator illuminates when the tow haul system has been engaged and is actively providing tow haul control for the vehicle. The Instrument Panel Cluster (IPC) receives the O/D off and tow haul indicator request from the PCM over the High Speed Controller Area Network (HS-CAN) .

If the tow haul indicator request message is missing or invalid for less than 5 seconds, the tow haul indicator remains in the last state (on or off), based upon the last known good message received. If the or tow haul indicator request message is missing for more than 5 seconds, the IPC sets DTC U0101:00 in continuous memory and defaults the tow haul indicator off.

This pinpoint test is intended to diagnose the following:

- Message center concern
- PCM

PINPOINT TEST AC : THE TOW HAUL MESSAGE CENTER INDICATOR IS NEVER/ALWAYS ON

AC1 VERIFY THE OPERATION OF THE DOOR AJAR MESSAGE CENTER WARNING INDICATOR

- Close all doors then open the LH front door.
- **NOTE:** *If multiple warnings are present, the message center cycles through each of the warnings. Allow the displays to cycle through each display while determining whether the door ajar message center warning indicator is active.*

Observe the message center.

Is the door ajar message center warning indicator active?

Yes	GO to AC2 .
No	GO to Pinpoint Test AF .

AC2 RETRIEVE AND RECORD THE IPC DTCS FROM THE SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCs, REFER to DTC Charts in this section.
No	GO to AC3 .

AC3 RETRIEVE AND RECORD THE DTCS FROM THE PCM KOEO SELF-TEST

- Check for recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Are any DTCs recorded?

Yes	REFER to Section 303-14. TEST the system for normal operation.
No	GO to AC4 .

AC4 VERIFY THE OPERATION OF THE TOW HAUL SYSTEM

- Verify the tow haul system turns off when turned off and turns on when turned on.

Does the tow haul system function correctly?

Yes	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.
No	REFER to Section 307-05.

Pinpoint Test AD: The Electronic Locking Differential (ELD) Message Center Indicator Is Never/Always On

Normal Operation

The Instrument Panel Cluster (IPC) receives the Electronic Locking Differential (ELD) status from the Transfer Case Control Module (TCCM) over the High Speed Controller Area Network (HS-CAN) . If the vehicle is shifted into ELD mode, the TCCM sends the IPC a message to illuminate the ELD message center indicator. If the vehicle is shifted out of ELD mode, the TCCM sends the IPC a message to turn off the ELD message center indicator, informing the driver the vehicle has shifted out of ELD mode.

If the ELD status message is missing or invalid for less than 5 seconds, the ELD warning indicator remains in the last state (on or off), based upon the last known good ELD status message received. If the ELD status message is missing for more than 5 seconds, the IPC sets DTC U0102:00 in continuous memory and defaults the ELD warning indicator off.

This pinpoint test is intended to diagnose the following:

- Message center concern
- ELD concern
- TCCM

PINPOINT TEST AD : THE ELD MESSAGE CENTER INDICATOR IS NEVER/ALWAYS ON

AD1 VERIFY THE OPERATION OF THE DOOR AJAR MESSAGE CENTER WARNING INDICATOR

- Close all doors then open the LH front door.
- **NOTE:** *If multiple warnings are present, the message center cycles through each of the warnings. Allow the displays to cycle through each display while determining whether the door ajar message center warning indicator is active.*

Observe the message center.

Is the door ajar message center warning indicator active?

Yes	GO to AD2 .
No	GO to Pinpoint Test AF .

AD2 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCs from the ~~IPC~~ self-test.

Is DTC U0102:00 recorded?

Yes	GO to Pinpoint Test AU .
No	GO to AD3 .

AD3 RETRIEVE THE RECORDED DTCS FROM THE TCCM SELF-TEST

- Check for recorded ~~TCCM~~ DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to Section 308-07A.
No	GO to AD4 .

AD4 CHECK THE 4X4 OPERATION

- Start the engine.
- Shift the transfer case into ~~4WD~~ mode.

Does the transfer case shift into ~~4WD~~ mode?

Yes	INSTALL a new TCCM . REFER to Section 308-07A.
No	REFER to Section 308-07A to diagnose 4WD concerns.

Pinpoint Test AE: The Door Ajar Message Center Warning Indicator Is Never/Always On

Normal Operation

The Instrument Panel Cluster (IPC) receives the door ajar status for the LH front, RH front, LH rear and RH rear door ajar switch from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . When the IPC receives a door ajar indication request from the BCM that one or more doors are open or ajar, the IPC turns on the appropriate door ajar message center warning indicator.

If the door ajar indication request message is missing for less than 5 seconds, the door ajar indication remains in its last state (on or off) based upon the last known good message received. If the door ajar status message is missing for more than 5 seconds, the IPC sets DTC U0140:00 and defaults the door ajar message center warning indicator active (on).

NOTE: *If DTC U0140:00 sets in the IPC, other symptoms may also be present such as high beam indicator, RH/LH turn indicators, instrument panel backlighting and Tire Pressure Monitoring System (TPMS) .*

This pinpoint test is intended to diagnose the following:

- Message center concern
- Interior lighting concern
- BCM

PINPOINT TEST AE : THE DOOR AJAR MESSAGE CENTER WARNING INDICATOR IS NEVER/ALWAYS ON

AE1 VERIFY THE OPERATION OF THE WRENCH AND ENGINE OVER-TEMPERATURE MESSAGE CENTER WARNING INDICATORS

- Ignition OFF.
- Disconnect: PCM [C175E](#) (3.7L), [C1381E](#) (5.0L and 6.2L) or [C1551E](#) (3.5L) .
- Ignition ON.
- **NOTE:** *If multiple warnings are present, the message center indicators rotate through each of the warnings. Allow the displays to rotate through each display while determining whether the powertrain malfunction (wrench) and over-temperature message center warning indicator is active.*

Observe the message center or integrated display area of the IPC with the PCM connected and disconnected.

Do the powertrain malfunction (wrench) and engine over-temperature warning message center indicators remain off when the PCM is connected and become active with the PCM disconnected?

Yes	GO to AE2 .
No	GO to Pinpoint Test AF .

AE2 CHECK THE INTERIOR LIGHTING OPERATION

- Open and close each door while monitoring the interior lamps.

Do the interior lamps operate correctly with each door?

Yes	GO to AE3 .
No	REFER to Section 417-02.

AE3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the continuous self-test.

Is DTC U0140:00 recorded?

Yes	GO to Pinpoint Test AY.
No	GO to AE4 .

AE4 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded BCM DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to Body Control Module (BCM) DTC Chart in Section 419-10 for correct navigation of the DTC.
No	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

Pinpoint Test AF: The Message Center Is Not Operating Correctly

Refer to Wiring Diagrams Cell [60](#) , Instrument Cluster for schematic and connector information.

Normal Operation

The message center functionality is controlled through the message center switch (part of the LH steering wheel switch), which is hardwired to the Instrument Panel Cluster (IPC) through input and return circuits. There are 3 message center switch buttons (base IPC) or 5 message center switch buttons (optional IPC). Each button operates a unique switch within the message center switch portion of the LH steering wheel switch assembly using a different resistance value associated with each specific button. The IPC sends out a reference voltage to the message center switch on the input circuit and monitors the voltage drop when a message center switch button is pressed. The voltage drop varies depending upon the resistance of the specific button pressed, providing indication to the IPC , which switch is pressed.

DTC Description	Fault Trigger Conditions
• B1A82:11 — External Trip Switch: Circuit Short To Ground	NOTE: <i>This DTC only sets when equipped with the base message center IPC and does not apply to the Raptor or optional message center IPC s.</i> A continuous and on-demand DTC that sets in the IPC when the IPC detects a short to ground on the message center switch input circuit.
• B1A82:13 — External Trip Switch: Circuit Open	A continuous and on-demand DTC that sets in the IPC when the IPC detects an open on the message center switch input circuit.
• B1A82:23 — External Trip Switch: Signal Stuck Low	A continuous memory DTC that sets in the IPC when the IPC detects the message center switch input pulled low (message center switch button stuck) but not shorted to ground.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Message center switch (part of the LH steering wheel switch)
- Clockspring
- IPC

PINPOINT TEST AF : THE MESSAGE CENTER IS NOT OPERATING CORRECTLY

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

AF1 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

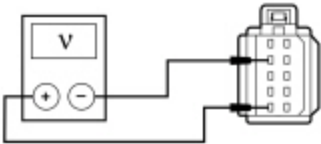
- Retrieve the recorded IPC DTCS from the self-test.

Are any DTCs recorded?

Yes	For DTC B1A82:23, INSTALL a new LH steering wheel switch. REFER to Section 211-05. CLEAR the DTCs. REPEAT the self-test. For DTC B1A82:11 or DTC B1A82:13, GO to AF2 . For all other DTCs, REFER to DTC Charts in this section.
No	For the base message center, GO to AF9 . For the Raptor or optional message center, GO to AF2 .

AF2 CHECK THE IPC OUTPUT AT THE LH STEERING WHEEL SWITCH

- Ignition OFF.
- Disconnect: LH Steering Wheel Switch [C2998](#).
- Ignition ON.
- Measure the voltage between the LH steering wheel switch [C2998](#) Pin 10, harness side and the LH steering wheel switch [C2998](#) Pin 7, harness side.



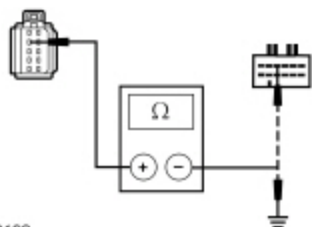
- N0120197

Is the voltage approximately 5 volts?

Yes	INSTALL a new LH steering wheel switch. REFER to Section 211-05. CLEAR the DTCs. REPEAT the self-test.
No	GO to AF3 .

AF3 CHECK THE MESSAGE CENTER SIGNAL CIRCUIT BETWEEN THE LH STEERING WHEEL SWITCH AND THE LOWER CLOCKSPRING FOR AN OPEN AND A SHORT TO GROUND

- Ignition OFF.
- Disconnect: Lower Clockspring [C218A](#).
- Measure the resistance between the LH steering wheel switch [C2998](#) Pin 7, harness side and the lower clockspring pin 3, component side; and between the LH steering wheel switch [C2998](#) Pin 7, harness side and ground.



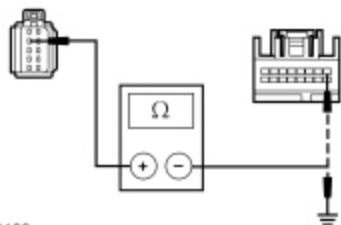
N0120198

Is the resistance less than 5 ohms between the LH steering wheel switch and the clockspring, and greater than 10,000 ohms between the LH steering wheel switch and ground?

Yes	GO to AF5 .
No	GO to AF4 .

AF4 CHECK THE MESSAGE CENTER SIGNAL CIRCUIT BETWEEN THE LH STEERING WHEEL SWITCH AND THE UPPER CLOCKSPring FOR AN OPEN AND A SHORT TO GROUND

- Remove the driver air bag. Refer to Section 501-20B.
- Disconnect: Upper Clockspring [C218B](#).
- Measure the resistance between the LH steering wheel switch [C2998](#) Pin 7, harness side and the upper clockspring [C218B](#) Pin 1, harness side; and between the LH steering wheel switch [C2998](#) Pin 7, harness side and ground.



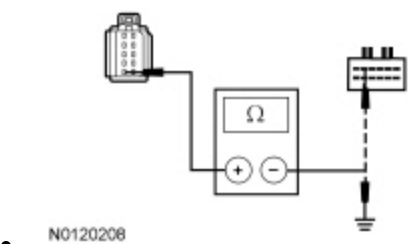
N0120199

Is the resistance less than 5 ohms between the LH steering wheel switch and the clockspring, and greater than 10,000 ohms between the LH steering wheel switch and ground?

Yes	INSTALL a new clockspring. REFER to Section 501-20B. CLEAR the DTC. REPEAT the self-test
No	INSTALL a new steering wheel. REFER to Section 211-04. CLEAR the DTC. REPEAT the self-test.

AF5 CHECK THE MESSAGE CENTER RETURN CIRCUIT BETWEEN THE LH STEERING WHEEL SWITCH AND THE LOWER CLOCKSPRING FOR AN OPEN

- Ignition OFF.
- Disconnect: Lower Clockspring [C218A](#).
- Measure the resistance between the LH steering wheel switch [C2998](#) Pin 10, harness side and the lower clockspring pin 2, component side.

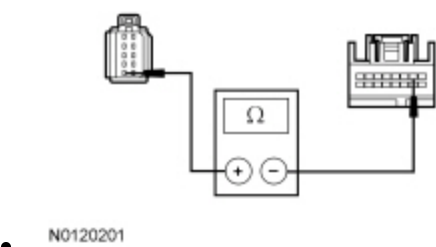


Is the resistance less than 5 ohms?

Yes	GO to AF7 .
No	GO to AF6 .

AF6 CHECK THE MESSAGE CENTER RETURN CIRCUIT BETWEEN THE LH STEERING WHEEL SWITCH AND THE UPPER CLOCKSPRING FOR AN OPEN

- Remove the driver air bag. Refer to Section 501-20B.
- Disconnect: Lower Clockspring [C218B](#).
- Measure the resistance between the LH steering wheel switch [C2998](#) Pin 10, harness side and the upper clockspring [C218B](#) Pin 2, harness side.

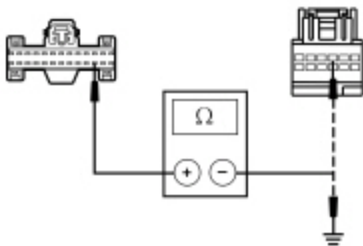


Is the resistance less than 5 ohms?

Yes	INSTALL a new clockspring. REFER to Section 501-20B. CLEAR the DTC. REPEAT the self-test
No	INSTALL a new steering wheel. REFER to Section 211-04. CLEAR the DTC. REPEAT the self-test.

AF7 CHECK THE MESSAGE CENTER SIGNAL CIRCUIT BETWEEN THE IPC AND THE LOWER CLOCKSPRING FOR AN OPEN AND A SHORT TO GROUND

- Disconnect: IPC [C220](#) .
- Measure the resistance between the IPC [C220](#) Pin 16, circuit CMC29 (GN/VT), harness side and the lower clockspring [C218A](#) Pin 3, circuit CMC29 (GN/VT), harness side; and between the IPC [C220](#) Pin 16, circuit CMC29 (GN/VT), harness side and ground.



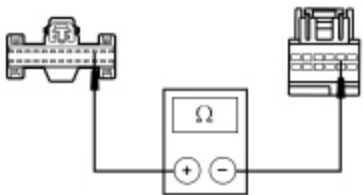
N0120209

Is the resistance less than 5 ohms between the IPC and the clockspring, and greater than 10,000 ohms between the IPC and ground?

Yes	GO to AF8 .
No	REPAIR circuit CMC29 (GN/VT) for an open or short to ground. CLEAR the DTC. REPEAT the self-test.

AF8 CHECK THE MESSAGE CENTER RETURN CIRCUIT BETWEEN THE IPC AND THE LOWER CLOCKSPRING FOR AN OPEN

- Measure the resistance between the IPC [C220](#) Pin 3, circuit RMC27 (WH/VT), harness side and the lower clockspring [C218A](#) Pin 2, circuit RMC27 (WH/VT), harness side.



N0120210

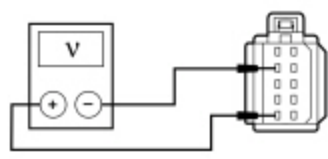
Is the resistance less than 5 ohms?

Yes	GO to AF10 .
No	REPAIR circuit RMC27 (WH/VT) for an open or short to ground. CLEAR the DTC. REPEAT the self-test.

AF9 CHECK FOR VOLTAGE FROM THE IPC

- Disconnect: LH Steering Wheel Switch [C2998](#).

- Ignition ON.
- Measure the voltage between the LH steering wheel switch [C2998](#) Pin 10, harness side and the LH steering wheel switch [C2998](#) Pin 7, harness side.



• N0120197

Is the voltage approximately 5 volts?

Yes	INSTALL a new LH steering wheel switch. REFER to Section 211-05. TEST the system for normal operation.
No	GO to AF10 .

AF10 CHECK FOR CORRECT IPC OPERATION

- Disconnect the **IPC** connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the **IPC** connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test AG: The Compass Is Inoperative (With 4.2" Message Center and 4.2" FCDIM))

Normal Operation

The compass is displayed in **IPC** and **FCDIM** when equipped with the 4.2" **FCDIM** , and in the Front Display Interface Module (FDIM) when equipped with the base Front Display Interface Module (FDIM) and the 8" Front Display Interface Module (FDIM) .

The **IPC** receives the compass display message from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The **BCM** receives the compass display message from the **GPSM** over the **MS-CAN** .

If the compass display message is missing or invalid for 5 seconds or less, the compass displays the last vehicle direction based upon the last known good compass display message received. If the compass display message is invalid or missing for 5 seconds or more, the **IPC** defaults the compass display to all dashes (---).

- DTC U0462:82 (Invalid Data Received from Compass Module: Alive/Sequence Counter Incorrect/Not Updated) — sets in the Instrument Panel Cluster (IPC) in continuous memory if the **IPC** does not receive updated compass data from the **BCM** for 5 seconds or longer.

This pinpoint test is intended to diagnose the following:

- Communication concern
- **GPSM**

PINPOINT TEST AG : THE COMPASS IS INOPERATIVE (WITH 4.2" MESSAGE CENTER AND 4.2" FCDIM))

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

AG1 CHECK THE COMPASS ZONE AND CALIBRATION SETTINGS THROUGH THE MESSAGE CENTER

- Ignition ON.
- Enter the vehicle settings in the message center and verify whether the compass zone and calibration setting menu is present.

Is the compass zone and calibration setting menu present?

Yes	DIAGNOSE the compass display in the 4.2" FCDIM . Refer to the appropriate section in Group 415 for the procedure.
No	GO to AG2 .

AG2 RETRIEVE THE RECORD DIAGNOSTIC TROUBLE CODES (DTCS) FROM THE GLOBAL POSITIONING SYSTEM MODULE (GPSM) SELF-TEST

- Carry out the **GPSM** self-test.

Are any DTCS recorded?

Yes	Refer to the appropriate section in Group 415 for the procedure..
No	GO to AG3 .

AG3 RETRIEVE THE RECORD DIAGNOSTIC TROUBLE CODES (DTCS) FROM THE BODY CONTROL MODULE (BCM) SELF-TEST

- Carry out the **BCM** self-test.

Are any DTCS recorded?

Yes	Section 419-10.
------------	-----------------

No	GO to AG4 .
----	-----------------------------

AG4 RETRIEVE THE RECORD DIAGNOSTIC TROUBLE CODES (DTCS) FROM THE INSTRUMENT PANEL CLUSTER (IPC) SELF-TEST

- Carry out the IPC self-test.

Are any DTCs recorded?

Yes	REFER to DTC Chart: IPC in this section.
No	INSTALL a new GPSM . Refer to the appropriate section in Group 415 for the procedure..

Pinpoint Test AI: The Outside Air Temperature Display Is Inoperative/Incorrect

Refer to Wiring Diagrams Cell [60](#) , Instrument Cluster for schematic and connector information.

Normal Operation

The outside air temperature is displayed in the Front Display Interface Module (FDIM) (with SYNC®) or Front Control/Display Interface Module (FCDIM) (without SYNC®). The Ambient Air Temperature (AAT) sensor is hardwired to the PCM through separate input and return circuits. The PCM provides a reference voltage to the AAT sensor and monitors the change in voltage resulting from changes in resistance as determined by outside air temperature. The PCM then messages the ambient air temperature data to the HVAC module through the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The BCM then filters the temperature data and sends the updated temperature status to the Front Display Interface Module (FDIM) (with SYNC®) or Front Control/Display Interface Module (FCDIM) (without SYNC®) over the Medium Speed Controller Area Network (MS-CAN) .

The HVAC module is programmed to update the messaged outside temperature data at different rates depending on several criteria to prevent false temperature displays due to a condition known as heat soaking. Heat soaking is where the outside air temperature is hotter in the location of the AAT sensor than the actual outside air temperature.

When the sensed outside temperature rises, the display updates slowly at varying rates based on vehicle speed. When the sensed outside temperature drops, the display updates more quickly following the drop experienced by the AAT sensor.

This pinpoint test is intended to diagnose the following:

- Communication concern
- AAT sensor
- PCM concern

PINPOINT TEST AI : PINPOINT TEST AI: THE OUTSIDE AIR TEMPERATURE DISPLAY IS INOPERATIVE/INCORRECT

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

AI1 RETRIEVE THE RECORDED DTCS FROM THE HVAC MODULE SELF-TEST

- Ignition ON.
- Check for recorded HVAC module DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0164, Refer to the appropriate section in Group 415 for the procedure.
No	If equipped with SYNC®, GO to AI2 . If not equipped with SYNC®, GO to AI3 .

AI2 RETRIEVE THE RECORDED DTCS FROM THE FDIM SELF-TEST

- Check for recorded ~~FDIM~~ DTCS from the self-test.

Are any DTCS recorded?

Yes	Refer to the appropriate section in Group 415 for the procedure.
No	GO to AI4 .

AI3 RETRIEVE THE RECORDED DTCS FROM THE FRONT DISPLAY INTERFACE MODULE (FDIM) (WITH SYNC®) OR FRONT CONTROL/DISPLAY INTERFACE MODULE (FCDIM) (WITHOUT SYNC®) SELF-TEST

- Check for recorded Front Display Interface Module (FDIM) (with SYNC®) or Front Control/Display Interface Module (FCDIM) (without SYNC®) DTCS from the self-test.

Are any DTCS recorded?

Yes	Refer to the appropriate section in Group 415 for the procedure.
No	GO to AI4 .

AI4 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check for recorded DTCS from the PCM ~~KOEO~~ self-test.

Are any ambient temperature sensor DTCS recorded?

Yes	REFER to Section 303-14.
No	GO to AI5 .

AI5 CHECK THE OUTSIDE AIR TEMPERATURE CONCERN

- Verify the customer outside air temperature display concern.

Is the outside air temperature display concern for an incorrect temperature display?

Yes	GO to AI6 .
No	INSTALL a new Front Display Interface Module (FDIM) (with SYNC®) or Front Control/Display Interface Module (FCDIM) (without SYNC®). REFER to the appropriate 415-00 section.

AI6 CHECK THE AMBIENT AIR TEMPERATURE INPUT TO THE PCM

- Using an infrared temperature scanner or similar tool, measure and record the temperature of the ambient air temperature sensor body.
- Ignition ON.
- Using a diagnostic scan tool, view the Parameter Identifications (PIDs).
- Select the ambient air temperature (AAT).
- Monitor the ambient air temperature and compare the value with the recorded temperature of the ambient air temperature sensor body.

Does the ambient air temperature match the recorded ambient air temperature sensor measurement within +/- 3° C (+/- 5° F)?

Yes	GO to AI7 .
No	GO to AI8 .

AI7 CHECK THE OUTSIDE AIR TEMPERATURE DISPLAY UPDATE

- Drive the vehicle above 33 km/h (20 mph) 1 to allow the outside air temperature display to update.

Did the outside air temperature display update toward the actual ambient air temperature?

Yes	The system is operating correctly at this time. Explain to the customer the operation of the system as it relates to the update strategy.
No	INSTALL a new Front Display Interface Module (FDIM) (with SYNC®) or Front Control/Display Interface Module (FCDIM) (without SYNC®). REFER to the appropriate 415-00 section.

AI8 CHECK THE OUTSIDE AIR TEMPERATURE (AAT) SENSOR AT THE PCM

- Ignition OFF.
- Disconnect: PCM [C175B](#) (3.7L Engine) .
- Disconnect: PCM [C1551B](#) (3.5L Engine) .
- Disconnect: PCM [C1381B](#) (5.0L and 6.2L Engines) .
- Measure:

3.5L Engine

Positive Lead	Measurement / Action	Negative Lead
C1551B Pin 51	Ω	C1551B Pin 18

- Measure:

3.7L Engine

Positive Lead	Measurement / Action	Negative Lead
C175B Pin 17	Ω	C175B Pin 56

- Measure:

5.0L and 6.2L Engines

Positive Lead	Measurement / Action	Negative Lead
C1381B Pin 17	Ω	C1381B Pin 56

- Compare the Ambient Air Temperature (AAT) sensor temperature with the resistance reading according to the table below.

Outside Air Temperature		Ambient Air Temperature Sensor Value
Celcius	Fahrenheit	Resistance (K Ohms)
-1.1° to 4.4°	30° to 40°	27.49 to 33.47
4.4° to 10°	40° to 50°	22.06 to 28.86
10° to 15.6°	50° to 60°	16.63 to 23.10
15.6° to 21.1°	60° to 70°	11.74 to 17.32
21.1° to 26.7°	70° to 80°	9.34 to 12.14
26.7° to 32.2°	80° to 90°	8.49 to 9.62
32.2° to 37.8°	90° to 100°	5.93 to 7.87
37.8° to 43.3°	100° to 110°	5.93 to 7.87

Does the resistance match the table specifications above?

Yes	GO to AI10 .
No	GO to AI9 .

AI9 CHECK THE AMBIENT AIR TEMPERATURE SENSOR RESISTANCE AT THE AMBIENT AIR TEMPERATURE SENSOR

- Disconnect: Ambient Air Temperature Sensor [C198](#).

- Measure:

Positive Lead	Measurement / Action	Negative Lead
Ambient air Temperature Sensor C198 pin 1, component side	Ω	Ambient air Temperature Sensor C198 pin 2, component side

- Compare the Ambient Air Temperature (AAT) sensor temperature with the resistance reading according to the table below.

Outside Air Temperature		Ambient Air Temperature Sensor Value
Celcius	Fahrenheit	Resistance (K Ohms)
-1.1° to 4.4°	30° to 40°	27.49 to 33.47
4.4° to 10°	40° to 50°	22.06 to 28.86
10° to 15.6°	50° to 60°	16.63 to 23.10
15.6° to 21.1°	60° to 70°	11.74 to 17.32
21.1° to 26.7°	70° to 80°	9.34 to 12.14
26.7° to 32.2°	80° to 90°	8.49 to 9.62
32.2° to 37.8°	90° to 100°	5.93 to 7.87
37.8° to 43.3°	100° to 110°	5.93 to 7.87

Does the resistance match the table specifications above?

Yes	The ambient air temperature sensor is operating correctly. REPAIR the circuits for high circuit resistance.
No	GO to AI11 .

AI10 CHECK THE AMBIENT AIR TEMPERATURE (AAT) SENSOR FOR OBSTRUCTIONS OR DEBRIS

- Check the area around the Ambient Air Temperature (AAT) sensor for any obstructions or debris such as leaves or paper.

Is the area around the Ambient Air Temperature (AAT) sensor free from any obstructions or debris?

Yes	GO to AI11 .
No	CLEAR all debris or obstructions as necessary to allow for the free flow of air around the Ambient Air Temperature (AAT) sensor.

AI11 CHECK THE AMBIENT AIR TEMPERATURE (AAT) SENSOR OPERATION

- Disconnect Ambient Air Temperature (AAT) Sensor C198 .

- Inspect the connector and wiring:
 - water intrusion
 - corrosion
 - damaged pins
 - pushed-out pins

Are there any signs of water intrusion, corrosion, damaged or pushed out pins?

Yes	REPAIR the Ambient Air Temperature (AAT) sensor connector or wiring for any water intrusion, corrosion, damaged or pushed out pins.
No	INSTALL a new ambient air temperature sensor.

Pinpoint Test AJ: A Hill Descent Control Message Is Inoperative/Always On

Normal Operation

The hill descent control system maintains vehicle speed on a downhill grade at operating speeds between 3 km/h (2 mph) and 32 km/h (20 mph). The system goes into standby mode on a flat or uphill terrain. The system deactivates at speeds above 64 km/h (40 mph). During periods of sustained use, the hill descent control system may automatically deactivate in order to cool the braking system.

When the hill descent control switch is pressed, a ground signal is sent to the PCM. The PCM then sends a message to the ABS module along the High Speed Controller Area Network (HS-CAN) requesting system activation. The ABS module uses inputs from the Restraints Control Module (RCM) and wheel speed sensors to determine the mode of operation.

The Instrument Panel Cluster (IPC) receives the hill descent control status from the ABS module over the HS-CAN . The message center also displays hill descent control messages to inform the driver they have entered or exited various states of the hill descent control function.

If the hill descent message is missing for less than 5 seconds, the hill descent active message center indicator remains in the last state (on or off) based on the last status message received. If the hill descent status message is missing for more than 5 seconds, the IPC sets DTC U0121:00 and defaults the hill descent active message center indicator off.

This pinpoint test is intended to diagnose the following:

- Message center concern
- ABS module

PINPOINT TEST AJ : A HILL DESCENT CONTROL MESSAGE IS INOPERATIVE/ALWAYS ON

AJ1 VERIFY THE OPERATION OF THE DOOR AJAR MESSAGE CENTER WARNING INDICATOR

- Close all doors then open the LH front door.
- Ignition ON.
- **NOTE:** *If multiple warnings are present, the message center cycles through each of the warning indicators. Allow the displays to cycle through each display while determining whether the door ajar message center warning indicator is active.*

Observe the message center.

Does the message center display DRIVER DOOR AJAR?

Yes	GO to AJ2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

AJ2 CHECK AND ENSURE THE HILL DESCENT CONTROL MODE FUNCTIONS

- Start the engine.
- Drive the vehicle between 3 km/h (2 mph) and 32 km/h (20 mph) on a downhill grade and activate the hill descent control system mode.

Are the brakes applying on a downhill grade?

Yes	GO to AJ3 .
No	REFER to Section 206-09.

AJ3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the IPC self-test.

Is DTC U0121:00 recorded?

Yes	GO to Pinpoint Test AV .
No	GO to AJ4 .

AJ4 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE SELF-TEST

- Check for recorded ABS module DTCS from the self-test.

Are any DTCS recorded?

Yes	REFER to Section 206-09.
No	INSTALL a new ABS module. REFER to Section 206-09.

Pinpoint Test AK: The CHECK FUEL FILL INLET Message Is Inoperative/Always On

Normal Operation

The PCM monitors the fuel tank evaporative emission system for significant leaks that occur following refueling of the vehicle. Once the PCM detects a fuel vapor leak, the PCM sends the Instrument Panel Cluster (IPC) a message over the communication network to turn on the CHECK FUEL FILL INLET message. The PCM only sets DTC P0457 following a successful cruise test, which is initiated when the vehicle is driven at a steady speed above 64 km/h (40 mph) for a duration of approximately 4-5 minutes. If the PCM is unable to successfully run the cruise test, the IPC does not receive the fuel inlet status and the CHECK FUEL FILL INLET message remains off.

If the **IPC** does not receive the fuel inlet status from the PCM for 5 seconds, the **IPC** sets DTC U0100:00 in continuous memory and defaults the CHECK FUEL FILL INLET message off.

NOTE: If DTC U0100:00 is set in the **IPC** , other observable symptoms may be an inoperative speedometer, temperature gauge or odometer.

- DTC P0457 (Evaporative Emission System Leak Detected) (Fuel Cap Loose/Off) — sets in the PCM if the PCM detects a significant fuel tank pressure change after refueling occurs.

This pinpoint test is intended to diagnose the following:

- Evaporative fuel system concern
- PCM
- **IPC**

PINPOINT TEST AK : THE CHECK FUEL FILL INLET MESSAGE IS INOPERATIVE/ALWAYS ON

AK1 CHECK THE MESSAGE CENTER OPERATION

- Close all doors.
- Ignition ON.
- Open the driver door while monitoring the message center displays.

Does the message center display DRIVER DOOR AJAR?

Yes	GO to AK2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

AK2 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check for recorded PCM DTCs from the Key ON Engine OFF (KOEO) self-test.

Is DTC P0457 recorded?

Yes	If the CHECK FUEL FILL INLET message is illuminated, REFER to Section 303-14. TEST the system for normal operation. If the CHECK FUEL FILL INLET message is not illuminated, GO to AK3 .
No	If the CHECK FUEL FILL INLET message is always on, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation. If the check fuel cap indicator is not illuminated, the system is operating normally at this time. If the fuel cap was left off and the CHECK FUEL FILL INLET message did not turn on, driving conditions may not have allowed for the PCM to run the cruise test and message the IPC to turn the CHECK FUEL FILL INLET message on.

AK3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded **IPC** DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0100:00, GO to Pinpoint Test AT . For all other DTCs, REFER to DTC Charts in this section.
No	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.

Pinpoint Test AL: An Off-Road Mode Message Is Inoperative/Always On (Raptor Only)

Normal Operation

The message center displays off-road mode messages to inform the driver they have entered or exited various states of the off-road mode function. When the off-road mode is active, the standard vehicle speed thresholds for the Instrument Panel Cluster (IPC) are lifted and the AdvanceTrac® settings are altered for optimized off-road driving. The IPC receives the off-road mode signal from the Transfer Case Control Module (TCCM) over the High Speed Controller Area Network (HS-CAN) .

If the off-road mode message is missing for less than 5 seconds, the IPC defaults the applicable off-road mode message center indicator to the last on/off state, based upon the last off-road mode status message received.

If the off-road mode status message is missing for greater than 5 seconds, the IPC sets DTC U0102:00 and defaults the applicable off-road mode message center indicator off.

This pinpoint test is intended to diagnose the following:

- Message center concern
- Four-Wheel Drive (4WD) concern
- TCCM

PINPOINT TEST AL : AN OFF-ROAD MODE MESSAGE IS INOPERATIVE/ALWAYS ON (RAPTOR ONLY)

AL1 CHECK THE MESSAGE CENTER OPERATION

- Close all doors.
- Ignition ON.
- Open the driver door while monitoring the message center displays.

Does the message center display DRIVER DOOR AJAR?

Yes	GO to AL2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

AL2 CHECK THE 4X4 OPERATION IN OFF-ROAD MODE

- Start the engine.
- Drive the vehicle in Four-Wheel Drive (4WD) and activate the off-road mode.

Has the vehicle entered off-road mode?

Yes	GO to AL3 .
------------	-----------------------------

No	REFER to Section 308-07A to diagnose 4X4 system concerns.
-----------	---

AL3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the **IPC** self-test.

Is DTC U0102:00 recorded?

Yes	GO to Pinpoint Test AU.
No	GO to AL4 .

AL4 RETRIEVE THE RECORDED DTCS FROM THE TCCM SELF-TEST

- Check for recorded DTCS from the **TCCM** self-test.

Are any DTCS recorded?

Yes	REFER to Section 308-07A.
No	INSTALL a new TCCM . REFER to Section 308-07A.

Pinpoint Test AM: The Chime Sounds When The Driver Door Is Ajar (No Key In Ignition And Headlamps Off)

Refer to Wiring Diagrams Cell [60](#) , Instrument Cluster for schematic and connector information.

Normal Operation

The Instrument Panel Cluster (IPC) receives the key-in-ignition warning status from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The key-in-ignition warning switch (part of the ignition switch) is hardwired to the **BCM** . When the key is inserted into the ignition lock cylinder, the key-in-ignition switch closes and supplies voltage to the **BCM** . The **BCM** monitors the door ajar input from the LH front door and the key-in-ignition switch status. When the key is inserted into the ignition lock cylinder and the LH front door is open or ajar, the **BCM** sends the **IPC** a chime request message to sound the key-in-ignition warning chime. The headlamps on warning chime sounds when the driver door is opened with the ignition off, and the headlamp switch is in the PARK or HEADLAMP ON position.

The **BCM** receives the headlamp switch status from the headlamp switch and the LH front door ajar status from the LH front door ajar switch through hardwired circuitry. The **BCM** communicates the headlamp on chime request to the **IPC** when the headlamps or parking lamps are on and the driver door is opened. The autolamps feature must not be selected for the headlamps on warning chime to operate.

- DTC B10F1:14 (Key In Switch: Circuit Short To Ground or Open) — a continuous and on-demand DTC that sets if the **BCM** detects an open on the key-in-ignition switch input circuit or if the **BCM** detects ignition START, RUN/START OR RUN/ACC inputs without the key-in-ignition input.

This pinpoint test is intended to diagnose the following:

- Exterior lighting concern
- Interior lighting concern
- Wiring, terminals or connectors
- Key-in-ignition warning switch (part of the ignition switch)

- BCM

PINPOINT TEST AM : THE CHIME SOUNDS WHEN THE DRIVER DOOR IS AJAR (NO KEY IN IGNITION AND HEADLAMPS OFF)

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

AM1 CHECK THE EXTERIOR LIGHTING OPERATION

- Turn the headlamp switch to the OFF position.
- Close all doors and wait one minute.
- Wait one minute.
- Observe the exterior lights.

Are the headlamps and parking lamps off?

Yes	GO to AM2 .
No	REFER to Section 417-01.

AM2 CHECK THE INTERIOR LAMPS OPERATION

- Open and close the drivers door.
- Observe the interior lamps.

Do the interior lamps operate correctly from the driver door?

Yes	GO to AM3 .
No	REFER to Section 417-02.

AM3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check the recorded DTCS from the BCM self-tests.

Are any DTCS present?

Yes	For DTC B10F1:14, GO to AM4 . For all other DTCS, REFER to Section 419-10.
No	GO to AM5 .

AM4 CHECK THE BCM INPUT PIDS FROM THE IGNITION SWITCH

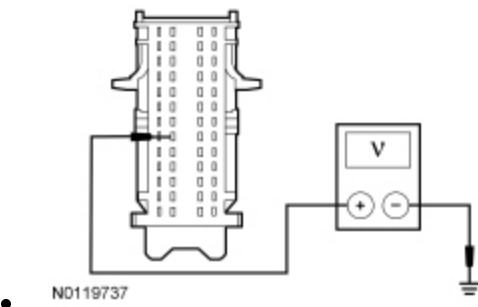
- Enter the following diagnostic mode on the scan tool: BCM DataLogger .
- Select and monitor the BCM ignition switch state (IGN_SW_STATE) while switching through the ignition states.

Do the BCM PIDs match the ignition switch position?

Yes	GO to AM8 .
No	REFER to Section 211-05 to diagnose the ignition switch outputs to the BCM .

AM5 CHECK THE KEY-IN-IGNITION INPUT CIRCUIT FOR VOLTAGE

- Ignition OFF.
- Remove the key from the ignition cylinder.
- Disconnect: BCM [C2280B](#) .
- Measure the voltage between the BCM [C2280B](#) Pin 34, circuit CDC30 (BU/GY), harness side and ground.

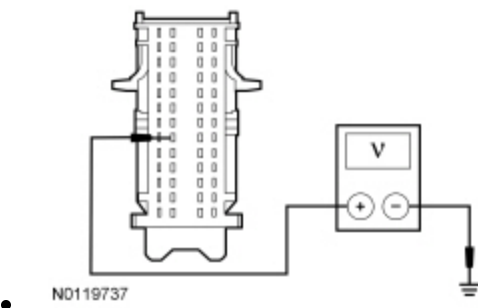


Is any voltage present?

Yes	GO to AM6 .
No	GO to AM8 .

AM6 CHECK KEY-IN-IGNITION INPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Without disconnecting the electrical connector, remove the ignition switch from the steering column. Refer to Section 211-05.
- Measure the voltage between the BCM [C2280B](#) Pin 34, circuit CDC30 (BU/GY), harness side and ground.

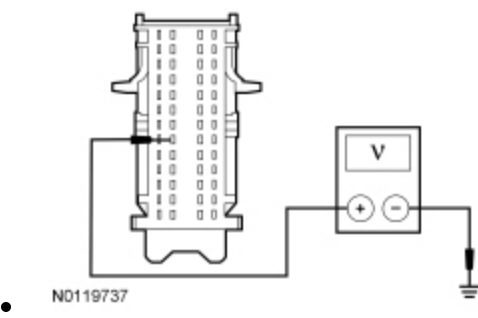


Is any voltage present?

Yes	GO to AM7 .
No	INSTALL a new ignition lock cylinder. TEST the system for normal operation.

AM7 CHECK KEY-IN-IGNITION INPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: Ignition Switch [C250](#).
- Measure the voltage between the BCM [C2280B](#) Pin 34, circuit CDC30 (BU/GY), harness side and ground.



Is any voltage present?

Yes	REPAIR circuit CDC30 (BU/GY) for a short to voltage. TEST the system for normal operation.
No	INSTALL a new ignition switch. REFER to Section 211-05. TEST the system for normal operation.

AM8 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test AN: The Headlamps On Warning Chime Is Inoperative

Normal Operation

The Body Control Module (BCM) receives the headlamp switch status from the headlamp switch and the LH front door ajar status from the LH front door ajar switch through hardwired circuitry. The headlamps on warning chime sounds when the LH front door is opened with the ignition off, and the headlamp switch is in the PARK or HEADLAMP ON position. The

BCM communicates the headlamp on chime request to the IPC over the High Speed Controller Area Network (HS-CAN) when the headlamps or parking lamps are on with the ignition off and the driver door is opened.

This pinpoint test is intended to diagnose the following:

- Interior lighting concern
- BCM
- IPC

PINPOINT TEST AN : THE HEADLAMPS ON WARNING CHIME IS INOPERATIVE

AN1 CHECK THE SAFETY BELT WARNING CHIME OPERATION

- Ignition OFF.
- Ignition ON.
- While monitoring the safety belt warning chime, transition the ignition switch from OFF to RUN.

Does the safety belt warning chime operate for approximately 6 seconds?

Yes	GO to AN2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

AN2 CHECK THE INTERIOR LAMPS OPERATION

- Open and close the driver door.
- Observe the interior lamps.

Do the interior lamps operate correctly from the driver door?

Yes	GO to AN3 .
No	REFER to Section 417-02.

AN3 CHECK THE KEY-IN-IGNITION WARNING CHIME OPERATION

- Ignition OFF.
- Insert the key into the ignition lock cylinder.
- Open the LH front door and observe the key-in-ignition warning chime operation.

Does the key-in-ignition warning chime sound?

Yes	GO to AN4 .
No	GO to Pinpoint Test AQ .

AN4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Are any DTCs recorded?

Yes	For DTC U0140:00, GO to Pinpoint Test AY . For all other DTCs, REFER to DTC Chart in this section.
No	GO to AN5 .

AN5 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded BCM DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to Section 419-10 for correct navigation of the DTC.
No	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

Pinpoint Test AO: The Key-In-Ignition Warning Chime Is Inoperative

Refer to Wiring Diagrams Cell [60](#) , Instrument Cluster for schematic and connector information.

Normal Operation

The Instrument Panel Cluster (IPC) receives the key-in-ignition warning status from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The key-in-ignition warning switch (part of the ignition switch) is hardwired to the BCM . When the key is inserted into the ignition lock cylinder, the key-in-ignition switch closes and supplies voltage to the BCM . The BCM monitors the door ajar input from the LH front door and the key-in-ignition switch status. When the key is inserted into the ignition lock cylinder and the driver door is open or ajar, the BCM sends the IPC a chime request message to sound the key-in-ignition warning chime.

- DTC B10F1:14 (Key In Switch: Circuit Short To Ground or Open) — a continuous and on-demand DTC that sets if the BCM detects an open on the key-in-ignition switch input circuit or if the BCM detects ignition START, RUN/START OR RUN/ACC inputs without the key-in-ignition input.

This pinpoint test is intended to diagnose the following:

- Interior lighting concern
- Wiring, terminals or connectors
- Key-in-ignition warning switch (part of the ignition switch)
- BCM
- IPC

PINPOINT TEST AO : THE KEY-IN-IGNITION WARNING CHIME IS INOPERATIVE

AO1 CHECK THE SAFETY BELT WARNING CHIME OPERATION

- Ignition OFF.
- Unbuckle the driver safety belt.
- Ignition ON.

- Monitor the safety belt warning chime when the ignition switch is transitioned from OFF to RUN for 6 seconds.

Does the safety belt warning chime operate for approximately 6 seconds?

Yes	GO to AO2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

AO2 CHECK THE INTERIOR LAMPS OPERATION

- Open and close the driver door.
- Observe the interior lamps.

Do the interior lamps operate correctly from the driver door?

Yes	GO to AO3 .
No	REFER to Section 417-02.

AO3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

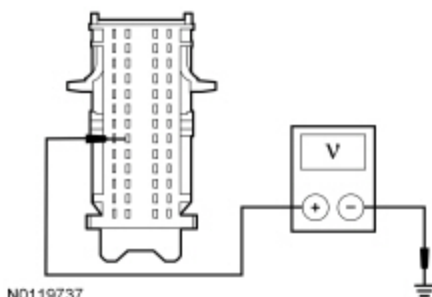
- Check the recorded DTCS from the BCM self-test.

Is DTC B10F1:14 present?

Yes	GO to AO4 .
No	GO to AO6 .

AO4 CHECK THE KEY-IN-IGNITION INPUT CIRCUIT FOR VOLTAGE

- Ignition OFF.
- Insert the key into the ignition cylinder.
- Disconnect: BCM [C2280B](#) .
- Measure the voltage between the BCM [C2280B](#) Pin 34, circuit CDC30 (BU/GY), harness side and ground.



Is the voltage greater than 10 volts?

Yes	GO to AO6 .
No	GO to AO5 .

AO5 CHECK THE IGNITION SWITCH

- Ignition OFF.
- Disconnect: Ignition Switch [C250](#).
- Carry out the ignition switch component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Is the ignition switch OK?

Yes	REPAIR circuit CDC30 (BU/GY) for an open. CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new ignition switch. REFER to Section 211-05. CLEAR the DTCs. REPEAT the self-test.

AO6 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test AP: The Memory Feedback Warning Chime Is Inoperative

Normal Operation

The Instrument Panel Cluster (IPC) receives the memory feedback chime request from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . When a successful memory set procedure is carried out, the Driver Seat Module (DSM) sends a chime request over the Medium Speed Controller Area Network (MS-CAN) to the **BCM** , which gateways the chime request to the **IPC** to sound the memory feedback chime.

This pinpoint test is intended to diagnose the following:

- DSM
- IPC

PINPOINT TEST AP : THE MEMORY FEEDBACK WARNING CHIME IS INOPERATIVE

AP1 CHECK THE KEY-IN-IGNITION WARNING CHIME OPERATION

- Ignition OFF.
- Insert the key into the ignition lock cylinder.
- Open the LH front door and observe the key-in-ignition warning chime operation.

Does the key-in-ignition warning chime operate correctly?

Yes	GO to AP2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

AP2 CHECK THE MEMORY SET FEATURE

- Verify the memory set procedure can successfully change the memory preferences.

Does the memory set procedure successfully change the memory preferences?

Yes	GO to AP3 .
No	REFER to Section 501-10.

AP3 RETRIEVE THE RECORDED DTCS FROM THE DSM SELF-TEST

- Check for recorded DTCS from the DSM self-test.

Is DTC U0140:00 recorded?

Yes	REFER to Section 501-10.
No	INSTALL a new DSM . REFER to Section 501-10. TEST the system for normal operation.

Pinpoint Test AQ: The Parking Brake Warning Chime Is Inoperative

Normal Operation

The Instrument Panel Cluster (IPC) receives the parking brake warning chime request from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The parking brake warning chime sounds when the ignition switch is on, the parking brake is engaged and the vehicle speed is greater than 5 km/h (3 mph). The parking brake warning chime stops sounding and resets if the parking brake is released, the ignition switch is not on, if the vehicle speed is below 5 km/h (3 mph), or after 90 seconds from the time the chime is activated.

NOTE: Make sure the parking brake warning chime operation is verified with the vehicle moving at least 5 km/h (3 mph) with the parking brake slightly applied (just enough to turn the brake warning indicator on) without damaging the rear brakes.

This pinpoint test is intended to diagnose the following:

- Brake warning indication concern
- BCM
- IPC

PINPOINT TEST AQ : THE PARKING BRAKE WARNING CHIME IS INOPERATIVE

AQ1 CHECK THE KEY-IN-IGNITION WARNING CHIME OPERATION

- Ignition OFF.
- With the ignition switch in the OFF position and the key in the ignition lock cylinder, open the LH front door and observe the key-in-ignition warning chime operation.

Does the key-in-ignition warning chime operate correctly?

Yes	GO to AQ2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

AQ2 CHECK THE BRAKE WARNING INDICATOR OPERATION

- Apply the parking brake.
- Ignition ON.
- Observe the brake warning indicator.

Does the brake warning indicator illuminate?

Yes	GO to AQ3 .
No	GO to Pinpoint Test M .

AQ3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCS from the self-test.

Are any DTCS recorded?

Yes	For DTC U0140:00, GO to Pinpoint Test AY . For all other DTCS, REFER to DTC Chart in this section.
No	GO to AQ4 .

AQ4 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded BCM DTCS from the self-test.

Are any DTCS recorded?

Yes	REFER to Section 419-10 for correct navigation of the DTC.
No	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

Pinpoint Test AR: The Safety Belt Warning Chime Is Inoperative/The Belt-Minder® Feature Does Not Operate Correctly

Normal Operation

The Instrument Panel Cluster (IPC) receives the safety belt and Belt-Minder® requests from the Restraints Control Module (RCM) over the High Speed Controller Area Network (HS-CAN) . The safety belt warning chime sounds for approximately 6 seconds when the driver safety belt is not fastened and the ignition lock cylinder is turned from the OFF/LOCK or ACC to the ON or START position.

The Belt-Minder® feature supplements the safety belt warning function and is enabled after the safety belt warning is complete. The Belt-Minder® simultaneously sounds the chime and illuminates the safety belt warning lamp in the IPC once the vehicle speed has exceeded 5 km/h (3 mph), and remains active for 5 minutes from the time it is started.

There are 2 Belt-Minder® strategies used and implemented by the IPC as commanded by the RCM . The 2 strategies are designed to provide an increased intensity level of the chime. Belt-Minder® A provides 6 one-second sequences of 0.5 second on/0.5 second off for the first minute of the Belt-Minder® warning. Belt-Minder® B provides 6 one-second sequences of 0.2 second on/0.2 second off then 0.3 second on/0.3 second off for the remainder of the Belt-Minder® warning.

NOTE: *Whenever the vehicle is operated using MyKey®, the RCM enables the Belt-Minder®.*

If the Belt-Minder® is deactivated using an administrator key and driven using a MyKey® key, the MyKey® system overrides the deactivation of Belt-Minder®. When MyKey® is in use, the driver cannot configure the Belt-Minder® off. Once the Belt-Minder® is activated, the Belt-Minder® continues to chime periodically (does not time out after 5 minutes) and the audio system is muted until the driver safety belt is fastened. Refer to Section 419-01B for more information on the MyKey® feature.

NOTE: *Make sure the Belt-Minder® chime operation is verified with the vehicle moving at least 5 km/h (3 mph).*

This pinpoint test is intended to diagnose the following:

- Safety belt warning indication concern
- Belt-Minder® deactivated
- Speedometer concern
- RCM
- IPC

PINPOINT TEST AR : THE SAFETY BELT WARNING CHIME IS INOPERATIVE/THE BELT-MINDER® FEATURE DOES NOT OPERATE CORRECTLY

AR1 CHECK THE KEY-IN-IGNITION WARNING CHIME OPERATION	
• Ignition OFF. • Open the LH front door and observe the key-in-ignition warning chime operation.	
Does the key-in-ignition warning chime operate correctly?	
Yes	GO to AR2 .

No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
----	---

AR2 CHECK THE DRIVER SAFETY BELT BUCKLE SWITCH

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **RCM** PIDs .
- Monitor the **RCM** driver safety belt buckle PID (SBELT_D_ST) while buckling and unbuckling the driver safety belt.

Does the PID display BUCKLED with the safety belt buckled and UNBUCKLED with the safety belt unbuckled?

Yes	GO to AR3 .
No	REFER to Section 501-20B.

AR3 CHECK THE SAFETY BELT WARNING INDICATOR FOR CORRECT OPERATION

- Ignition ON.
- Within one minute, buckle then unbuckle the LH front safety belt.
- With the engine running, verify the safety belt warning indicator illuminates with the safety belt unbuckled and turns off when buckled.

Does the safety belt warning indicator operate correctly?

Yes	GO to AR4 .
No	GO to Pinpoint Test U .

AR4 CHECK THE SPEEDOMETER OPERATION

- Verify the speedometer is operating correctly.

Does the speedometer operate correctly?

Yes	GO to AR5 .
No	GO to Pinpoint Test F .

AR5 CHECK THE BELT-MINDER® CONFIGURATION

NOTE: Verify an administrator key is used for this test and **not** a MyKey® programmed key. When a MyKey® programmed key is used, the Belt-Minder® automatically turns on and cannot be disabled (if previously disabled by an administrator, the Belt-Minder® remains disabled when an administrator key is used).

- Verify the Belt-Minder® is activated or configured on for the seating position in question. Refer to Belt-Minder® Deactivating/Activating in this section to configure without a scan tool.

Is the Belt-Minder® activated for the seating position in question?

Yes	If the Belt-Minder® is always on above 5 km/h (3 mph) with the safety belt unfastened, the system is operating normally. If the Belt-Minder® is always on with the safety belt fastened or never on with the safety belt unfastened above 5 km/h (3 mph), INSTALL a new BCM . REFER to Section 501-20B. TEST the system for normal operation.
No	ACTIVATE the Belt-Minder® for the seating position in question. REFER to Belt-Minder® Deactivating/Activating in this section. TEST the system for normal operation.

Pinpoint Test AS: The Turn Signal On Warning Chime Is Inoperative

Normal Operation

The turn signal on warning chime sounds when the vehicle travels more than 3.2 km (2.0 miles) with the turn signal on. The Body Control Module (BCM) communicates the turn signal switch status to the IPC over the High Speed Controller Area Network (HS-CAN) . The PCM communicates the odometer count data to the IPC over the HS-CAN .

This pinpoint test is intended to diagnose the following:

- Odometer operation
- Turn signal indication concern
- BCM
- IPC

PINPOINT TEST AS : THE TURN SIGNAL ON WARNING CHIME IS INOPERATIVE

AS1 CHECK THE KEY-IN-IGNITION WARNING CHIME OPERATION

- Ignition OFF.
- With the ignition switch off and the key in the ignition lock cylinder, open the LH front door and observe the key-in-ignition warning chime operation.

Does the key-in-ignition warning chime operate correctly?

Yes	GO to AS2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

AS2 CHECK THE TURN SIGNAL INDICATORS OPERATION

- Ignition ON.
- Verify the turn signal indicators operation.

Do the turn signal indicators operate correctly?

Yes	GO to AS3 .
No	GO to Pinpoint Test R .

AS3 CHECK THE ODOMETER FOR CORRECT OPERATION

- Drive the vehicle and observe the odometer for correct operation.

Does the odometer operate correctly?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	GO to Pinpoint Test H.

Pinpoint Test AT: DTC U0100:00

Normal Operation

- DTC U0100:00 (Lost Communication With ECM /PCM "A": No Sub-Type Information) — sets in continuous memory in the Instrument Panel Cluster (IPC) if data messages received from the PCM over the High Speed Controller Area Network (HS-CAN) are missing for 5 seconds or longer. For a complete list of all network messages, refer to Section 418-00.

NOTE: Once the IPC sets DTC U0100:00, multiple gauges or indicators are likely to be inoperative in the IPC. Other modules that require data from the IPC all set communication DTCs and may also exhibit system function concerns.

This pinpoint test is intended to diagnose the following:

- PCM
- IPC

PINPOINT TEST AT : DTC U0100:00

AT1 VERIFY THE SCAN TOOL CAN COMMUNICATE WITH THE PCM

- Connect the scan tool.
- Make sure a vehicle session can be established using the scan tool.

Can a vehicle session be established?

Yes	GO to AT2 .
No	REFER to Section 418-00 to diagnose no communication with the PCM.

AT2 CHECK THE IPC CONTINUOUS MEMORY DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: IPC Self-Test .
- Clear the DTCs.
- Wait 10 seconds.
- Repeat the IPC self-test.

Is DTC U0100:00 retrieved again?

Yes	GO to AT3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AT3 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check for recorded DTCs from the PCM Key ON Engine OFF (KOEO) self-test.

Is DTC P0562 or P0563 recorded?

Yes	REFER to Section 303-14.
No	GO to AT4 .

AT4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCs from the IPC self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test BD . For DTC U3003:17, GO to Pinpoint Test BE .
No	GO to AT5 .

AT5 CHECK FOR DTC U0100:00 SET IN OTHER MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0100:87 set in the Remote Function Actuator (RFA) module (if equipped) or DTC U0100:00 in the Restraints Control Module (RCM) ?

Yes	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test AU: DTC U0102:00

Normal Operation

- **NOTE:** DTC U0102:00 may set in the Instrument Panel Cluster (IPC) on all ~~4WD~~ vehicles and on ~~2WD~~ vehicles when equipped with ~~ELD~~ .

DTC U0102:00 (Lost Communication With Transfer Case Control Module: No Sub-Type Information) — sets in continuous memory in the Instrument Panel Cluster (IPC) if data messages received from the Transfer Case Control Module (TCCM) over the High Speed Controller Area Network (HS-CAN) are missing for 5 seconds or longer. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- ~~TCCM~~
- ~~IPC~~

PINPOINT TEST AU : DTC U0102:00

AU1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AU2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AU2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the ~~TCCM~~ pass the network test?

Yes	GO to AU3 .
No	REFER to Section 418-00.

AU3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the ~~IPC~~ self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test BD . For DTC U3003:17, GO to Pinpoint Test BE .
No	GO to AU4 .

AU4 RECHECK THE IPC DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the IPC self-test.

Is DTC U0102:00 still present?

Yes	GO to AU5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AU5 CHECK FOR DTC U0102:00 SET IN OTHER MODULES

- Clear all DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0102:00 set in the ABS module?

Yes	INSTALL a new TCCM . REFER to Section 308-07A. CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test AV: DTC U0121:00

Normal Operation

- DTC U0121:00 (Lost Communication With Anti-Lock Brake System (ABS) Control Module: No Sub-Type Information) — sets in continuous memory in the Instrument Panel Cluster (IPC) if data messages received from the ABS module over the High Speed Controller Area Network (HS-CAN) are missing for 5 seconds or longer. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- ABS module
- IPC

PINPOINT TEST AV : DTC U0121:00

AV1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AV2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AV2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the ABS module pass the network test?

Yes	GO to AV3 .
No	REFER to Section 418-00.

AV3 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE SELF-TEST

- Check for recorded DTCS from the ABS module self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	REFER to Section 206-09.
No	GO to AV4 .

AV4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the ~~IPC~~ self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test BD . For DTC U3003:17, GO to Pinpoint Test BE .
No	GO to AV5 .

AV5 RECHECK THE IPC DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the Programmable Module Installation (PMI) or the ~~PMI~~ may not have been carried out.*

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the IPC self-test.

Is DTC U0121:00 still present?

Yes	INSTALL a new ABS module. REFER to Section 206-09. CLEAR the DTCs. REPEAT the <u>IPC</u> self-test. If DTC U0121:00 is still present, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

Pinpoint Test AW: DTC U0131:00

Normal Operation

- DTC U0131:00 (Lost Communication With Power Steering Control Module: No Sub-Type Information) — sets in the Instrument Panel Cluster (IPC) if data messages received from the Power Steering Control Module (PSCM) over the High Speed Controller Area Network (HS-CAN) are missing for greater than 5 seconds. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- PSCM
- IPC

PINPOINT TEST AW : DTC U0131:00

AW1 VERIFY THE CUSTOMER CONCERN	
• Ignition ON. • Verify that there is an observable symptom present.	
Is an observable symptom present?	
Yes	GO to AW2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
AW2 CHECK THE COMMUNICATION NETWORK	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: Network Test . • Carry out the network test.	
Does the <u>PSCM</u> pass the network test?	

Yes	GO to AW3 .
No	REFER to Section 418-00.

AW3 RETRIEVE THE RECORDED DTCS FROM THE PSCM MODULE SELF-TEST

- Check for recorded DTCS from the PSCM module self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	REFER to Section 211-00A or Section 211-00B.
No	GO to AW4 .

AW4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the IPC self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test BD . For DTC U3003:17, GO to Pinpoint Test BE .
No	GO to AW5 .

AW5 RECHECK THE IPC DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Clear the DTCS. Repeat the IPC self-test.

Is DTC U0131:00 still present?

Yes	GO to AW6 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AW6 CHECK FOR DTC U0131:00 SET IN OTHER MODULES

- Clear all DTCS.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCS from all modules.

Is DTC U0131:00 set in the ABS module?

Yes	INSTALL a new PSCM (part of the steering gear). REFER to Section 211-02. CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test AX: DTC U0137:00

Normal Operation

- DTC U0137:00 (Lost Communication With Trailer Brake Control Module: No Sub Type Information) — sets in continuous memory if data messages received from the Trailer Brake Control (TBC) module over the High Speed Controller Area Network (HS-CAN) are missing for 5 seconds or longer. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- ~~TBC~~ module
- ~~IPC~~

PINPOINT TEST AX : DTC U0137:00

AX1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AX2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AX2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the ~~TBC~~ module pass the network test?

Yes	GO to AX3 .
No	REFER to Section 418-00.

AX3 RETRIEVE THE RECORDED DTCS FROM THE TBC MODULE SELF-TEST

- Check for recorded TBC module DTCs from the self-test.

Is DTC B1317 or DTC B1318 recorded?

Yes	REFER to Section 206-10.
No	GO to AX4 .

AX4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test BD . For DTC U3003:17, GO to Pinpoint Test BE .
No	GO to AX5 .

AX5 RECHECK THE IPC DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the IPC self-test.

Are any DTCs present?

Yes	For DTC U0137:00, INSTALL a new TBC module. REFER to Section 206-10. CLEAR the DTCs. REPEAT the IPC self-test. If DTC U0137:00 is still present, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

Pinpoint Test AY: DTC U0140:00

Normal Operation

- DTC U0140:00 (Lost Communication With Body Control Module: No Sub-Type Information) — sets in continuous memory in the Instrument Panel Cluster (IPC) if data messages received from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) are missing for 5 seconds or longer. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- BCM
- IPC

PINPOINT TEST AY : DTC U0140:00

AY1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AY2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AY2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the BCM pass the network test?

Yes	GO to AY3 .
No	REFER to Section 418-00.

AY3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded DTCS from the BCM self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	REFER to Section 419-10.
No	GO to AY4 .

AY4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the IPC self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test BD . For DTC U3003:17, GO to Pinpoint Test BE .
No	GO to AY5 .

AY5 RECHECK THE IPC DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the IPC self-test.

Is DTC U0140:00 still present?

Yes	GO to AY6 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AY6 CHECK FOR DTC U0140 SET IN OTHER MODULES

- Clear all DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0140 set in the Restraints Control Module (RCM) , Driver Seat Module (DSM) or the Driver Door Module (DDM) ?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test AZ: DTC U0151:00

Normal Operation

- DTC U0151:00 (Lost Communication With Restraints Control Module: No Sub-Type Information) — sets in continuous memory in the Instrument Panel Cluster (IPC) if data messages received from the Restraints Control Module (RCM) over the High Speed Controller Area Network (HS-CAN) are missing for 5 seconds or longer. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- RCM
- IPC

PINPOINT TEST AZ : DTC U0151:00

AZ1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AZ2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AZ2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the RCM pass the network test?

Yes	GO to AZ3 .
No	REFER to Section 418-00.

AZ3 RETRIEVE THE RECORDED DTCS FROM THE RCM SELF-TEST

- Check for recorded DTCS from the RCM self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	REFER to Section 501-20B.
No	GO to AZ4 .

AZ4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the IPC self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test BD . For DTC U3003:17, GO to Pinpoint Test BE .
No	GO to AZ5 .

AZ5 RECHECK THE IPC DTCS

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the IPC self-test.

Is DTC U0151:00 still present?

Yes	GO to AZ6 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AZ6 CHECK FOR DTC U0151:00 SET IN OTHER MODULES

- Clear all DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0151:00 set in the Occupant Classification System Module (OCSM) ?

Yes	INSTALL a new RCM . REFER to Section 501-20B. CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test BA: DTC U0154:00

Normal Operation

- DTC U0154:00 (Lost Communication With Occupant Classification Sensor [OCS]: No Sub-Type Information) — sets in the Instrument Panel Cluster (IPC) if data messages received from the Occupant Classification System Module (OCSM) over the High Speed Controller Area Network (HS-CAN) are missing. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- OCSM
- IPC

PINPOINT TEST BA : DTC U0154:00

BA1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to BA2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

BA2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the OCSM pass the network test?

Yes	GO to BA3 .
No	REFER to Section 418-00.

BA3 RETRIEVE THE RECORDED DTCS FROM THE OCSM SELF-TEST

- Check for recorded DTCS from the OCSM self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	REFER to Section 501-20B.
No	GO to BA4 .

BA4 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the IPC self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test BD . For DTC U3003:17, GO to Pinpoint Test BE .
No	GO to BA5 .

BA5 RECHECK THE IPC DTCS

- Clear the DTCS. Repeat the IPC self-test.

Is DTC U0154:00 still present?

Yes	GO to BA6 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

BA6 CHECK FOR DTC U0154:00 SET IN OTHER MODULES

- Clear all DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0154:00 set in the Restraints Control Module (RCM) module?

Yes	INSTALL a new QCSM . REFER to Section 501-20B. CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test BB: DTC U0159:00

Normal Operation

- DTC U0159:00 (Lost Communication With Parking Assist Control Module "A": No Sub-Type Information) — sets in continuous memory in the Instrument Panel Cluster (IPC) if data messages received from the Parking Aid Module (PAM) over the High Speed Controller Area Network (HS-CAN) are missing for 5 seconds or longer. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- PAM
- IPC

PINPOINT TEST BB : DTC U0159:00

BB1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to BB2 .
-----	-----------------------------

No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
-----------	--

BB2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the PAM pass the network test?

Yes	GO to BB3 .
No	REFER to Section 418-00.

BB3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCs from the IPC self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test BD . For DTC U3003:17, GO to Pinpoint Test BE .
No	GO to BB4 .

BB4 RECHECK THE IPC DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the IPC self-test.

Is DTC U0159:00 still present?

Yes	INSTALL a new PAM . REFER to Section 413-13. CLEAR the DTCs. REPEAT the IPC self-test. If DTC U0159:00 is still present, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

Normal Operation

- DTC U0212:00 (Lost Communication With Steering Column Control Module: No Sub-Type Information) — sets in continuous memory in the Instrument Panel Cluster (IPC) if data messages received from the Steering Column Control Module (SCCM) over the High Speed Controller Area Network (HS-CAN) are missing for 5 seconds or longer. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- SCCM
- IPC

PINPOINT TEST BC : DTC U0212:00

BC1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to BC2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

BC2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the SCCM pass the network test?

Yes	GO to BC3 .
No	REFER to Section 418-00.

BC3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded DTCS from the IPC self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test BD . For DTC U3003:17, GO to Pinpoint Test BE .
No	GO to BC4 .

BC4 RECHECK THE IPC DTCS

- Clear the DTCS.

- Wait 10 seconds.
- Repeat the **IPC** self-test.

Is DTC U0212:00 still present?

Yes	GO to BC5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

BC5 CHECK FOR DTC U0212:87 SET IN OTHER MODULES

- Clear all DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0212:87 set in the Body Control Module (BCM) ?

Yes	INSTALL a new SCCM . REFER to Section 211-04. CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test BD: DTC U3003:16

Refer to Wiring Diagrams Cell [60](#) , Instrument Cluster for schematic and connector information.

Normal Operation

- DTC U3003:16 (Battery Voltage: Circuit Voltage Below Threshold) — sets in the Instrument Panel Cluster (IPC) in continuous memory if the **IPC** detects battery voltage below 9 volts on the voltage supply circuit for 5 seconds or longer.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- High circuit resistance
- **IPC**

PINPOINT TEST BD : DTC U3003:16

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector

NOTE: Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.

BD1 RECHECK THE IPC DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **IPC Self-Test** .
- Clear the DTCs.
- Wait 10 seconds.
- Repeat the **IPC** self-test.

Is DTC U3003:16 still present?

Yes	GO to BD2 .
No	The system is operating correctly at this time. The DTC may have been set due to a previous low battery voltage condition.

BD2 CHECK FOR CHARGING SYSTEM DTCS IN THE PCM

- Enter the following diagnostic mode on the scan tool: **PCM Self-Test** .
- Carry out the PCM Key ON Engine OFF (KOEO) self-test.
- Retrieve the continuous memory DTCs from all modules.

Are any charging system related DTCs set in the PCM?

Yes	REFER to Section 414-00.
No	GO to BD3 .

BD3 CHECK THE BATTERY CONDITION AND STATE OF CHARGE

- Check the battery condition and verify the battery is fully charged. Refer to Section 414-01.

Is the battery OK and fully charged?

Yes	GO to BD4 .
No	REFER to Section 414-01.

BD4 CHECK THE IPC VOLTAGE PID

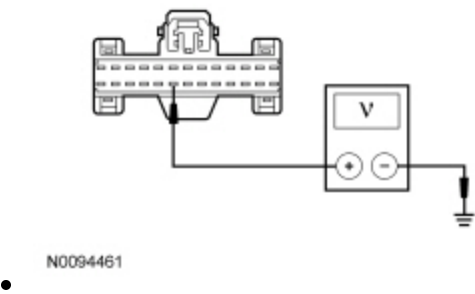
- Ignition ON.
- Measure and record the voltage at the battery.
- Enter the following diagnostic mode on the scan tool: **IPC DataLogger** .
- Monitor the **IPC** voltage PID (MODULE_VOLT).

Is the voltage within 0.2 volt of the recorded battery voltage?

Yes	GO to BD7 .
No	GO to BD5 .

BD5 CHECK THE IPC VOLTAGE SUPPLY

- Ignition OFF.
- Disconnect: **IPC** [C220](#) .
- Ignition ON.
- Measure the voltage between the **IPC** [C220](#) Pin 21, circuit SBP11 (BU/RD), harness side and ground.

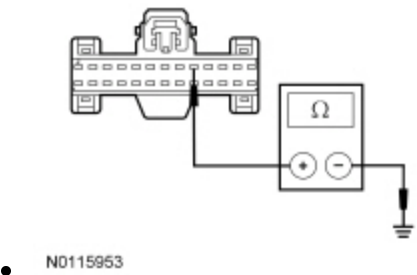


Is the voltage within 0.2 volt of the recorded battery voltage?

Yes	GO to BD6 .
No	REPAIR circuit SBP11 (BU/RD) for high resistance. CLEAR the DTC. REPEAT the self-test.

BD6 CHECK THE IPC GROUND CIRCUIT

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the **IPC** [C220](#) Pin 5, circuit GD133 (BK), harness side and ground.



Is the resistance less than 5 ohms?

Yes	GO to BD7 .
No	REPAIR circuit GD133 (BK) for high resistance. CLEAR the DTCs. REPEAT the self-test.

BD7 CHECK FOR CORRECT IPC OPERATION

- Disconnect the **IPC** connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the **IPC** connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. REFER to Instrument Panel Cluster (IPC). TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test BE: DTC U3003:17

Refer to Wiring Diagrams Cell [60](#) , Instrument Cluster for schematic and connector information.

Normal Operation

- DTC U3003:17 (Battery Voltage: Circuit Voltage Above Threshold) — sets in the Instrument Panel Cluster (IPC) in continuous memory if the **IPC** detects battery voltage above 16 volts on the voltage supply circuit for 5 seconds or longer.

This pinpoint test is intended to diagnose the following:

- Charging system concern
- **IPC**

PINPOINT TEST BE : DTC U3003:17

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: <i>DTC U3003:17 may be stored in the module memory due to past battery charging or vehicle jump starting events.</i>

BE1 CHECK FOR LOW VOLTAGE DTCS SET IN OTHER MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is a low voltage DTC set in more than one module?

Yes	REFER to Section 414-00 to diagnose an overcharging condition.
No	GO to BE2 .

BE2 CHECK THE BATTERY VOLTAGE

- Turn off all interior/exterior lights and accessories.
- Start and run the engine at approximately 2,000 rpm for 3 minutes while monitoring the battery voltage.

Does the battery voltage rise to 15.5 volts or higher?

Yes	REFER to Section 414-00 to diagnose an overcharging condition.
No	GO to BE3 .

BE3 RECHECK FOR DTC U3003:17

- Turn the engine off.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Clear the continuous memory DTCs.
- Carry out the ~~IPC~~ self-test.

Is DTC U3003:17 present?

Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.
No	The system is operating normally at this time. The DTC may have been set previously during battery charging or while jump starting the vehicle.

Pinpoint Test BF: The Low Fuel Message Center Indicator Is Never/Always On

Normal Operation

The low fuel message center indicator illuminates when the remaining fuel level is 1/16 tank (with an administrator key) or 1/8 tank (with a MyKey® programmed key) or less.

This pinpoint test is intended to diagnose the following:

- Fuel gauge indication concern
- ~~IPC~~

PINPOINT TEST BF : THE LOW FUEL MESSAGE CENTER INDICATOR IS NEVER/ALWAYS ON

BF1 CHECK THE MESSAGE CENTER WARNING INDICATOR FUNCTION

- Close all doors, hood and liftgate.
- Ignition ON.
- **NOTE:** *If multiple warnings are present, the message center cycles through each of the warning indicators. Allow the message center to cycle through each display while determining whether the door ajar message center warning indicator is active.*

Open the driver door while monitoring the door ajar message center warning indicator.

Does the driver door ajar message center warning indicator turn off with the driver door closed and illuminate with the driver door open?

Yes	GO to BF2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

BF2 CHECK THE FUEL GAUGE OPERATION

- Verify the fuel gauge is operating correctly and is not registering empty (E) or full (F) all the time.

Does the fuel gauge operate correctly?

Yes	GO to BF3
No	GO to Pinpoint Test B .

BF3 CHECK THE DTE REMAINING MILEAGE DISPLAYED

- Enter the dealer test mode. Refer to Dealer Test Mode in Inspection and Verification in this section.
- **NOTE:** *The display below uses xxxx's to represent a numeric or alpha-numeric value. The value may display the same amount of characters represented by the xxxx's or there may be more/less depending on the input value.*

Scroll through the displays until FUEL LEVEL% STATUS xxx (base ~~IPC~~) or FUEL LVL FILT FLPS IN xxxx (optional ~~IPC~~) is displayed

Does the fuel level value indicate 41 (with a MyKey®) or 26 (with an administrator programmed key) or less?

Yes	If the low fuel message center indicator is never on, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation. If the low fuel message center warning indicator is always on, the system is operating correctly at this time.
No	If the low fuel message center warning indicator is always on, INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation. If the low fuel message center warning indicator is never on, the system is operating correctly at this time. The low fuel message center indicator illuminates when the fuel level is at or below 1/16 tank (with an administrator key) or 1/8 tank (with MyKey® programmed key).

Pinpoint Test BG: The ControlTrac 4WD Gauge Is Inoperative

Normal Operation

The ControlTrac 4WD gauge indicates the level of power applied to each wheel in a continuous display over the vehicle. The Instrument Panel Cluster (IPC) uses multiple inputs to determine the ControlTrac 4WD gauge display. The first is an engine torque message sent to the IPC from the PCM over the High Speed Controller Area Network (HS-CAN) . The second is an AWD torque message sent from the Transfer Case Control Module (TCCM) to the IPC over the HS-CAN . The third is wheel speed data sent from the PCM to the IPC over the HS-CAN .

As power is applied to the wheels, the display area around the wheels begin to light up and fill in. The lowest power is displayed closest to the wheel. As the amount of power increases, the display begins to fill rearward and/or forward of the vehicle.

If the engine torque data, wheel speed data or AWD torque message is missing for 5 seconds or less or if the IPC receives invalid engine torque data, wheel speed data or AWD torque for 5 seconds or less, the IPC defaults the display area fill around the wheels off.

If the IPC does not receive the engine torque data or wheel speed data from the PCM for more than 5 seconds, the IPC sets DTC U0100:00 in continuous memory and defaults the display area fill around the wheels off.

If the IPC does not receive the AWD torque message from the TCCM for 5 seconds, the IPC sets DTC U0401:00 in continuous memory and defaults the display area fill around the wheels off.

NOTE: If DTC U0100:00 sets in the IPC , other observable symptoms may be an inoperative tachometer, temperature gauge or odometer.

This pinpoint test is intended to diagnose the following:

- 4WD concern
- IPC

PINPOINT TEST BG : THE CONTROLTRAC 4WD GAUGE IS INOPERATIVE

BG1 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST	
• Check for recorded IPC DTCS from the self-test.	
Are any DTCS recorded?	
Yes	REFER to DTC Charts in this section.
No	GO to BG2 .
BG2 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST	
• Check for recorded PCM DTCS from the Key ON Engine OFF (KOEO) self-test.	
Are any DTCS recorded?	
Yes	For all 4WD related DTCS, REFER to Section 308-07A. For all other DTCS, REFER to Section 303-14. TEST the system for normal operation.
No	GO to BG3 .

BG3 RETRIEVE THE RECORDED DTCS FROM THE TCCM SELF-TEST

- Check for recorded PCM DTCs from the TCCM self-test.

Are any DTCs recorded?

Yes	REFER to Section 308-07A.
No	GO to BG4 .

BG4 CHECK THE DISPLAY MODE MENU SETTING

- In the gauge display mode, check the ControlTrac 4WD gauge.

Does the ControlTrac 4WD gauge display?

Yes	The ControlTrac 4WD gauge is operating correctly at this time. The display mode was not set to correct display mode.
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

Pinpoint Test BH: Incorrect Or Fluctuating DTE

NOTE: The DTE display and the fuel gauge both use the fuel level input from the fuel tank to provide their respective functions. If the fuel gauge doesn't function correctly, both the fuel gauge and the DTE display are affected.

NOTE: The actual DTE can be higher or lower than the DTE displayed in the message center due to changes in driving conditions. It is important to understand how the DTE is calculated and the factors that impact the DTE display when determining how to address any DTE concerns.

Normal Operation

The DTE is calculated in the IPC using the Running Average Fuel Economy (RAFE), which is the fuel economy over the last 480 km (300 miles), and the fuel level input from the fuel sender(s) to determine how many miles the vehicle can be driven based on the remaining fuel in the tank. The DTE can vary in the short term by up to 50 miles, but is usually within 10 miles.

NOTE: Even if the fuel economy is relatively constant, the DTE can be off over a 50 mile range by -24% to +38%.

The IPC defaults to around 18 mpg when the battery is initially connected and changes based on driving habits and conditions. Generally, the driver should be getting somewhere between 13-21 mpg.

NOTE: If a concern exists related to an erratic DTE display, check the message center menu for a DTE calculation setting and change to normal use, if available. Changing to Normal uses a longer driving history and stabilizes DTE display.

Since the DTE is calculated and averaged over a longer period of time (480 km [300 miles]), varying driving conditions can have a significant impact on the current or short term DTE as opposed to the displayed DTE. This difference often leads to customer complaints of incorrect or invalid DTE. The following list provides some (not all) of the driving conditions that may lead to an incorrect or fluctuating DTE concern:

- Changing between towing/not towing
- Changing driving between city and highway
- Allowing the vehicle to idle for long periods of time
- Using the remote start feature frequently to allow the vehicle to warm up, particularly when parked on a grade
- Parking or driving on grades
- Inconsistent use of gasoline or E85 fuels
- Over-fueling or not filling the tank completely (partial refueling)

To better illustrate the affects of how driving conditions can affect DTE , please refer to the 2 following examples. The first example below illustrates how the following observations are normal and expected since the low fuel reminder is triggered from a fuel volume and not from a fixed distance to empty.

- If while driving, the low fuel reminder (low fuel indicator and low fuel warning message) displays when the DTE equals 94.4 km (59 miles) and the driver adds 11.36 L (3 gallons) of fuel, the new DTE may become 124.8 km (78 miles). After continued driving, the low fuel reminder may now display when the DTE equals 83.2 km (52 miles).

The next example (below) illustrates what occurs when idling on an incline. In this example, the customer should be made aware of how the condition will correct after a few minutes of idling on a level surface.

- If the customer stops and parks the vehicle on an incline in a driveway, then in the morning remote starts the vehicle, allowing the engine to idle, the DTE may now equal 184 km (115 miles). As the customer drives, the low fuel reminder displays when the DTE equals 148.8 km (93 miles). Finally, after 5 more minutes of driving, the DTE is back to 80 km (50 miles).

This pinpoint test is intended to diagnose the following:

- Fuel gauge concern
- Changes in driving conditions, such as:
 - towing.
 - extended idle times.
 - parking on any grades.
 - alcohol in the fuel.
 - refueling patterns.
- IPC

PINPOINT TEST BH : INCORRECT OR FLUCTUATING DTE

BH1 REVIEW AND INVESTIGATE THE FOLLOWING DRIVING CONDITIONS WITH THE CUSTOMER

NOTE: Make sure to record the Running Average Fuel Economy (RAFE) prior to disconnecting the battery, removing the IPC or BCM fuse 11 (10A). If any of the previous conditions occur, the vehicle must be driven over time to allow the DTE calculations to adjust.

- Determine whether the driver changes the following driving conditions:
 - Change between towing/not towing, city/highway driving.
 - Use of extended idle times.
 - Use of remote start frequently.
 - Park/drive on grades.
 - Use E85 fuel exclusively or switch to non-Ethanol fuel.
 - Full refill the fuel tank or not fill the tank completely (partial refueling).

Does the driver change driving conditions?

Yes	Advise the customer how changing driving conditions can affect the current <u>DTE</u> as opposed to the <u>DTE</u> displayed.
No	GO to BH2 .

BH2 CHECK THE FUEL GAUGE FOR CORRECT OPERATION

- Ignition ON.
- **NOTE:** *If necessary interview the customer to find out if there are any fuel gauge indication concerns.*

Verify the fuel gauge is functioning and correctly indicating the fuel level.

Is the fuel gauge operating correctly?

Yes	GO to BH3 .
No	GO to Pinpoint Test B .

BH3 CHECK THE DTE CALCULATION SETTING IN THE MESSAGE CENTER

- Ignition ON.
- Using the message center navigate to the DTE calculation setting and check to see if Towing History Used is selected.

Is Towing History Used selected for the DTE calculation?

Yes	If the customer is not towing a trailer all the time on hilly terrains, explain that using this setting may cause the <u>DTE</u> display to appear erratic.
No	GO to BH4 .

BH4 COMPARE THE DISPLAYED DTE WITH THE IPC INPUTS

- **NOTE:** *The vehicle VIN must be entered to enable the OASIS tab. Navigate to PTS > OASIS > HvBOM> Window Sticker to find the fuel tank the vehicle was built with.*

Locate and record the fuel tank capacity (example: 28 gallons) using the as-built record.

- Enter the following diagnostic mode on the scan tool: IPC DataLogger .
- Select the IPC fuel gauge (FUELLEVEL) active command.
 - Record the current fuel level percent displayed on the scan tool.
- Select the IPC DTE (DIST_EMP) PID .
 - Record the instantaneous DTE .
- Determine the current DTE .
 - Multiply the fuel tank capacity by the fuel percent to determine how many gallons of fuel remain in the tank (example: 28 gallons X 50% [.50] = 14 gallons of remaining fuel).

Multiply the calculated fuel remaining in the tank by the RAFE (example: 14 gallons X 20 MPG = 448 km (280 miles) DTE).
- Compare the DTE displayed in the message center with the current DTE .

Does the current DTE match the displayed DTE ?

Yes	The system is operating correctly. REVIEW with the customer the system operation and the conditions that may affect DTE .
No	The vehicle configuration is most likely incorrect for the fuel tank size. INSTALL As-Built data from Professional Technician Society (PTS) following scan tool instructions under Module Programming>As-Built.

Pinpoint Test BI: The The Shift To Park Message Is Never Or Always Displayed

Normal Operation

The IPC displays a shift to park message when the vehicle is turned off with the shifter out of the PARK (P) position. The IPC monitors the park position detect switch input through the park detect signal circuit. When the selector lever is out of the PARK (P) position, the park detect switch closes to ground and the IPC detects the reference voltage pulled low. The IPC monitors the ignition state and when the ignition is OFF with the selector lever out of PARK (P), the IPC displays the shift to park message.

This pinpoint test is intended to diagnose the following:

- Message center concern
- PRNDL indication concern
- Communication concern
- IPC

PINPOINT TEST BI : THE THE SHIFT TO PARK MESSAGE IS NEVER OR ALWAYS DISPLAYED

BI1 CHECK THE MESSAGE CENTER OPERATION

- Close all doors.
- Ignition ON.
- Open the driver door while monitoring the message center displays.

Does the message center display DRIVER DOOR AJAR?

Yes	GO to BI2 .
No	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. TEST the system for normal operation.

BI2 CHECK THE PRNDL DISPLAY OPERATION

- Ignition ON.
- Shift the transmission through the ranges and back to PARK (P).

Does the PRNDL display the correct gear including PARK (P)?

Yes	GO to BI3 .
No	DIAGNOSE the PRNDL display. GO to Pinpoint Test X .

BI3 RETRIEVE THE RECORDED DTCS FROM THE IPC SELF-TEST

- Check for recorded IPC DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to DTC Charts in this section.
No	GO to BI4 .

BI4 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded BCM DTCs from the self-test.

Are any DTCs recorded?

Yes	Section 419-10.
No	GO to BI5 .

BI5 CHECK FOR CORRECT IPC OPERATION

- Disconnect the IPC connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the IPC connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. REFER to Instrument Panel Cluster (IPC). TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.